

DCP 700
Digital Control Programmer
Product Manual

57-77-25-07

DCP700

Digital Control Programmer

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**DCP 700
Digital Control Programmer****Installation**

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Inspecting for Shipping Damage and Checking for Accessories

If the DCP hasn't been removed from its shipping carton, inspect the carton for damage, and remove the DCP. Inspect the DCP for any obvious shipping damage and report any damage due to transit to the carrier. Check that a bag containing mounting hardware, and another bag containing stick-on labels for identification of PV and set point units are included in the carton with the DCP.

Reviewing Data Before Installation

While installing the DCP is straightforward, we recommend that you review the following data before installation.

Ambient Conditions and Design Limits During Operation

Temperature: 32 to 122°F (0 to 50°C)

Relative humidity: 10 to 90%

Vibration: 0 to 0.02G

Power:

Voltage 120 Vac $\pm$ 12Vac

Frequency 50/60Hz $\pm$ 1Hz

Consumption 40 VA

Mounting angle: $\pm$ 10°

Stray rejection:

Common mode 120 db down @ 250Vac

Normal mode 60 db down @ 2x span

Weight: Approximately 13 lb. (6kg)

Mounting Dimensions

(See Figure 2-1).

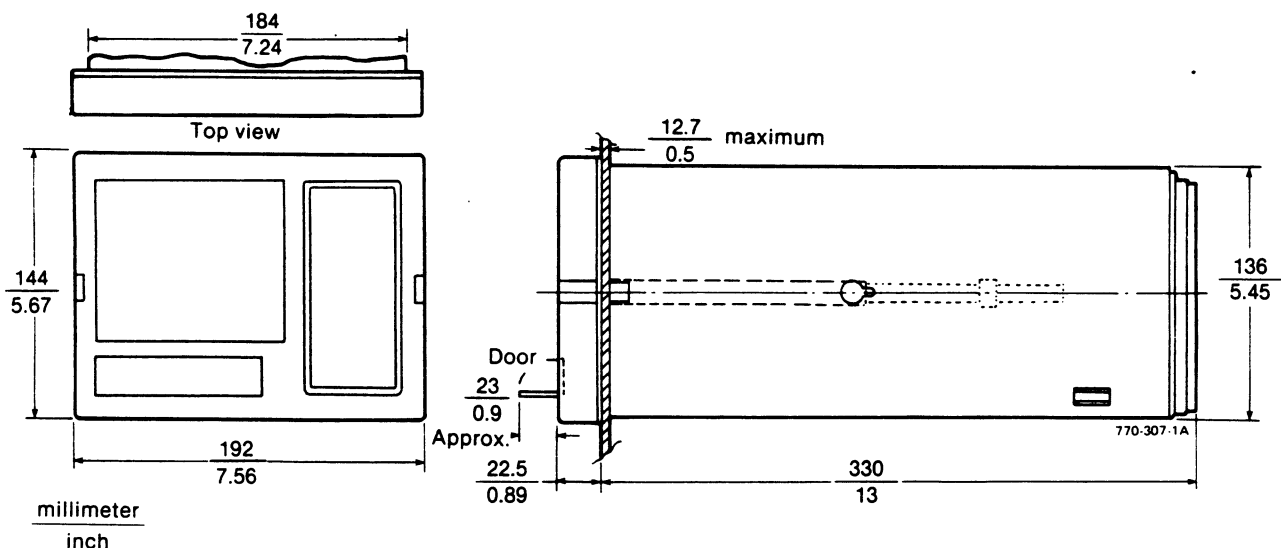


Figure 2-1—Mounting Dimensions

Electrical Noise

NOTE: For additional information about electrical noise, refer to STD-518-1977 of the Institute of Electrical and Electronic Engineers.

Electrical noise is unwanted electrical signals in the input and output wires of an instrument. If superimposed on a valid electrical signal, this noise can impair the performance of your DCP.

Most electrical noise problems can be eliminated by proper routing and shielding of external wires connected to the DCP and by suppressing the noise at the source (usually external equipment connected to the DCP).

The following is a list of wires, identified by the signal carried, that must be shielded and that can be routed (bundled) together.

NOTE: Bundle numbers are for reference only.

Signal carried	Twist	Shield	Bundle Number	Bundle combos
Power	No	No	1	4, 7, 8
PV	Yes	Yes	2	None
Event coils	Yes	Yes	3	5
Event contacts	No	No	4	1, 7, 8
External contacts	Yes	Yes	5	3
Current output	Yes	Yes	6	None
Time Prop. output	No	No	7	1, 4, 8
Ground	No	No	8	1, 4, 7

Suppressing Noise at the Source

The following are methods of using suppression devices on external equipment connected to the DCP.

- If a dc relay actuates the contacts to operate the program, install a diode (Table 2-1) across the relay coil.
- Do not use the extra contacts on the relays to switch loads larger than 5mA, 24Vdc.
- If an event relay is used to switch an inductive load, connect a resistor capacitor network (Table 2-1) across the contacts, and a metal-oxide varistor (Table 2-1) across the wires to the load (Figure 2-2, 2-3).

TABLE 2-1—Devices for noise suppression

Type	Part number	
	Honeywell	Commercial
Diode	30723408-002	1N4003
R-C Network	30371852-001	47 ohm-0.1mfd
Metal-oxide varistor (Max. steady state voltage: 25 Vac rms). (Max. transient energy: + 6 Joules).	30732481-003	GEV39ZA6

Supressing Noise at the Source (continued)

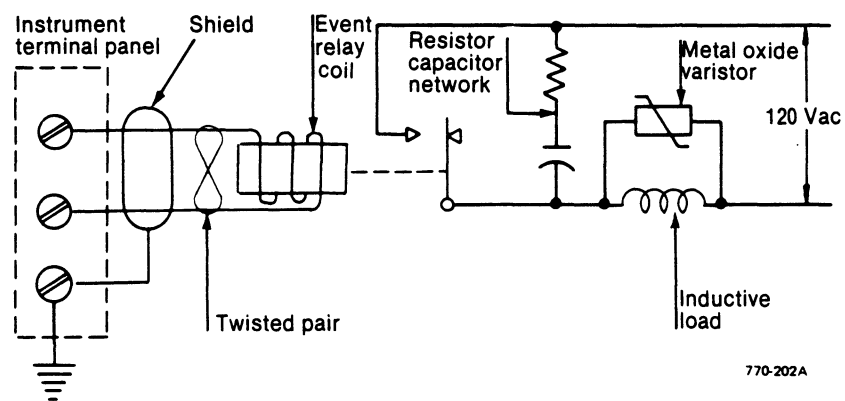


Figure 2-2—Switching an inductive load

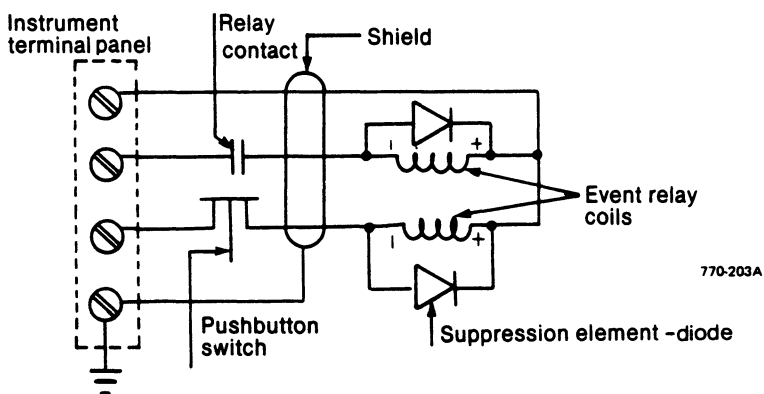


Figure 2-3—Contacts in an event relay coil circuit

Optional Operational Check

If the displays do not conform to the results specified in Steps 3 through 7 in the figure below, contact the nearest Honeywell Process Control Division branch office for assistance.

1. Connect terminals 15 and 16 on the rear terminal block to 120VAC 50/60 Hz power, and connect Terminal 14 to ground then supply power to the DCP.

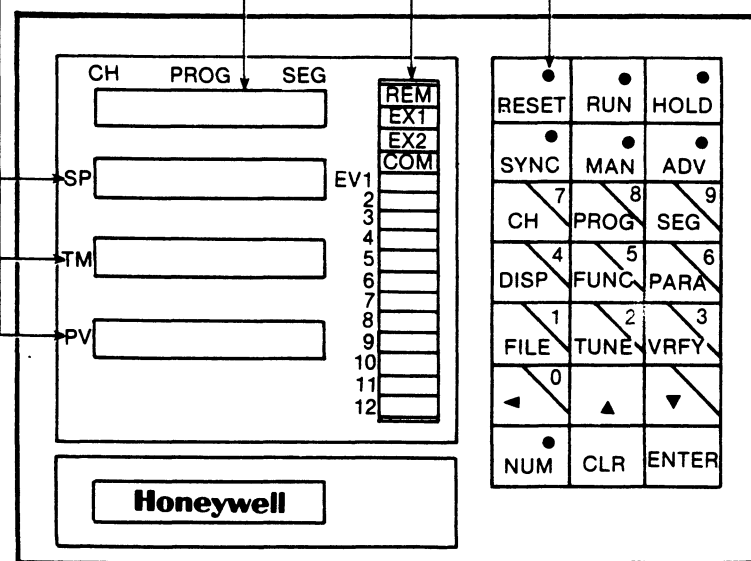
2. Select CH 1; then press **RESET**, **PROG**, **ENTER**, **RUN**.

3. Make sure the displays in the SP, TM, and PV windows are flashing the numbers 1 through 9, then a decimal, and that all of the numbers are formed correctly.

4. Make sure the eight displays in the upper window are flashing the numbers 1 through 9 then the letter M, and that the numbers and letter are formed correctly.

5. Make sure that a rectangular indicator is flashing next to or under each of the identifiers in this row.

6. Make sure there is an indicator flashing on each of the seven keys with the symbol "•" in the upper right corner.



NOTES: 1) In the following step, do not press RESET until you have pressed the keys you want to check as this causes Program 0 to stop.
2) The keys in the list (except RESET) can be pressed randomly.

7. Press each key and while pressing, make sure the number appearing in the SP window corresponds to the one next to the key in the following listing:

Key	Code	Key	Code
RUN	2	FILE	13
HOLD	3	TUNE	14
SYNC	4	VRFY	15
MAN	5	←	16
ADV	6	▲	17
CH	7	▼	18
PROG	8	NUM	19
SEG	9	CLR	20
DISP	10	ENTER	21
FUNC	11	RESET	(none)
PARA	12		

Figure 2-4—Optional operational check

Mounting the DCP

Equipment you will need

- Tools for making the panel cutout
- Medium-bladed screwdriver
- Mounting hardware furnished with the DCP.

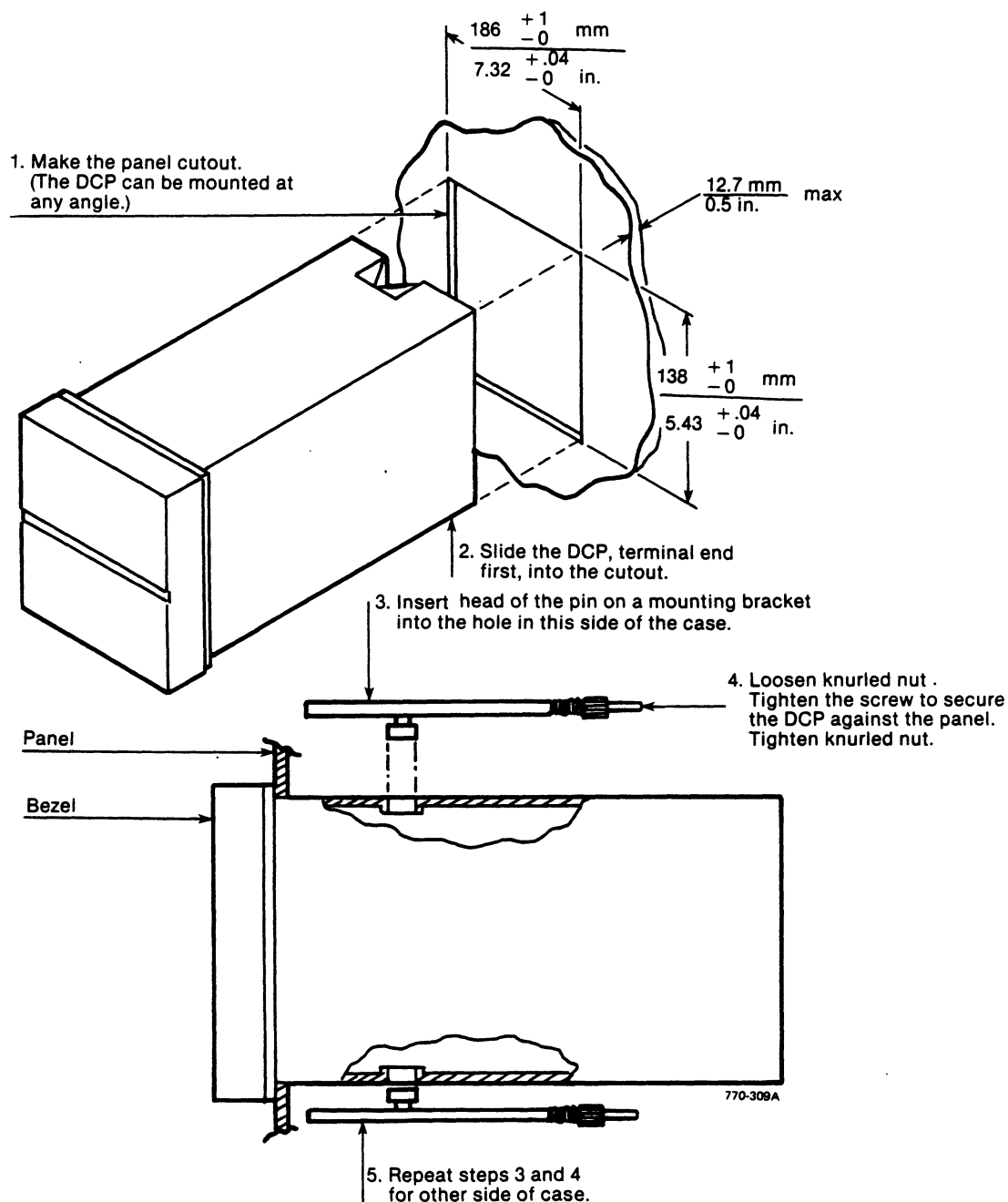


Figure 2-5—Mounting the DCP

External Wiring

Connections to the DCP are shown in Figures 2-6 through 2-16. Information is included in the figures to guide you in connecting external equipment, and, where needed, to inform you that programming is required to allow the external equipment to function as intended. Wiring connections are shown in the following figures:

- | | |
|--|-------------|
| ■ Power and ground | Figure 2-6 |
| ■ Input | Figure 2-7 |
| ■ Current proportioning control or programmer output | Figure 2-8 |
| ■ Time proportioning output | Figure 2-9 |
| ■ External power supply | Figure 2-10 |
| ■ Events | Figure 2-11 |
| ■ Auxiliary output | Figure 2-12 |
| ■ External operation contacts | Figure 2-13 |
| ■ Program number selection | Figure 2-14 |
| ■ Alarm outputs | Figure 2-15 |
| ■ Communication | Figure 2-16 |
| ■ Composite | Figure 2-17 |

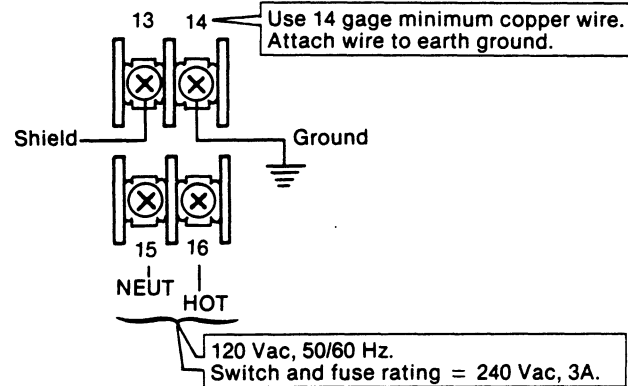
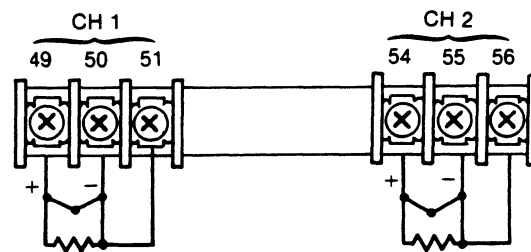


Figure 2-6—Power and ground

770-310A



For RTD, use three copper wires of equal resistance twisted together and shielded.

For RH, connect the dry bulb to the CH1 terminals and the wet bulb to CH2 terminals

For current and voltage inputs, use two copper wires twisted together and shielded.

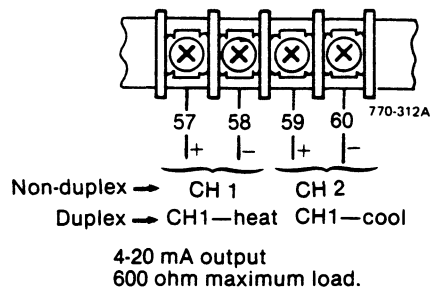
For thermocouple inputs, use twisted and shielded leadwires of the correct calibration.

NOTE: Connect shields to Terminal 13
Do not ground other end.

770-311A

Figure 2-7—Input

External Wiring (continued)



NOTE: Connect shields to Terminal 13
Do not ground other end.

Figure 2-8—Current proportioning control or programmer output

NOTE: User-supplied relay required for each event circuit used. Sum of event, alarm, and time proportional controller relays must not exceed 15.

- Rating 24 Vdc, 75 mA. Order separately Honeywell 30731674-001 relay and 30731675-001 socket for plug-in or 30736332-001 for solder in.

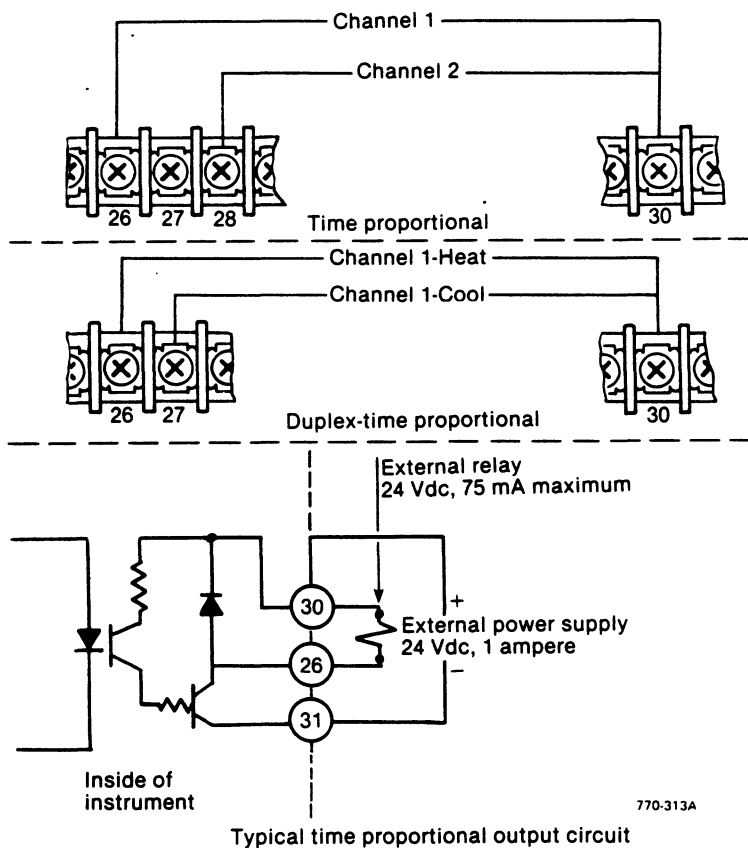
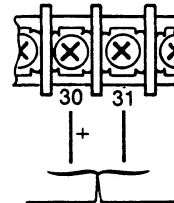


Figure 2-9—Time proportioning output

External Wiring (continued)

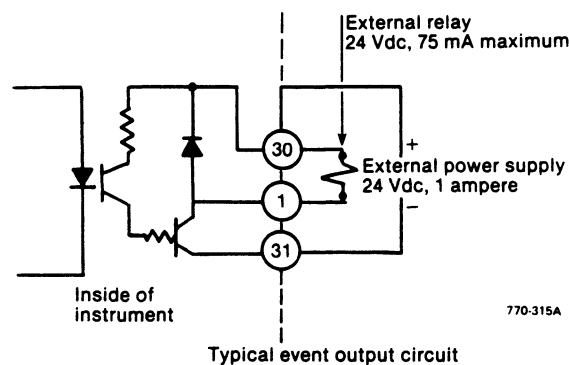
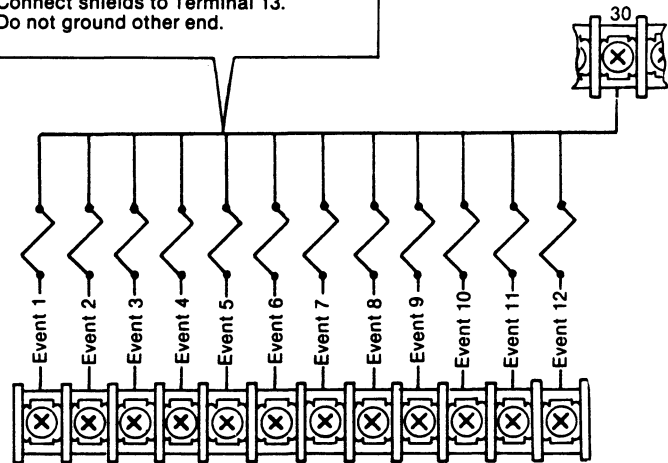


User-supplied power supply required for operation of the event and time proportional control relays, and alarm outputs.
Rating = 24 Vdc $\pm$ 5%, 1A
(Honeywell 30683383-001 can be ordered separately to meet this requirement)

770-314A

Figure 2-10—External power supply

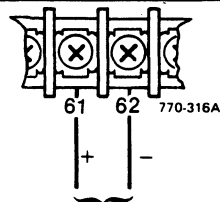
- 1) User-supplied relay required for each event circuit used.
Sum of event, alarm, and time proportional controller relays must not exceed 15.
• Rating 24 Vdc, 75 mA. Order separately Honeywell 30731674-001 relay and 30731675-001 socket for plug-in or 30736332-001 for solder in.
- 2) Type of event (time-, PV-, and DEV-based) is established during programming.
- 3) Connect shields to Terminal 13. Do not ground other end.



770-315A

Figure 2-11—Events

External Wiring (continued)



4-20 mA output
600 ohm maximum load

NOTE: 1) Connect shields to Terminal 13.
Do not ground other end.

2) Output is linear and can be configured
(See Configuration section) to represent one of the following (for either channel of a two channel DCP):
Set point
Process variable
Output

Figure 2-12—Auxiliary (Recorder) output

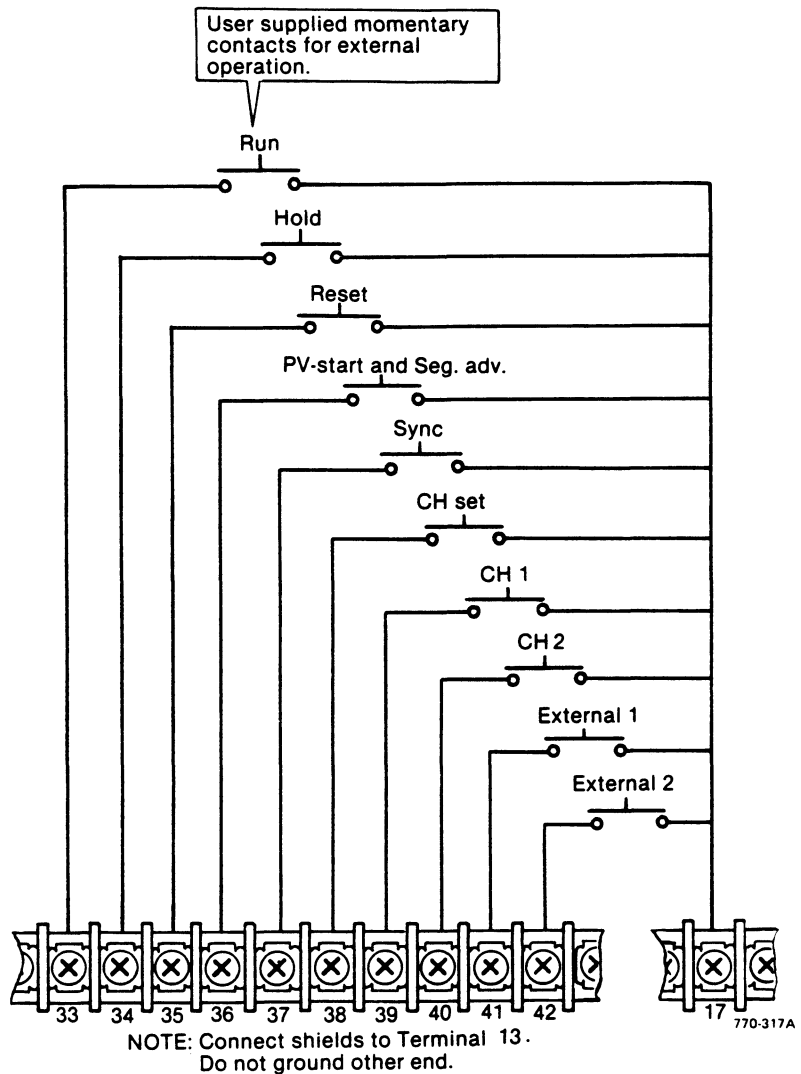
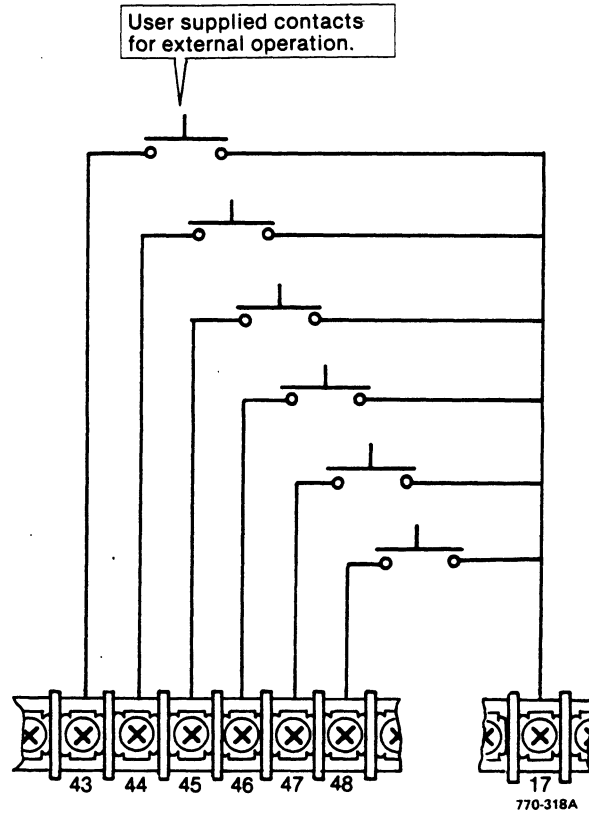


Figure 2-13—External operation contacts

External Wiring (continued)

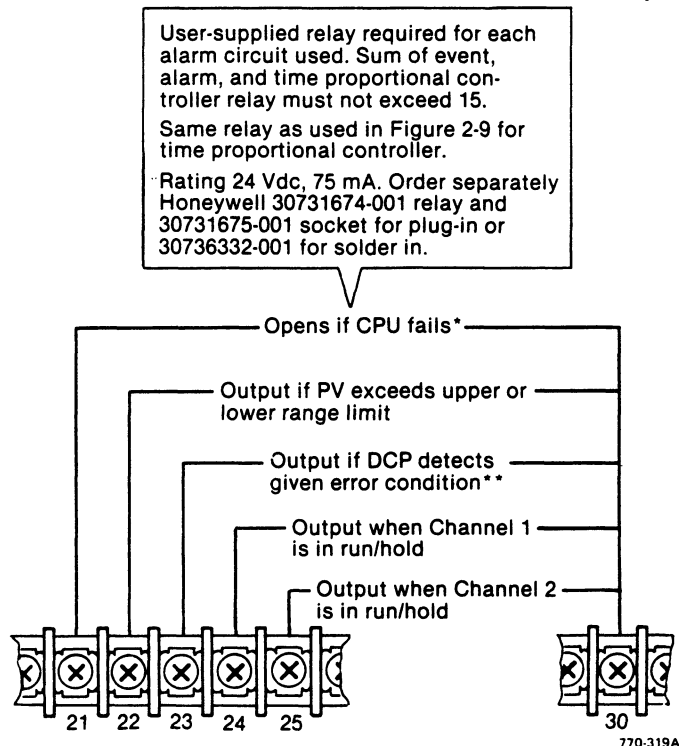


NOTE: Connect shields to Terminal G.
Do not ground other end.

Prog	Closed contacts	Prog	Closed contacts
1	17-43	21	17-43, 17-48
2	17-44	22	17-44, 17-48
3	17-43, 17-44	23	17-43, 17-44, 17-48
4	17-45	24	17-45, 17-48
5	17-43, 17-45	25	17-43, 17-45, 17-48
6	17-44, 17-45	26	17-44, 17-45, 17-48
7	17-43, 17-44, 17-45	27	17-43, 17-44, 17-45, 17-48
8	17-46	28	17-46, 17-48
9	17-43, 17-46	29	17-43, 17-46, 17-48
10	17-47	30	17-47, 17-48
11	17-43, 17-47	31	17-43, 17-47, 17-48
12	17-44, 17-47	32	17-44, 17-47, 17-48
13	17-43, 17-44, 17-47	33	17-43, 17-44, 17-47, 17-48
14	17-45, 17-47	34	17-45, 17-47, 17-48
15	17-43, 17-45, 17-47	35	17-43, 17-45, 17-47, 17-48
16	17-44, 17-45, 17-47	36	17-44, 17-45, 17-47, 17-48
17	17-43, 17-44, 17-45, 17-47	37	17-43, 17-44, 17-45, 17-47, 17-48
18	17-46, 17-47	38	17-46, 17-47, 17-48
19	17-43, 17-46, 17-47	39	17-43, 17-46, 17-47, 17-48
20	17-48		

Figure 2-14—External program number selection

External Wiring (continued)



NOTES: 1) Output is 24 Vdc, 75 mA.
Can operate the same type relay used for events. See Figure 2-11.
2) Connect shields to Terminal 13.

*Note that the logic for the CPU failure output differs from the other outputs. The signal is turned off, indicating that CPU operation has ceased, if the watchdog timer circuit times out.

**Error conditions include:

- Error in external contact operation
- "IF EXT/ONEXT" function cannot run because program data was not entered (ERROR 04)
- External 1 or 2 signal has not been detected for execution of RAMP-E segment (ERROR 06 or ERROR 07)
- Time event cannot be executed because area of timer block has been filled (ERROR 08)
- Shed time has elapsed and communication with host has been interrupted

Figure 2-15—Alarm outputs

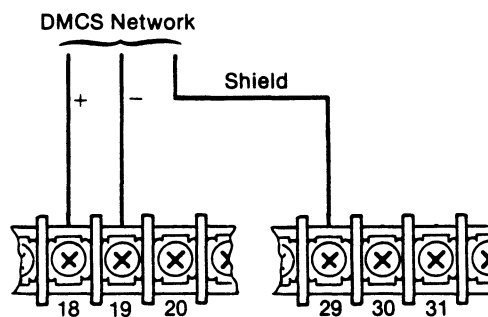


Figure 2-16—Communication cable

External Wiring (continued)

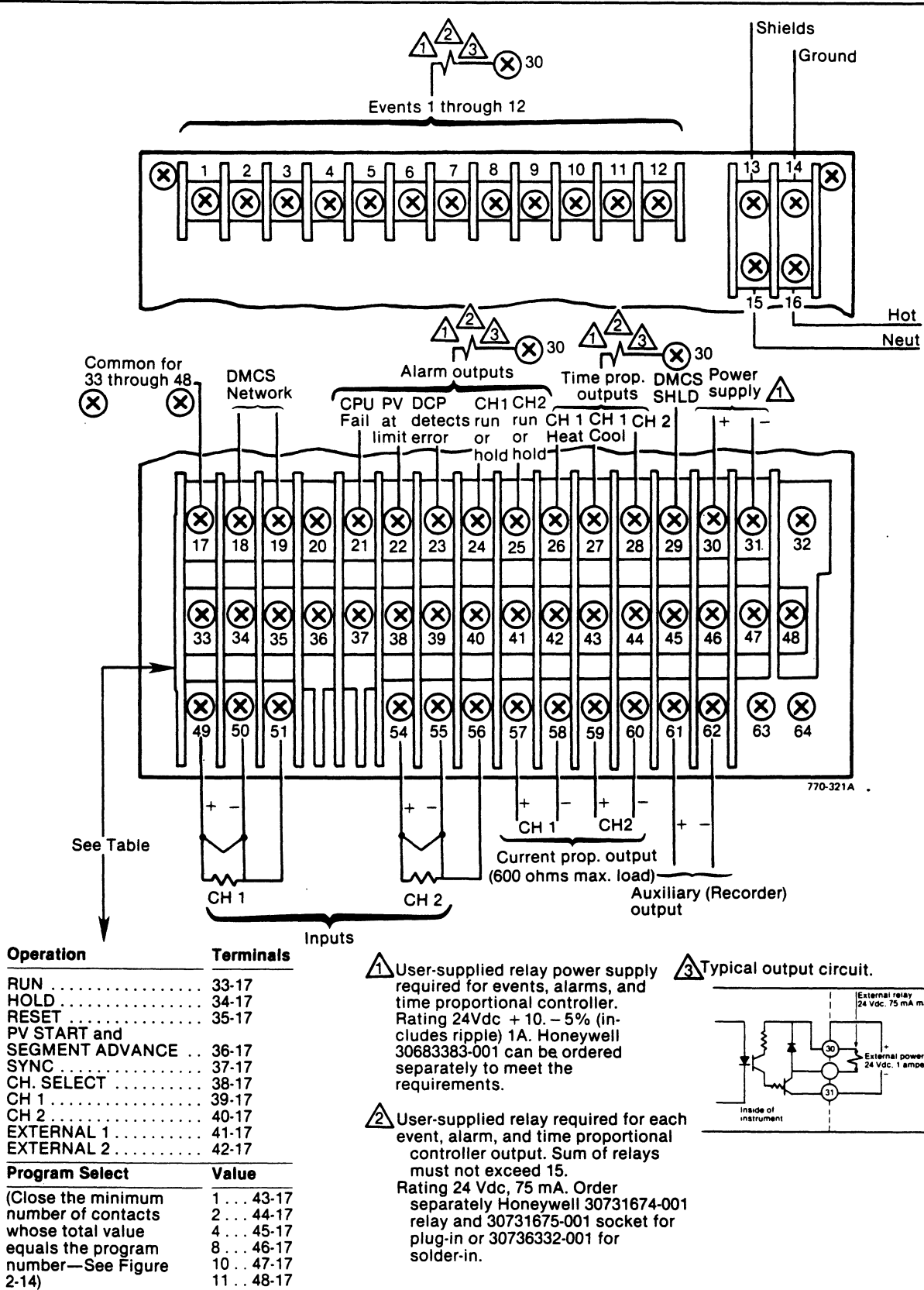


Figure 2-17—Composite external wiring diagram

DCP 700 Digital Control Programmer

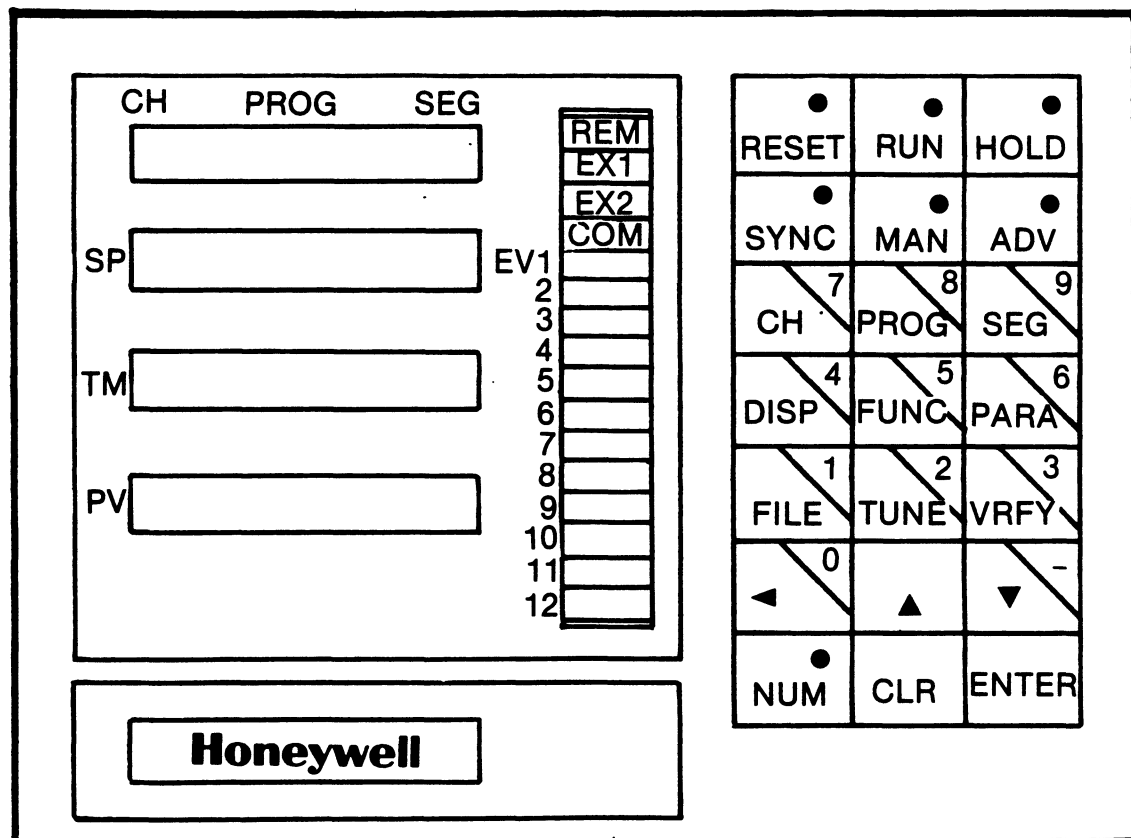
Operator Interface

Introduction

The DCP's operator interface comprises these three main elements:

- Keyboard
- Digital Displays
- Light Emitting Diodes (LEDs)

The keyboard, displays, and LEDs combine to form an interactive interface that is easy to use.



770-308

Figure 3-1—Keyboard, displays, and LEDs make up the DCP Operator Interface

Keyboard

The keyboard is a matrix of membrane type keys with audible feedback to confirm key activation. Some keys also include LEDs for visual indication of operating status.

Review the following key descriptions so you will be able to quickly interpret a key's function by its label.

Key Description

RESET When lit, DCP is in ready mode – no program is running and DCP output is value set under Configuration parameter RDY-OUT.

In ready mode, exits programming configuration, or tuning modes. In Run or Hold mode, press with  shift key to return DCP to ready mode – program returns to segment 1. If SYNC LED is lit, both channels are reset. Does not work if DCP is in manual mode.

SYNC When lit, DCP is in synchronous mode and any action initiated in one channel is also initiated in the other.

Alternately switches DCP between synchronous and asynchronous modes. In asynchronous mode, any action initiated is valid only in the channel selected.

RUN When lit, DCP is in run mode – a program is running. Start a program from ready mode, restart a program or jump to another program from hold mode. If SYNC LED is lit, both channels are started. Use with  or  to make a PV (Hot) start – see Operation section.

MAN When lit, DCP is in manual mode of operation. Alternately switches DCP between manual and automatic modes of operation. In manual mode, use ,  and  keys to adjust output value one digit at a time or number keys to change all digits at the same time.

HOLD When lit, DCP is in hold mode – a program is being held in channel and segment displayed.

Stops all timed channel operation such as soak deviation and time event, and maintains set point at present value. If SYNC LED is lit, both channels are stopped – unless one channel was not in run mode.

ADV When lit, time base for program is accelerated by factor configured for the ADV-RATE parameter.

Alternately switches DCP between configured time unit and accelerated time base.

Use with  to interrupt segment execution and advance program to next segment.

Key Description — Continued

CH Alternately selects Channel 1 or Channel 2. In numerical mode, enter 7 as integer for keyed in value. DISP In run or hold mode, alternately select time remaining in a segment or % output for indication in TM display. Use to exit verify, tune, or configuration operations. Does not work in manual mode of operation In numerical mode, enter 4 as integer for keyed in value. Used with to display cycle number being run.	PROG Select program number through numerical keys. Use with to put DCP in programming mode. In numerical mode, enter 8 as integer for keyed in value. FUNC Initiate function prompts for individual segment settings during programming. In numerical mode, enter 5 as integer for keyed in value.	SEG Select segment number through numerical keys. In numerical mode, enter 9 as integer for keyed in value. In programming mode, select next segment in sequence. PARA Initiate parameter prompts for setting during configuration. In numerical mode, enter 6 as integer for keyed in value.
FILE Initiate upload or download of programs with memory module installed. In numerical mode, enter 1 as integer for keyed in value. Multifunction shift key is used in conjunction with other keys to enter value and change modes of operation. In numerical mode, enter 0 as integer for keyed in value. NUM When lit or blinking, DCP is in numerical mode. Initiates numerical mode so values can be entered using numerical function keys.	TUNE In Programmer Controller models, initiate tuning prompts for setting of tuning constants for up to 9 PID combinations. In Programmer Only models, initiate prompts for setting of input range limits and codes. In numerical mode, enter 2 as integer for keyed in value. Multifunction decrement key scrolls through prompts and is used in conjunction with other keys to enter value and initiate various DCP operations. CLR Clear incorrect keyed in value before ENTER is pressed.	VRFY Enter verify mode to check/edit stored program data. In numerical mode, enter 3 as integer for keyed in value. Multifunction increment key scrolls through prompts and is used in conjunction with other keys to enter values and initiate various DCP operations. In numerical mode, enter minus sign for keyed in value. ENTER Initiates storage of entered values in memory.

Displays

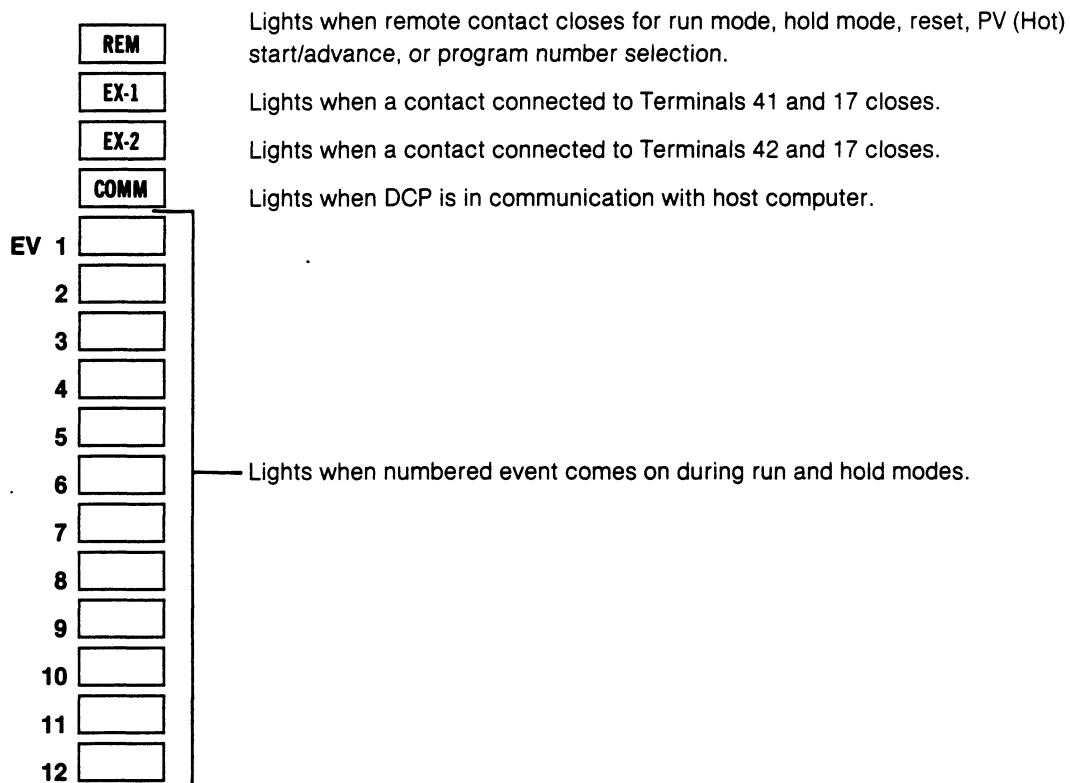
A 16-segment alpha-numeric display and three 7-segment numeric displays provide indication to aid programming, monitoring, and operating activities.

Review the following display descriptions to become familiar with display indications.

Display Descriptions

Display	What It's Called	Description
CH PROG SEG 	Alpha-Numeric	Normally shows channel number, program number, and segment number. Normal display is pre-empted by prompts when you enter mode for programming, tuning, configuring, verifying or filing operations.
SP 	Set Point	Normally shows value of set point in run and hold modes and is blank in ready mode when the RESET LED is lit. Normal display is pre-empted by pertinent parameter value when you enter mode for programming, tuning, configuring, verifying or filing operations.
TM 	Time	Normally shows time remaining in a segment or value of output in percent as selected in run and hold modes, nothing in ready mode when the RESET LED is lit, or value of output in percent in manual mode. Normal display is pre-empted by pertinent parameter value when you enter mode for programming, tuning, configuring, verifying or filing operations.
PV 	Process Variable	Normally shows value of process variable. Normal display is pre-empted by pertinent parameter value when you enter mode for programming, tuning, configuring, verifying or filing operations.

Light Emitting Diodes (LEDs) | The 16 LEDs provide status indication for pertinent DCP operations as noted below.



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Programming

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Introduction

The term "programming" is being used here to identify the process for entering the individual ramp and soak segment data to generate the required set point versus time profile (also called a program).

While the procedure for programming is straight-forward and aided by prompts, we suggest that you read this Programming section as well as the Tuning and Configuration sections, and complete the Program Worksheet included in the back of this manual before programming your DCP. This will acquaint you with the many unique functions that the DCP can perform, and provide a record of the operating parameters that are relevant to a given process.

Note that some of these parameters won't apply to your specific controller or application. Therefore, you'll probably leave some blank spaces on your worksheet.

This section identifies what you can program, how to enter a program, cycle a program, copy a program, delete a program, verify/edit a program, and upload/download programs.

Software Version Note

Occasionally, the software of a microprocessor-based controller is revised to provide additional features or to correct "BUGS." Thus, whenever you have a question concerning the operation of a unit, be sure you specify your DCP's software version.

To determine the software version of a DCP 700 controller, simultaneously press the **CLR** and **RESET** keys when the unit is in reset. The key number will appear in the alphanumeric display, e.g. DCP721. The software version will appear in the SP and T M displays. The display will return to normal when **RESET** is pressed.

What You Can Program

Basically, you can program all the data that is relevant to each ramp and soak segment for a given set point versus time profile. Specific data that can be programmed is listed here by the segment, function and parameter prompts which will appear in the DCP's alphanumeric display, when the DCP is in the program mode.

Segment Prompt	Parameter Prompt	Setting Range	Comments
SOAK	SET POINT	Within high and low range limits	For current or voltage input, see Configuration section for data about setting Linear Range Limits.
	DURATION	0.00 to 255.59	Time unit is configurable as hours/minutes or minutes/seconds for each program. See Configuration section.
G.SOAK	SET POINT	Within high and low range limits	Same as for SOAK, SET POINT.
	DURATION	0.00 to 255.59	Same as for SOAK, DURATION.
	GS-BAND	1 to 100	Establishes guaranteed soak band in units of PV indication. Soak timer will not start until PV is within G-S BAND. However, once the soak time is started it will not stop even if the PV goes out of the established guaranteed soak band.
RAMP-T	RAMPRATE	1 to 9999 or 1 to 999.9 PV unit/Time unit	Type of input determines position of decimal point. • Thermocouple & RTD: 1 to 999.9 • Linear: 1 to 9999 NOTE: Do not use RAMP-T as first or last segment of a program.
RAMP-X	SET POINT	Within high and low range limits	Can use this ramp to exceed the 999.9 rate limit for a RAMP-T. NOTE: Do not use RAMP-X as first segment of a program. Same as for SOAK, SET POINT.
	DURATION	0.00 to 255.59	Same as for SOAK, DURATION.
RAMP-E	EXT NO	1 or 2	NOTE: Do not use RAMP-E as the first or last segment of a program. To change the SP on an external command, select 1 for the contact across Terminals 41 and 17, or 2 for the contact across Terminals 42 and 17.
	RAMPRATE	1 to 9999 or 1 to 999.9 PV unit	Select a value by which the SP is to be increased or decreased each time the contact closes. Thus, this is a stepped rate through a pulsed contact and not a continuous ramp signal. When the SP reaches the value of the SP in the next segment, that segment will start.
Function Prompt	Parameter Prompt	Setting Range	Comments
TIME-EVT	EVT-NO	1 to 12	Specify number for Time Event. See Configuration section for more details about events (Page 5-8).
	ON-TIME	0.00 to 255.59	Same as for SOAK, DURATION.
	OFF-TIME	0.00 to 255.59	Same as above.
PV-EVT	EVT-NO	1 to 12	Specify number for process variable event. See Configuration section for more details about events (Page 5-8).
	PV-HIGH	Within high and low range limits	Establish "HIGH" set point. See Comments for SOAK, SET POINT.
	PV-LOW	Within high and low range limits	Establish "LOW" set point. See Comments for SOAK, SET POINT.

Function Prompt	Parameter Prompt	Setting Range	Comments
DEV-EVT	EVT-NO	1 to 12	Specify number for deviation event. See Configuration section for more details about events (Page 5-8).
	DEV-HIGH	0 to Span	Establish "HIGH" set point. See Comments for SOAK, SET POINT.
	DEV-LOW	0 to Span	Establish "LOW" set point. See Comments for SOAK, SET POINT.
CYCLE	CYCLE	0 to 9999	Establish number of times a program or portion of program is to be cycled. See How to Cycle a Program for details (Page 4-12).
PV-BIAS	PV-RATIO	0.000 to 9.999* *While the lower Ratio setting value is "0", it is unlikely that you would ever use a ratio setting of "0" since it would result in a PV value of "0".	Changes PV input to designated segment by multiplying it by the established ratio value. Set value applies for complete program unless different value is set in another segment. If no value is set, the PV ratio/bias set under configuration applies. See Configuration section for more data (Page 5-3).
	PV-BIAS	$\frac{\text{Low-High Range Limit}}{2}$ to $\frac{\text{High-Low Range Limit}}{2}$	Changes PV input to designated segment by adding the established bias value to it. Set value applies for complete program unless different value is set in another segment. If no value is set, the PV ratio/bias value set under configuration applies. See Configuration section for more data (Page 5-3).
PID-NO	PID-NO	1 to 9	Select number of PID combination to be used for controller during the designated segment execution. See Tuning section for more data. If no PID numbers are selected: Channel 1 uses PID 1 For Duplex operation (heat and cool) Channel 1 (heat) uses PID 1 Channel 1 (cool) uses PID 2 Channel 2 uses PID 5
IF EXT	EXT NO	1 or 2	Select 1 for the contact connected across Terminals 41 and 17 and 2 for the contact connected across Terminals 42 and 17. When contact closes during designated segment run, program changes to new segment in the same or different program <i>when segment run finishes</i> . The changeover segment and program numbers are set in the next two parameter prompts.
	PROG NO	1 to 39	Select number of program that contains segment which DCP is to change to if remote contact closes at designated external terminal.
	SEG NO	1 to 99	Select segment number that DCP is to change to if remote contact closes at designated external terminal.

Function Prompt	Parameter Prompt	Setting Range	Comments
ON EXT	EXT NO	1 or 2	Select 1 for the contact connected across Terminals 41 and 17 and 2 for the contact connected across Terminals 42 and 17. When contact closes during designated segment run or following segment run, program changes to new segment in the same or different program <i>immediately</i> . The changeover segment and program numbers are set in the next two parameter prompts.
	PROG NO	1 to 39	Select number of program that contains segment which DCP is to change to when remote contact closes at designated terminal.
	SEG NO	1 to 99	Select segment number that DCP is to change to when remote contact closes at designated external terminal.
END	END	--	Use in segment with IF EXT or ON EXT to designate end of program if remote contact doesn't close. When segment run finishes and there is no contact closure, program execution ends.

How to Enter a Program

This procedure assumes that you have completed a program worksheet, or know all the data relevant to your set point versus time profile. Before entering a program, peruse the following programming checklist.

Programming Checklist

- Be sure the first segment is a soak type. However, you can effectively bypass this requirement by entering a soak duration time of "0".
 - Do you know the configuration of all twelve events? If you don't know, see the Configuration section.
 - Include any data for time-based events with the segment in which the on-time is to occur.
 - The set point of a PV or DEV-based event can be entered with any segment and is effective until a new set point is entered in a subsequent segment. To make the PV or DEV event ineffective in a subsequent segment, use the high and low range limits as the High and Low set points in that segment.
 - Do you know the values of the tuning constants for all nine PID groups (programmer controller) or the value of the set point output limits (programmer only)? If you don't know, see the Tuning section.
 - Always include a PID group (programmer/controllers only) with the first segment since it will be effective until a different PID group is entered in a subsequent segment.
- Remember: If no PID numbers are selected:
- Channel 1 will use PID 1 and channel 2 will use PID 5
 - For Duplex operation (heat and cool)
 - Channel 1 (heat) will use PID 1
 - Channel 1 (cool) will use PID 2
 - Channel 2 will use PID 5
- Do you need a different PV ratio/bias for a segment during program execution? If you do, be sure to enter values with PV-BIAS prompt during the programming of the segment. Otherwise, the PV ratio/bias values entered during configuration apply.
 - Do you know range values assigned for linear input (4-20 mA dc or 1-5 Vdc). See the Configuration section for details.
 - Include "CYCLE" number with appropriate segment to cycle a whole or a portion of a program. See How to Cycle a Program in this section for details.
 - Include IF EXT, ON EXT, or END function with the segment in which you want it to occur.
 - Be sure last segment is either a SOAK or a RAMP-X type.
 - If your DCP has two channels, you can create another set point versus time profile (program) under the same program number for the second channel. However, you must enter the program in channel 1; then enter the program in channel 2 before pressing **PROG** to store the programs. To add a program for channel 2 under a program number that has a program stored under it for one channel, you must first delete the program.
 - If your DCP has Relative Humidity actuation, you must create a program for the dry bulb and RH under the same channel number.

Entering a Program

The following steps outline the basic key sequences to use to enter any of the data listed under the "What You Can Program" paragraph. Obviously, some of the values and displays are for example only to illustrate the response you can expect when you enter a particular parameter. While not every segment, function and parameter prompt is detailed in these steps, the same general key sequence for one applies for all. Prompts and error codes will help guide you through the programming steps and keep you from entering the wrong data.

Display Designations:

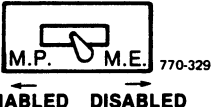
A-N — Alphanumeric (upper display)

SP — Set Point

TM — Time

PV — Process Variable

NOTE: If power is interrupted while you are entering a program, the DCP will not let you get back into the program you were entering after power is restored. However, you can get around this by making a partial General Reset. Pull down the module door on the front of the DCP and be sure the memory protect switch lever is in the M.P. position. Locate the General Reset switch that is accessible through the hole marked G. R. to the left of the memory protect switch. While pressing the [RESET] key, use a pen or pencil to press the G. R. switch — an 8. appears in the SP display. Remove the pen or pencil and continue pressing the [RESET] key until you hear a beep (takes about 5 seconds) and the [SYNC] LED lights. This clears the partially entered program, but leaves all other previously stored and complete programs intact. You can now re-enter the program you were putting in when the power failed.

Step	Key	Displays	Numerical Mode	Comments
1				Pull down module door on front of DCP and be sure switch lever is in far right position (M.E.). MEMORY PROTECT SWITCH  Be sure the MAN LED is out. You cannot enter a program with the DCP in the manual mode.
2	PARA	A-N TIME UNIT SP 0 or 1		If DCP is in another mode and does not acknowledge PARA , press DISP and then PARA . 0 in display means program time base is minutes/seconds; 1 means time base is hours/minutes. Is time unit correct one for program? YES—go to Step 2 (3) NO—go to Step 2 (1) to change time unit.
2 (1)	ENTER	A-N TIME UNIT SP 0	Key in desired time unit code.	0 = minutes/seconds 1 = hours/minutes
2 (2)	ENTER	A-N TIME UNIT SP 1*		Program time base will be hours/minutes.
2 (3)	DISP	A-N 1-1-1*		Exit configuration mode.
3	PROG ENTER	A-N 1-0-1 A-N 1-1-1*	Key in desired program number: 1 to 39	NUM LED blinks to show that DCP is in numerical mode. If you key in wrong number, press CLR to erase and key in correct number before you press ENTER .
4	◀ PROG	A-N @ 1-1-1 SP 3540 TM FrEE		In program mode, number in SP display is available (free) bytes of memory. If you get an ERROR code, see the Error Code list in appendix and take appropriate action.

*Typical display—shown for reference only

Step	Key	Displays	Numerical Mode	Comments
5	▲	A-N SOAK		Do you want this to be a SOAK segment? NOTE: SOAK and G.SOAK are only prompts that can be displayed in first segment, since it must be a soak type. YES—Go to Step 6 NO—Alternately press ▲ or ▼ to step forward or backward through segment prompts—SOAK, G.SOAK, RAMP-T, RAMP-X, and RAMP-E to find desired segment type. Proceed to Step 6 but substitute parameter prompts listed for desired segment in "What You Can Program" paragraph (Page 4-2) for alphanumeric displays shown in Step 6.
6	ENTER	A-N SET POINT SP 0	Key in desired set point value.	NUM LED blinks. If you get an ERROR code, see Error Code list in Appendix and take appropriate action. Use CLR to erase wrong values before ENTER is pressed.
	ENTER	A-N DURATION SP 350* TM 0.00*	Key in desired soak duration time.	350 is entered as set point value.
	ENTER	A-N DURATION SP 350* TM 2.00*		2 hours is entered as time of soak duration.
7				Do you want to include other FUNCTIONS in this segment? YES—Go to Step 7 (1). NO—Go to Step 8.
7 (1)	FUNC	A-N TM-EVT		NOTE: TM-EVT, PV-EVT, and DEV-EVT will not appear if you have not configured any of the events. See Configuration section for details. Use ▲ or ▼ to step forward and backward through function prompts—TM-EVT, PV-EVT, DEV-EVT, CYCLE, PV-BIAS, PID-NO, IF EXT, ON EXT, and END—to find desired functions. See "What You Can Program" paragraph (Page 4-2) for appropriate parameter prompts and enter data in similar fashion as shown for TM-EVT in Steps 7 (1) through 7 (5).
7 (2)	ENTER	A-N EVT-NO SP 0	Key in appropriate event number.	NUM LED blinks. Use CLR to erase wrong values before ENTER is pressed.

*Typical display—shown for reference only

Step	Key	Displays	Numerical Mode	Comments
7 (3)	ENTER	A-N ON-TIME SP 1* TM 0.00	Key in time when event must come on.	1 is entered as event number.
7 (4)	ENTER	A-N OFF-TIME SP 1* TM 1.05* PV 0.00	Key in time when event must go off.	1 hour and 5 minutes is entered as time when event comes on.
7 (5)	ENTER	A-N OFF-TIME SP 1* TM 1.05* PV 2.00*		NUM LED off. 2 hours is entered as time when event goes off. Do you want to enter another FUNCTION? YES—Press FUNC and see Comments for Step 7 (1). NO—Go to Step 8.
8	SEG	A-N @ 1-1-2 SP 3529 TM FrEE		Segment number is automatically increased by one (+ 1). You can delete an entered segment at this time by pressing CLR and SEG together—segment decreases by one (– 1).
9				Repeat steps 5 through 8 until all program data is entered for channel 1. If you have a two channel DCP, press CH instead of SEG after last segment data for channel 1 is entered. Then, repeat Steps 5 through 8 until all program data is entered for channel 2.
10	PROG	A-N 1-1-1		When data for last segment is entered, press PROG instead of SEG to store the program. NOTE: Pressing RESET aborts programming procedure (i.e., the program is lost). See other paragraphs in this section to copy, delete, verify/edit, or upload/download a program. NOTE: If ALARM 96 message appears, some program data has been lost. To recover from ALARM 96, press ENTER to get prompt for lost data to appear, key in data values, and press ENTER . This clears ALARM 96 and completes the programming operation.

*Typical display—shown for reference only

Programming Example

The following steps present the key sequences used and the displays encountered to enter the profile data shown in Figure 4-1 into the DCP as program number 14.

Example assumes that memory protect switch is in the far right position.

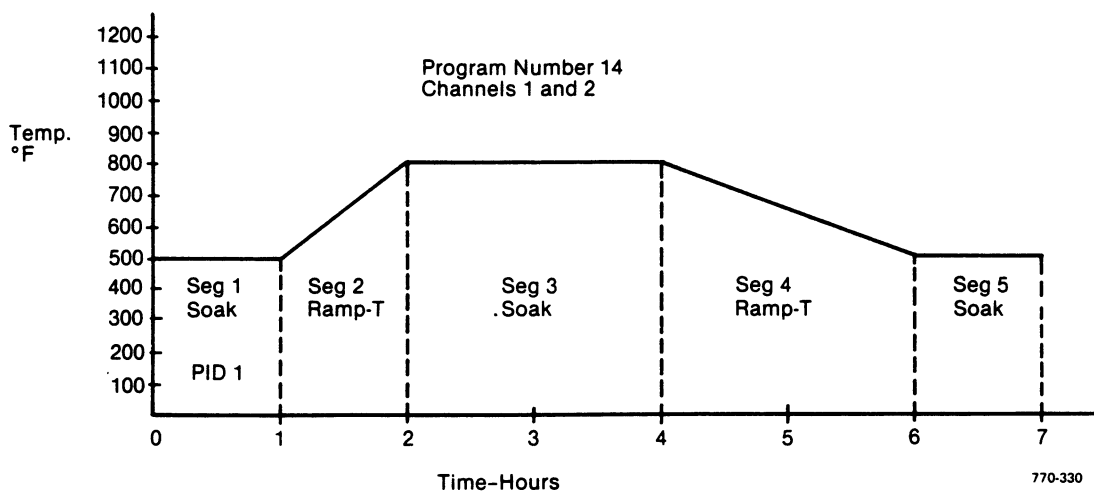


Figure 4-1—Set point versus time profile for programming example

Step	Key	Displays	Numerical Mode (NUM LED blinks; key in value)
1	PARA	A-N TIME UNIT SP 0	
2	ENTER	A-N TIME UNIT SP 0	FILE 1
3	ENTER	A-N TIME UNIT SP 1	
4	DISP	A-N 1-4-1	
5	PROG	A-N 1-0-1	FILE 1, DISP 4
6	ENTER	A-N 1-14-1	
7	◀ PROG	A-N @ 1-14-1 SP (bytes) TM FrEE	
8	▲	A-N SOAK	
9	ENTER	A-N SET POINT SP 0	FUNC 0, ◀ 0, ◀ 0
10	ENTER	A-N DURATION SP 500 TM 0.00	FILE 1, ◀ 0, ◀ 0
11	ENTER	A-N DURATION SP 500 TM 1.00	
12	FUNC	A-N TM-EVT	
13	▲ (5)*	A-N PID NO	
14	ENTER	A-N PID NO SP 0	FILE 1

*Number in parentheses is number of times key was pressed to read parameter shown.

Step	Key	Displays	Numerical Mode (NUM LED blinks; key in value)
15	ENTER	A-N PID NO. SP 1	
16	SEG	A-N @ 1-14-2 SP (bytes) TM FrEE	
17	▲ (3)*	A-N RAMP-T	
18	ENTER	A-N RAMPRATE SP 0.0	VRFY ³ , ◀ ⁰ , ◀ ⁰ , ◀ ⁰
19	ENTER	A-N RAMPRATE 300.0	
20	SEG	A-N @ 1-14-3 SP (bytes) TM FrEE	
21	▲	A-N SOAK	
22	ENTER	A-N SET POINT SP 0	PROG ³ , ◀ ⁰ , ◀ ⁰
23	ENTER	A-N DURATION SP 800 TM 0.00	TUNE ² , ◀ ⁰ , ◀ ⁰
24	ENTER	A-N DURATION SP 800 TM 2.00	
25	SEG	A-N @ 1-14-4 SP (bytes) TM FrEE	
26	▲ (3)*	A-N RAMP-T	
27	ENTER	A-N RAMPRATE 0.0	FILE ¹ , FUNC ³ , ◀ ⁰ , ◀ ⁰
28	ENTER	A-N RAMPRATE 150.0	
29	SEG	A-N @ 1-14-5 SP (bytes) TM FrEE	
30	▲	A-N SOAK	
31	ENTER	A-N SET POINT SP 0	FUNC ³ , ◀ ⁰ , ◀ ⁰
32	ENTER	A-N DURATION SP 500 TM 0.00	FILE ¹ , ◀ ⁰ , ◀ ⁰
33	ENTER	A-N DURATION 500 1.00	
34	CH	A-N @ 2-14-1 SP 1073 TM FrEE	
35	Repeat Steps 8 through 33 for Channel 2		
36	PROG	A-N 1-14-1	

*Number in parentheses is number of times key was pressed to reach parameter shown

How to Cycle a Program*

You can get a portion of a program, a whole program, or a combination of a portion of a program and a whole program to cycle (repeat) for a maximum of 9999 times. This is accomplished by using the cycle function prompt during segment programming to enter the appropriate cycle number information. Note that the total number of cycles will equal the cycle number entered plus one (1). Refer to the "Entering a Program" procedure in this section.

Getting a Portion of a Program to Cycle

Identify the segments within the program that represent the beginning and end of the portion to be cycled. When programming the segment that marks the beginning of a cycle, enter the required segment data and, then, press **FUNC** to access the CYCLE prompt. Enter 0 as the number of cycles.

When programming the segment that marks the end of a cycle, enter the required segment data and, then, press **FUNC** to access the CYCLE prompt. Enter the integer that represents the number of times the portion is to be cycled—must be greater than zero (0).

Getting a Whole Program to Cycle

After entering the required segment data for the last segment of a program, press **FUNC** to access the CYCLE prompt. Enter integer (greater than 0) that represents the number of times the program is to be cycled. Press **PROG** to complete programming or **CH** to continue programming in Channel 2.

Getting a Combination of a Portion of a Program and a Whole Program to Cycle

You can combine the previous procedures in the same program to cycle a portion of a program as well as the whole program (including the portion). Follow the previous individual procedures to enter the cycle data in the appropriate segments of the same program. You can include up to two portions of a program to cycle within the whole program cycle. However, if two portions are set and segment 2 is the beginning segment for one portion, enter 0 in segment 1 instead of segment 2. Otherwise, the program portions will be cycled continuously. Also, if three portions are set within a program, the program portions will be cycled continuously.

*Program used in this context stands for the set point versus time profile data entered on a per channel basis.

How to Copy a Program

Use the following steps to copy the contents of a stored program to any unused program number.

Be sure the memory protect switch lever is in the far right position (M.E.).

Step	Key	Displays	Numerical Mode	Comments
1	 	A-N COPY SP 0	Key in number of stored program to be copied.	 LED blinks. DCP is in COPY mode. If you key in wrong number, press  and key in correct number. If you get an ERROR code, see Error Code List in Appendix and take appropriate action.
2		A-N COPY SP 1* TM 0	Key in number of unused program that stored program is to be copied to.	If you get an ERROR code, see Error Code List in Appendix and take appropriate action.
3		A-N 1-1-3* SP 300* TM 1.42* PV 300*		Program has been copied, you can use verify procedure in this section to check the copied program.

*Typical display—shown for reference only

How to Delete a Stored Program

Use the following steps to delete a stored program from memory.
Be sure the memory protect switch lever is in the far right position (M.E.).

Step	Key	Displays	Numerical Mode	Comments
1	CLR PROG	A-N CLR PROG SP 0	Key in number of stored program to be deleted.	NUM LED blinks. Cannot do this when DCP is in Manual mode.
2	ENTER	A-N DEL OK? SP 32*		Prompt asks if it's okay to delete displayed program? If answer is YES—Go to Step 3. If answer is NO—Press RESET to exit Clear Program mode.
3	ENTER	A-N 1-1-3* 300* 1.42* 300*		Program is deleted from memory. To check, try to verify contents of deleted program—ERROR 2 message will appear.

*Typical display—shown for reference only

How to Delete all Stored Programs/Make a General Reset

Use the following steps to delete all of the stored programs as well as configuration and tuning (PID) data.
Be sure the memory protect switch lever is in the far right position (M.E.).*

Step	Key	Displays	Comments
1			Pull down module door on front of DCP and locate hole marked G.R. to left of memory protect switch.
2	RESET  (Hold)	SP 8.	While pressing RESET , use pen or pencil to press G.R. switch.
3	RESET 	All displays light for an instant	Remove the pen or pencil and continue pressing RESET until you hear a beep (takes about 5 seconds) and the SYNC LED lights, if not lit before.
4			All stored programs have been cleared from memory. Also, all configuration and tuning data have been reset to initial/factory values listed in Tables 5-2 and 6-1, respectively.

*With switch in the M.P. position, you will only make a partial General Reset. A partial General Reset leaves all stored programs, configuration data, and tuning (PID) data intact, but clears all the other areas in the random access memory.

How to Verify/Edit a Stored Program

Use the following steps to verify/edit data in a stored program. You can change any value displayed in the verify mode, but you cannot add or delete any segment or segment function. If you want to add or delete any segment function, you must delete the program and then, reprogram it with the changes.

If you want to change values, be sure memory protect switch lever is in far right position (M.E.).

Step	Key	Displays	Numerical Mode	Comments
1	PROG	A-N 1-0-3* SP 185 TM 0.15* PV 185*	Key in program number to be verified.	NUM LED blinks. Select program number to be verified—this is not necessary if program is running. NOTE: If a program is running and you select a different one for verification, program that is running will continue to run while you verify the other one.
2	ENTER	A-N 1-4-1* SP 190 TM 0.13 PV 1.90*		Can verify data in program number 4.
3	SEG	A-N 1-4-0* SP 190* TM 0.12 PV 190*	Key in segment number to be verified.	NUM LED blinks. Selects segment to be verified—this is not necessary if program is running and desired segment is being executed.
4	ENTER	A-N 1-4-5* SP 192* TM 0.10* PV 192*		Can verify data in segment 5 of program number 4.
5	VRFY	A-N 1-MIN-*		In verify mode, display shows channel number being verified and time unit of program. NOTE: VRFY will not operate if DCP is in manual mode (MAN LED lit).
6	▲	A-N SOAK5* SP 110* TM 12.00*		Do you want to change a parameter value? YES: Go to Step 6 (1). NO: Go to Step 7.
6 (1)	ENTER	A-N SET POINT SP 110* TM 12.00*		Value in SP display blinks to show that it can be changed. If you don't want to change this value, press ▲ to advance to next parameter [Step 6 (4)] or ◀ to return to segment prompt (Step 6).
6 (2)	ENTER	A-N SET POINT SP 0 TM 12.00*	Key in desired set point value.	NUM LED blinks.

*Typical display—shown for reference only

Step	Key	Displays	Numerical Mode	Comments
6 (3)		A-N SET POINT SP 120* TM 12.00*		In RUN or HOLD mode, a change in value for one of the following parameters takes effect immediately if segment is being executed: SOAK SP, G.SOAK SP, G.SOAK BAND, RAMP-T, RAMP-E SP, PV/DEV EVENT (except Event Number) and PV RATIO/BIAS. Otherwise, change is valid and will be used during the next program run. NOTE: If message ERROR 17 blinks, move memory protect switch lever to right, press  , key in new value, and press  .
6 (4)		A-N DURATION SP 120* TM 12.00*		Value in TM display blinks to show that it can be changed. If you don't want to change this value, press  to return to segment prompt.
6 (5)		A-N DURATION 120* 0.00	Key in new duration time.	
6 (6)		A-N DURATION SP 120* 10.00*		Duration time is changed to 10 minutes.
6 (7)		A-N SOAK 5* SP 120* TM 10.00*		
7		A-N RAMP-T 6* SP 500.0*		Do you want to change parameter value? YES—Go to Step 6 (1) and follow general key sequence given to enter a new value. NO—Press  to scroll through all the stored data and find segment and/or function parameter you want to change. Follow general key sequence in Steps 6 (1) through 6 (7) to change selected values.
8		A-N 1-4-7* SP 255* TM 1.09* PV 254*		Return to previous mode of operation. You can also press  to exit verify mode.

*Typical display—shown for reference only

How to Download Programs to a Memory Module

Use the following steps to download programs stored in the DCP's memory to a plug-in memory module. Program status may be reset, run, or hold.

Step	Key	Displays	Numerical Mode	Comments
1		A-N 1-2-1*		Pull down small door on front of DCP and insert memory module into slot provided. The memory protect switch may be left in the protect M.P. position.
2	FILE	A-N FILE-NO SP 0		If a different SP display appears, memory module already contains programs. Get a new module or erase stored programs by loading new programs over them. If memory module is new, a zero (0) appears in SP display; otherwise, the file number (1 to 9999) appears.
3	▼ FILE	A-N FILE-NO SP 0	Key in a number (1 to 9999) to identify filed programs.	NUM LED blinks. Use the same number (other than 0) that appeared in Step 2, or give the file a new number.
4	ENTER	A-N FILE-NO or DUMP OK SP 1*		NUM LED stops blinking.
5	ENTER	A-N PROG-DUMP		PROG-DUMP blinks. Stored programs, parameter (configuration) data and adjustment data are downloaded to memory module.
6		A-N 1-2-1*		Return to normal A-N display means that programs have been loaded. If ERROR message blinks, refer to list of Error Codes in Appendix and take appropriate action.
7				Remove module and close door. Enter new programs into DCP's memory through keyboard or upload different programs from another memory module.

*Typical display—shown for reference only

How to Upload Programs to the DCP's Memory

Use the following steps to upload a program or all programs stored in a memory module into the DCP's memory.

Step	Key	Displays	Numerical Mode	Comments
1		A-N 1-2-1*		Pull down small door on front of DCP and insert memory module into slot provided. Be sure memory protect switch lever is moved to right—M. E. position. Also, RESET LED must be lit—DCP must be in ready mode.
2	FILE	A-N FILE-NO SP 1*		If file number in SP display is incorrect, remove module and insert correct one. Repeat Step 2.
3	▲ FILE	A-N PRG-LOAD		- Do you want to load all programs stored in memory module into DCP's memory? Yes—Go to Step 3(1). No—See next question. - Do you want to load only one program? Yes—Go to step 4(1) No—See next question - Do you want to load only parameters (configuration) data? Yes—Go to Step 5(1) No—See next question. - Do you want to load only adjustment data? Yes—Go to Step 6 No—See other questions.
3 (1)	▲ OR ▼	A-N ALL-PROG		Steps through prompts until ALL-PROG appears in A-N display.
3 (2)	ENTER	A-N ALL-PROG		All programs stored in memory module will be loaded into DCP's memory. All programs in DCP are cleared (erased) before the programs in memory are uploaded into DCP. Parameters and tuning constants in DCP memory remain unchanged.
4 (1)	▲ OR ▼	A-N ONE-PROG SP 0 TM 0		Steps through prompts until ONE-PROG appears in A-N display.
4 (2)	ENTER	A-N ONE-PROG SP 0 TM 0	Key in number of module program to be loaded. For example, number 4.	
4 (3)	ENTER	A-N ONE-PROG SP 4* TM 0	Key in number of DCP program (1 to 39) to which module program is to be loaded. For example, number 22.	If ERROR message appears, refer to list of Error Codes in Appendix and take appropriate action. DCP program number selected must be clear (no program with that number in memory) before uploading one program.
4 (4)	ENTER	A-N ONE-PROG SP 4* TM 22*		ONE-PROG blinks. Starts uploading program 4 as program 22 in DCP's memory.
5 (1)	▲ OR ▼	A-N PAR-DATA		Step through prompts until PAR-DATA appears in A-N display
5 (2)	ENTER	A-N PAR-DATA		PAR-DATA blinks. Starts uploading parameter data into DCP's memory.
6	▲ OR ▼	A-N ADJ-DATA or OPTION		Ignore ADJ-DATA or OPTION prompt when it appears in A-N display—You can <i>not</i> upload adjustment data.

*Typical display—shown for reference only

Appendix

Programming Error Messages

Message	Meaning	Corrective Action
ERROR 1	You are trying to enter data in, or copy data to, a program number that has been used.	Select a new program number or delete the program data.
ERROR 2	You are trying to verify or copy an unused program number.	Select a new program number.
ERROR 3	You are trying to run an unused program number.	Select a new program number.
ERROR 4	You did not enter the program number and segment number to run when IF EXT/ON EXT was programmed.	Enter the program number and segment number.
ERROR 5	You are trying to use a 2-channel Relative Humidity model (DCP722) as a single channel for RH only.	You can use channel 1 for "Dry Bulb" temperatures only, but you cannot use channel 2 alone for % RH since the dry bulb set point (for channel 1) is needed to calculate the Wet Bulb set point for channel 2.
ERROR 6	The external signal has not been detected by the RAMP-E segment running in Channel 1.	Verify program data for Ramp-E segment. Check appropriate external contact (1 or 2), and repair or replace as required
ERROR 7	The external signal has not been detected by the Ramp-E segment running in Channel 2.	Verify program data for Ramp-E segment. Check appropriate external contact (1 or 2) and repair or replace as required.
ERROR 8	Signals that too many time event calculation blocks are being used at the same time. The software provides 12 time event calculation blocks. Therefore, that is the maximum number of time events that can be "active" at the same time.	Reconfigure events to allocate space in the event calculation blocks. Note that one time event number (e.g. Time Event No. 1) can use up all 12 blocks if it is turned on and off 12 times in the same segment.
ERROR 10	The low limit you are entering is greater than the high limit.	Decrease low limit or increase the high limit.
ERROR 11	You are using RAMP-T or RAMP-E as the last segment in the program.	Use RAMP-X or a soak segment.
ERROR 14	Data you are entering for the event is incorrect for the type.	Select event number with the correct type or change type for that number.
ERROR 15	Insufficient memory remaining to store the data.	Delete one or more stored programs.
ERROR 16	Your program has more than ninety-nine segments.	Reduce the number of segments to ninety-nine or less.

Appendix (continued)

Programming Error Messages (continued)

Message	Meaning	Corrective Action
ERROR 17	The program you are trying to change is protected.	Change the position of the Program-Protection-Switch.
ERROR 18	The set point is greater than the high range limit.	Reduce the set point as required.
ERROR 19	The set point is lower than the low range limit.	Increase the set point as required.
ERROR 20	There is no set point that meets the PV Start requirements.	Select another method of starting.
ERROR 21	Incorrect remote program number.	Select another program number.
ERROR 22	Incorrect remote channel number.	Select another channel number.
ERROR 23	You are trying to delete a program that is running or held.	Reset the program or wait for the program to finish.
ERROR 24	You made an error in the downloading procedure.	Repeat the downloading procedure.
ERROR 25	Program number you are trying to load is not in the cartridge.	Select the correct program number.
ERROR 26	Program number you are trying to load already has data stored under it.	Delete data under the program number and repeat the loading procedure.
	Insufficient memory available.	Delete one or more programs.
ERROR 27	You made an error in the loading procedure.	Repeat loading procedure.
ERROR 28	Illegal remove operation actuated during program loading operation.	Disable remote operation and repeat loading procedure.
ERROR 29	You cannot download data while check-sum is being run.	Download data after check-sum run.
ERROR 30	DCP detects an EEPROM write error	Check that PV input actuation is correct. Must short the EEPROM protect connector pins to complete the attempted operation.
ERROR 31	Incorrect zero and/or span calibration values.	Check calibration values and change as required.

NOTE: If you get extraneous ERROR messages that cannot be cleared, perform the General Reset procedure described on Page 4-14.

Considerations when Using Events

Time-based

- To make a time-based event stop at the end of a segment, enter the on-time in the segment in which the event is to start; then enter an off-time of "0".
- When the RESET LED is on, all time-based event timers are set to "0", and there are no outputs from the events.
- When the HOLD LED is on, the timers for all time-based events that are on are stopped, but the event outputs are maintained.
- If a time-based event with an on-time of "0" is included in a guaranteed soak segment, the event timer will not start until the soak timer starts. However, there will be an event output immediately upon completion of the previous segment, regardless of the relation of the PV to the SP.
- If a program is advanced to a new segment or is shifted to a previous segment, the event timers will be in the state they would be in had the program reached that segment in a normal manner.

PV-based

- A PV-based event entered in a segment is valid until it is entered again with different SPs in a subsequent segment.
 - If you do not want a PV-based event effective in a subsequent segment, enter it in that segment with the SP range limits as the HIGH and LOW SPs.
 - If a PV-based event with the same number is entered more than once in the same segment or in the programs for CH1 and CH2 with the same or different SPs, the last entry will be used when the program is running.
- When the program is verified, all of the entries will appear. If any entry is changed, that entry will be used immediately when the program is run.
- If a bias is put on the PV, the biased PV is compared to the PV-event SPs.
 - When the RESET LED is on, there is no output from a PV-based event, regardless of the relation between the PV and the event SP.
 - Output is present when the PV is greater than the PV-HIGH-SP or less than the PV-LOW-SP.
 - There is no output when the PV is
 - (1) less than the difference between the PV-HIGH-SP and the event hysteresis.
 - (2) greater than the difference between the PV-LOW-SP and the event hysteresis.

DEV-based

- A DEV-based event entered in a segment is valid until it is entered again in a subsequent segment. If you do not want a DEV-based event effective in a subsequent segment, enter it in that segment with the input or SP range limits as the HIGH and LOW SPs.

- If a DEV-based event with the same number is entered more than once in the same segment or in the programs for CH1 and CH2 with the same or different SPs, the last entry will be used when the program is running.

When the program is verified, all of the entries will appear. If any entry is changed, that entry will be used immediately when the program is run.

- If a bias is put on the PV, the biased PV is used when the deviation is calculated from the biased PV.

- When the RESET LED is on, there is no output from a DEV-based event, regardless of the relation between the PV and the event SP.

- Output is present when the difference between the PV and the SP is greater than the DEV-HIGH-SP or DEV-LOW-SP. The output will remain until the difference between PV and the SP is less than the difference between the DEV-HIGH-SP and the event hysteresis, or the difference between the DEV-LOW-SP and the event hysteresis.

DCP 700
Digital Control Programmer

Configuration

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What You Can Configure

Parameters discussed in this section are those that you can select (configure) to tailor the set point program and process variable to fit your process needs.

Which parameters are available depends on the model of the DCP. You can use the "Display" portion of "How to Configure" to determine which parameters are available on your DCP.

- The parameters you can configure are listed in Table 5-1 by name with their setting range or identifying code and an explanation of their use. Use this table and the parameter portion of the worksheet in the manual to determine and list the value for the parameters needed for your application.
- Refer to How to Configure paragraph for the key sequences to use to display and enter parameter data.
- Refer to Table 5-2 for list of prompts and the initial/factory values entered for each parameter.
- Refer to the Example of Event Configuration paragraph for the specific steps involved in configuring an Event.

Table 5-1—Parameters

Selection	Selection or Range	Code	Explanation
Time unit	minute/second	0	Establishes the time unit for a program. Must be entered after entering the program number when entering a program.
	hours/minutes	1	
Advance rate	1 to 120		This is the factor by which the program and event timers are increased when ADV is pressed. For example, if you have a soak segment with a duration of one hour, an advance rate of 60 causes it to be one minute. NOTE: Guaranteed soak segments become plain soak segments.
Auxiliary output	(All DCPs)		Provides a 4 to 20 mA output on Terminals 61 and 62 to represent the variable you select.
	CH 1 – Set point	1	
	– Process variable	2	
	– Output	3	
	CH 2 – Set point or RH Set point	4	
	– Process variable or RH process variable (CH 2)	5	
	– Output	6	
	(With relative humidity) Wet bulb set point (CH 2) Process variable as RH	7 8	
Input filter	CH 1 – 0 to 255 sec.		Establishes the amount of filtering to "quiet" a "noisy" input. The higher the value selected, the more the filtering. Can be stored at any time the DCP is powered. Use "0" for no filtering.
	CH 2 – 0 to 255 sec.		
Event configuration	Not used	0	Establishes the type for each of the twelve events. Once the type has been selected for an event it is effective on all programs stored in the DCP. Select a code for each of the twelve events.
	Time-based	1	
	PV-based	2	
	DEV-based	3	
Event hysteresis	CH 1 – 1 to 200 CH 2 – 1 to 200		Establishes a "deadband" in set point units where no switch action occurs.
External ramp monitoring time	CH 1 – 0 to 255:59 CH 2 – 0 to 255:59		If you designated a Ramp-E in the program to be pulsed only when an external contact closes, this parameter establishes a time-delay before Alarm code 11 or 12 appears to signify the contact didn't close. Note that the remaining monitoring time appears in the TM display during the execution of a Ramp-E segment.
Set point bias	CH 1 – 0 to Span/2 CH 2 – 0 to Span/2		When any program is running or held, the bias is performed on the program and event set points. The stored bias is effective until reset to a "0" value. If you don't want bias, set to "0"

Table 5-1—Parameters (continued)

Selection	Selection or Range	Code	Explanation
PV ratio/bias	CH 1 – Ratio – 0.000 to 9.999* – Bias – ± 0 to $\pm$ Span/2 CH 2 – Ratio – 0.000 to 9.999* – Bias ± 0 to $\pm$ Span/2 *While the lower ratio setting range value is "0", it is unlikely that you would ever use a ratio setting of "0" since it would result in a PV value of "0".		The PV is first multiplied by the ratio; then the bias is performed. If no PV ratio/bias was included in the program (see Programming), this PV ratio/bias is performed on the PV when the RESET, RUN or HOLD indicator is lit. If a PV ratio/bias was included in the program, this PV ratio/bias is performed on the PV only when the RESET indicator is lit.
External PV (Hot) start	CH 1–SP = or > PV (upward) SP = or < PV (downward) CH 2–SP = or > PV (upward) SP = or < PV (downward)	0 1 0 1	Establishes the set point (upward or downward) at which the program will start relative to PV when the contacts connected to Terminals 17 and 36 are closed.
Maximum linear range limits	Range 1 – ± 19999 ± 1999.9 ± 199.99 ± 19.999 ± 1.9999 Range 2 – ± 19999 ± 1999.9 ± 199.99 ± 19.999 ± 1.9999	0 1 2 3 4 0 1 2 3 4	On DCPs with current or voltage inputs, this code establishes the maximum set point limits. Notice that with each increase in the code, the decimal moves one place to the left. This is to give better resolution with smaller input ranges.
Linear Range values	Range 1 – Hi-rng limit Lo-rng limit Range 2 – Hi-rng limit Lo-rng value		Scales the set point to the input. Set between the maximum plus and minus range limits selected above. Hi-rng value must be greater than the Lo-rng value.
Square root	CH 1 CH 2 No --- Yes --- No No Yes No No Yes Yes Yes	0 1 0 1 2 3	On DCPs with current or voltage inputs, causes the square root function to be performed on the PV.
Dropout value	CH 1 – 0.00 to 10.0% CH 2 – 0.00 to 10.0%		A value at which the controller output will go to zero if the output reaches it.
Output relay interval	CH 1 – Heat–4 to 600 sec Cool–4 to 600 sec CH 2 –4 to 600 sec		On DCPs with a time proportional controller, sets the operating interval of the output relay. Set in 4 sec. intervals.
Current proportional controller output action	CH 1 CH 2 Reverse --- Direct --- Reverse Reverse Direct Direct Reverse Direct Direct Reverse	0 1 0 1 2 3	Reverse action results in a decrease in the output when the input is above the set point. Direct action results in an increase in the output when the input is above the set point.

Table 5-1 — Parameters (continued)

Selection	Selection or Range		Code	Explanation
PID Type	<u>CH 1</u>	<u>CH 2</u>		With PID A, the proportional band, reset, and rate act on the error.
	PIDA	---	0	
	PIDB	---	1	With PID B, the proportional band and rate act on the PV, and the reset acts on the error.
	PIDA	PIDA	0	
	PIDB	PIDB	1	
	PIDA	PIDB	2	
	PIDB	PIDA	3	
Control operation initial value	<u>CH 1</u>	<u>CH 2</u>		Establishes the initial controller output when the RUN key is pressed.
	0.0	---	0	
	50.0	---	1	Do not set if you have duplex-time proportional control as the initial output is automatically set to 50.0%.
	0.0	0.0	0	
	50.0	50.0	1	
	0.0	50.0	2	
	50.0	0.0	3	
Ready mode control output	<u>CH 1</u>	<u>CH 2</u>		Establishes the controller output when the RESET LED is lit.
	0.0 to 100%	0.0 to 100%		
Duplex-time output relay deadband	- 5.0 to 5.0%			Sets the deadband between the heat and cool relays.
RH corrections	Pressure - 500 to 1000 mmHg			
	Wind velocity - 2.5 metres per second minimum		0	Used on DCPs with relative humidity to compensate for atmospheric pressure and wind velocity at the sensor location.
	- 0.6 to 2.4 metres per second		1	
	- 0.5 metres per second maximum		2	
Communication address	0 to 254			Established by the system requirements to give this DCP 700 a unique address. The assigned address must be within the numbers 1 to 31. Cycle power after address is entered.

How to Configure

Display

NOTE: Only those parameters pertinent to your DCP can be displayed. For example, "Range limits" is available only if your DCP has a linear, voltage, or current input, and "RH corrections" is available only if your DCP has relative humidity actuation.

1. Press **PARA** to display the parameter-prompt "TIMEUNIT" in the upper display
2. Press **▲** to display the remaining parameter-prompts in order starting at the beginning of Table 5-2, or press **▼** to display the remaining prompts in order starting at the end of the table.

NOTE: Be sure that the memory-protect-switch is in the far right position (M.E.).

3. Press **DISP** or **◀**, **DISP** to exit the parameter setting mode.

Store

To store a parameter value or code, use the following steps after the correct parameter-prompt is displayed.

NOTE: If there is just one piece of data (i.e. the TM and PV displays are blank) start the sequence at step 2.

Step	Key	Action
1	ENTER	Data-prompt appears in upper display and the number in the SP display blinks.
2	ENTER	SP display shows "0", and the NUM LED blinks.
3	Number keys for code or value	Numbers appear in the SP display.
4	ENTER	NUM LED goes out.
5	■ If there isn't a number in the TM display, go to Step 17. ■ If there is a number in the TM display and you want to change it, go to Step 6; otherwise go to Step 10.	
6	▲	Data-prompt appears in upper display and the number in the TM display blinks.
7	ENTER	TM display shows "0" and the NUM LED blinks.
8	Number keys for code or value.	Numbers appear in the TM display.
9	ENTER	NUM LED goes out, and the code or value is stored.
10	■ If there isn't a number in the PV display, go to Step 15. ■ If there is a number in the PV display and you want to change it, go to Step 11; otherwise go to Step 15.	
11	▲	Data-prompt appears in upper display and the number in the TM display blinks.
12	ENTER	PV display shows "0" and the NUM LED blinks.
13	Number keys for code or value.	Numbers appear in the PV display.
14	ENTER	NUM LED goes out.
15	◀	Parameter-prompt reappears.
16	■ If you want to display another parameter, go to Step 17. ■ If you want to get out of the parameter change mode, go to Step 18.	
17	▲ or ▼	Next parameter-prompt appears. Go to step 1 or 2.
18	DISP	Resets to previous operating mode.

*After pressing **ENTER** (steps 2, 7, & 12) the display will become 0, indicating the expected entry of a number.

Table 5-2—Parameter Prompts and Initial/Factory Entered Values

Parameter	Upper display		Initial/Factory Entered Code or value in other displays*		
	Parameter prompt	Data- prompt	SP	TM	PV
Time unit	TIMEUNIT		0		
Advance rate	ADV-RATE		1		
Recorder output	RECORDER		1		
Input filter	FILTER				
Channel 1		FILTER1	1		
Channel 2		FILTER2		1	
Event conf	EVT-CONF				
Events		EVT1—4	0		
Events		EVT5—8		0	
Events		EVT9-12			0
Event hysteresis	EVT-PARA				
Channel 1		EVT-HYS1	2.0 or 2		
Channel 2		EVT-HYS2		2.0 or 2	
External ramp monitoring time	RAMP-PAR				
Channel 1		ALM-TM1	0		
Channel 2		ALM-TM2		0	
Set point bias	SP-BIAS				
Channel 1		SP-BIAS1	0		
Channel 2		SP-BIAS2		0	
PV ratio & bias.	PVBIAS1				
Channel 1 — Ratio		PVRATIO1	1.000		
— Bias		PVBIAS1		0	
	PVBIAS2				
Channel 2— Ratio		PVRATIO2	1.000		
— Bias		PVBIAS2		0	
External PV (Hot) start	PV-START				
Channel 1		PVSTART1	0		
Channel 2		PVSTART2		0	
Range limits					
Channel 1	RANGE-1				
Limits		DEC-PT1	0		
Hi-rng value		RANGE-H1		10000	
Lo-rng value		RANGE-L1			0
Channel 2	RANGE-2				
Limits		DEC-PT2	0		
Hi-rng value		RANGE-H2		10000	
Lo-rng value		RANGE-L2			0

Table 5-2—Parameter Prompts and Initial/Factory Entered Values (continued)

Parameter	Upper display		Initial/factory entered code or value in other displays*		
	Parameter prompt	Data-prompt	SP	TM	PV
Square root	ROOT-CAL				
Apply to PV		USE-CH	0		
Dropout value					
Channel 1		DROP-OT1	0		
Channel 2		DROP-OT2		0	
Output relay interval	TP-INTVL				
Channel 1					
Nonduplex		INTAVL-1	4		
Duplex-Heat		INTAVL-1	4		
Duplex-cool		INTAVL-2		4	
Channel 2		INTAVL-3			4
Output directiont	CTL-ACT		0		
PID Type	PID-TYPE		0		
Control Operation Initial Value	INIT-VAL		0.0**		
Ready mode control output	RDY-OUT	RDY-OUT1	0**		
		RDY-OUT2		0	
RH corrections	RH-PARA				
Pressure		PRESSURE	760		
Wind velocity		WIND-VEL		0	
Time-prop relay deadband	DEADBAND		0.0		
Communication address	COMMPAR		0		

*DCP also reverts to these codes and values after a General Reset is performed.

**If DCP has duplex time proportional control, initial value (or ready mode output) for Channel 1 controller output is 50.0%.

Example of Event Configuration

Assume the following;

- All events are configured Code 0 (not to be used).
- Events 1, 2, 3 and 4 are to be reconfigured as time based events (Code 1).
- Events 5, 6, 7 and 8 are to be reconfigured as process variable-based events (Code 2).
- Events 9, 10, 11 and 12 are to be reconfigured as deviation-based events (Code 3).

The following steps configure the events.

NOTE: Make sure the memory-protect-switch is in the far right position.

Step	Key	Action																								
1	PARA	"TIMEUNIT" appears in upper display.																								
2	▲	Until "EVT-CONF" appears in the upper display, and the SP, TM, and PV displays show a 0. Note 1: The DCP 700 does not show leading 0's. If EVT 1 and 2 are not used and 3 and 4 are configured as TIME, this shows up as <table><tr><td></td><td></td><td>1</td><td>1</td></tr></table> in the SP display. If EVT 1 and 2 are configured as TIME and 3 and 4 as not used, this shows up as <table><tr><td>1</td><td>1</td><td>0</td><td>0</td></tr></table> in the SP display.			1	1	1	1	0	0																
		1	1																							
1	1	0	0																							
		<table><tr><th>Display</th><th>SP</th><th>TM</th><th>PV</th></tr><tr><td>Event</td><td>1</td><td>5</td><td>9</td></tr><tr><td>Event</td><td>2</td><td>6</td><td>10</td></tr><tr><td>Event</td><td>3</td><td>7</td><td>11</td></tr><tr><td>Event</td><td>4</td><td>8</td><td>12</td></tr></table> <table><tr><td></td><td></td><td></td><td>0</td></tr></table>	Display	SP	TM	PV	Event	1	5	9	Event	2	6	10	Event	3	7	11	Event	4	8	12				0
Display	SP	TM	PV																							
Event	1	5	9																							
Event	2	6	10																							
Event	3	7	11																							
Event	4	8	12																							
			0																							
3	ENTER	"EVT1-4" appears in upper display, "0" blinks in the SP display.																								
4	ENTER	"0" stops blinking and the NUM LED blinks signifying number keys allowed.																								
5	1 (4 times)	The four 1s appear in the SP display.																								
6	ENTER	The NUM LED goes out and the ones are stored.																								
7	▲	"EVT5-8" appears in upper display, and the "0" in the TM display blinks.																								
8	ENTER	"EVT5-8" appears in upper display. "0" appears in the TM display and the NUM LED blinks.																								
9	2 (4 times)	The four 2s appear in the TM display.																								
10	ENTER	The NUM LED goes out and the twos are stored.																								
11	▲	"EVT9-12" appears in upper display, and the "0" in the PV display blinks.																								
12	ENTER	"EVT9-12" appears in upper display. "0" appears in the PV display, and the NUM LED blinks.																								
13	3 (4 times)	The four 3s appear in the PV display.																								
14	ENTER	The NUM LED goes out and the threes are stored.																								
15	◀ or DISP	"EVT-CONF" reappears and you can now display another parameter. Displays return to their condition before PARA was pressed.																								

**DCP 700
Digital Control Programmer****Tuning**

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Introduction

Controller tuning consists of determining the appropriate values for the Proportional Band, Rate Time, and Reset Time tuning parameters along with the Output and Reset Limiting information that will give you the control you want.

Depending on the characteristics of the deviation of the process variable from the set point, these parameters interact to alter the controller's output and produce changes in the value of the process variable. Since each parameter responds to a specific characteristic of the deviation, you may not need a combination of proportional band, rate, and reset time. Your selection depends on the process characteristics and the desired controller response. Refer to the appendix in this section for more information regarding Proportional Band, Rate, and Reset.

Tuning the Controller

In the controller, the tune function is to enter, verify or correct PID tuning parameters. You can enter up to 9 sets of tuning parameters into your controller and use any of these 9 sets when you are programming the DCP (see programming section).

The tuning parameters include:

PID Data

- proportional band
- reset time
- rate time

Controller Output Limits

- high limit
- low limit
- output rate limiter

Reset Limits

- high limit
- low limit

First, refer to Table 6-1 and determine what data you will need for each set of tuning parameters. This table gives you the tuning parameter's definition and range of setting allowed for each plus the initial setting. If you want to change these settings, write the information in the blocks provided on the Configuration Work Sheet (see Figure 6-1) so you will have all the information together when you want to enter it.

Second, enter the data into the controller using keystroke sequences presented later in this section.

For each set of tuning parameters (which appear consecutively under the **TUNE** key), you will call up three types of information

- PID DATA
- OUTPUT LIMITS
- RESET LIMITS

and use a keystroke subroutine to enter the data under each of these types of information.

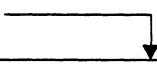
If at any time an *error* code appears in the upper display, refer to the programming section for definitions of errors.

TABLE 6-1—Tuning Parameters, Range of Setting, initial Value, and Definition

Tuning Parameter	Range of Setting	Initial Value	Definition
PID Parameters			
Proportional Band	0.1 to 1000.0%	10.0%	Proportional band regulates the controller's output in proportion to the error signal: the difference between the process variable and set point.
Reset Time	0.1 to 1800 seconds	120 seconds	Reset time regulates the controller's output at a rate proportional to the size of a static error. The amount of corrective action depends on the value of proportional band. 0 = no reset time (see Manual Reset below)
Rate Time	1 to 600 seconds	30 seconds	Rate regulates the controller's output in proportion to the rate of change in the process variable. The amount of corrective action depends on the value of the proportional band. 0 = no rate time
Output Limits			
High Limit	105.0% maximum	100.0%	This is the highest value of output above which you do not want the controller output to go. Do not set lower than the low output limit.
Low Limit	– 0.50% minimum	0.0%	This is the lowest value of output below which you do not want the controller output to go. Do not set higher than the high output limit.
Output Rate Limiter	0.0 to 100.0%	100%	This is the value of the output beyond which you do not want the rate action to regulate the controller output.
Reset Limits			
High Limit	105.0% maximum	100.0%	This is the highest value of output beyond which you want no reset action to occur. Set within the configured range of the high and low output limits.
Low Limit	– 0.5% minimum	0.0%	This is the lowest value of output below which you want no reset action to occur. Set within the configured range of the high and low output limits.

Manual Reset

You can have manual reset by entering "0" for Reset Time and using the Reset High Limit for the value to eliminate offset.

Write your selections under the PID number 

	PID—Number								
	1	2	3	4	5	6	7	8	9
PID Parameters									
Proportional Band									
Reset Time									
Rate Time									
Controller Output Limits									
High Limit									
Low Limit									
Output Rate Limit									
Reset Limits									
High Limit									
Low Limit									

Figure 6-1—Sample from configuration work sheet

Procedure The following procedure explains how to enter Tune Mode and access the PID-Para, Output Limits, or Reset Limits.

Entering Tune Mode

Press **TUNE** key. The upper display shows

PID-PAR 1

The values shown in the SP, TM and PV displays are the PID data for your first set of tuning parameters (number shown in the rightmost digit).

Press **▲** key. The upper display shows

OUT-LIM 1

The values shown in the SP, TM, and PV displays are the output limits for your first set of tuning parameters.

Press **▲** key. The upper display shows

RST-LIM 1

The values shown in the SP, TM, and PV displays are the reset limits for your first set of tuning parameters.

Continuously pressing **▲** or **▼** keys will display the PID parameters, output limits, and reset limits for all 9 sets of tuning parameters.

To access and change any of the parameters shown in the SP, TM, or PV displays refer to

Changing PID Parameter Data—Page 6-5

Changing Output Limits—Page 6-6

Change Reset Limits—Page 6-7

Changing PID Parameter Data

Refer to your configuration record sheet for the values for each set of PID parameter data.

Press **TUNE** then **▲** or **▼** until the upper display shows **PID PARA#** for any of the 9 sets. (# the number appears in the rightmost digit of the upper display.)

Table 6-2 gives the key sequence for entering or changing the value of
Proportional band,
Reset time, and
Rate time

TABLE 2

Press	Action	Comments
ENTER (Note 1)	Upper display shows PROP-BND#. SP display shows blinking proportional band value.	# = PID number
ENTER	NUM LED is blinking. SP display shows 0.0.	Indicates numbered keys allowed. Display cleared for entry.
Number Keys 0 through 9	Until SP display shows desired proportional band value.	Press CLR to clear any errors.
ENTER	NUM LED goes out. SP display value entered into memory.	
▲	Upper display shows INTG-TM#. TM display shows blinking Reset Time value.	# = PID number
ENTER	NUM LED is blinking. TM display shows 0.	Indicates numbered keys allowed. Display cleared for entry.
Number Keys 0 through 9	Until TM display shows desired Reset Time value.	0 = No Reset Press CLR to clear any errors.
ENTER	NUM LED goes out. TM display value entered into memory	
▲	Upper display shows RATE-TM# PV display shows blinking Rate Time value.	# = PID number
ENTER	NUM LED blinking. PV display shows 0.	Indicates numbered keys allowed. Display cleared for entry.
Number Keys 0 through 9	Until PV display shows desired Rate Time value.	0 = No Rate Press CLR to clear any errors.
ENTER	NUM LED goes out. PV display value entered into memory.	
▲ or ▼	To cycle through the 3 parameters if any changes need be made.	
◀	Upper display will return to PID-PAR#	# = PID number
DISP or RESET (Note 2)	To return to control.	

NOTE 1: After you press **ENTER** press **▲** or **▼** to access INTG-TM# or RATE-TM#.

NOTE 2: To continue to another PID parameter, output limit, or reset limit, press **▲ or ▼**.

Changing Output Limits

Refer to your configuration record sheet for the values for each set of output limits.

Press **TUNE** then **▲** or **▼** until the upper display shows **OUT-LIM#** for any of the 9 sets. (# the number appears in the rightmost digit of the upper display.)

Table 3 gives the key sequence for entering or changing the value of
High output limit,
Low output limit, and
Output rate

TABLE 6-3

Press	Action	Comments
ENTER (Note 1)	Upper display shows LIM-HIG#. SP display shows blinking High Output Limit value.	# = PID number
ENTER	NUM LED is blinking. SP display shows 0.0.	Indicates numbered keys allowed. Display cleared for entry.
Number Keys 0 through 9	Until SP display shows desired High Output Limit value.	Press CLR to clear any errors.
ENTER	NUM LED goes out. SP display value entered into memory.	
▲	Upper display shows LIM-LOW#. TM display shows blinking Low Output Limit value.	# = PID number
ENTER	NUM LED is blinking. TM display shows 0.0.	Indicates numbered keys allowed. Display cleared for entry.
Number Keys 0 through 9	Until TM display shows desired Low Output Limit value.	Press CLR to clear any errors.
ENTER	NUM LED goes out. TM display value entered into memory.	
▲	Upper display shows OUTFATE# PV display shows blinking Output Rate value.	# = PID number
ENTER	NUM LED is blinking. PV display shows 0.0.	Indicates numbered keys allowed. Display cleared for entry.
Number Keys 0 through 9	Until PV display shows desired Output Rate value.	Press CLR to clear any errors.
ENTER	NUM LED goes out. PV display value entered into memory.	
▲ or ▼	To cycle through the 3 parameters if any changes need be made.	
◀	Upper display will return to OUT-LIM#	# = PID number
DISP or RESET (Note 2)	To return to control.	

NOTE 1: After you press **ENTER** press **▲** or **▼** to access LIM-LOW# or OUTFATE#.

NOTE 2: To continue to another PID parameter, output limit, or reset limit, press **▲ or ▼**.

Changing Reset Limits

Refer to your configuration record sheet for the values for each set of reset limits.

Press **TUNE** then **▲** or **▼** until the upper display shows **RST-LIM#** for any of the 9 sets. (# the number appears in the rightmost digit of the upper display.)

Table 6-4 gives the key sequence for entering or changing the value of
 High reset limit
 Low reset limit

TABLE 6-4

Press	Action	Comments
ENTER (Note 1)	Upper display shows LIM-HIG#. SP display shows blinking High Reset Limit value.	# = PID number
ENTER	NUM LED is blinking. SP display shows 0.0.	Indicates numbered keys are allowed. Display cleared for entry.
Number Keys 0 through 9	Until SP display shows desired High Reset Limit value.	Press CLR to clear any errors.
ENTER	NUM LED goes out. SP display value entered into memory.	
▲	Upper display shows LIM-LOW#. TM display shows blinking Low Reset Limit value.	# = PID number
ENTER	NUM LED is blinking. TM display shows 0.0.	Indicates numbered keys allowed. Display cleared for entry.
Number Keys 0 through 9	Until TM display shows desired Low Reset Limit value.	Press CLR to clear any errors.
ENTER	NUM LED goes out. TM display value entered into memory.	
▲ or ▼	To cycle between High and Low Reset Limit.	
◀	Upper display will return to RST-LIM#	# = PID number
DISP or RESET (Note 2)	To return to control.	

NOTE 1: After you press **ENTER**, press **▲** or **▼** to access LIM-LOW#.

NOTE 2: To continue to another PID parameter, output limit, or reset limit, press **▲ or ▼**.

Tuning the Programmer

In the programmer the TUNE function is to set the output range of the set point (programmable output range). By using this function you can make the output range conform to the input range of the controller to which it is connected.

Procedure

The following procedure enables you to enter tune mode and access the High Limits, Low Limits, and Output Type for range 1 or range 2.

Entering Tune Mode

Press **TUNE** key. The upper display shows

RANGE-1

The values shown in the SP, TM, and PV displays are the high and low range limits and output type code for range one.

Press **▲ or ▼** key. The upper display shows

RANGE-2

The values shown in the SP, TM, AND PV displays are the set point high and low range limits, and output code for range two.

Changing the High Range Limit, Low Range Limit, and Output Type

To access and change any of the parameters shown in the SP, TM, or PV display.

Press **TUNE** then **▲ or ▼** key as shown above until the upper display shows

RANGE-1

or

RANGE-2

Table 6-5 gives the key sequence for entering or changing the value of
High range limit,
Low range limit, and
Output type code

If at any time an error code appears in the upper display, refer to the programming section for the definition of the error code.

TABLE 6-5

Press	Action	Comments
ENTER	Upper display shows RANGE-H#. SP display shows blinking High Range Limit value for channel indicated by a number in the rightmost digit.	# = Channel 1 or 2
ENTER	NUM LED is blinking. SP display shows 0.	Indicates numbered keys allowed. Display cleared for entry.
Number Keys 0 through 9	Until SP display shows desired High Range Limit value.	High range must be greater than or equal to low range. Press CLR to clear any errors.
ENTER	NUM LED goes out. SP display value entered into memory.	
▲	Upper display shows RANGE-L#. TM display shows blinking Low Output Limit value.	# = Channel 1 or 2
ENTER	NUM LED is blinking. TM display shows 0.	Indicates numbered keys allowed. Display cleared for entry.
Number Keys 0 through 9	Until TM display shows desired Low Output Limit value.	Low range value must be less than or equal to high range. Press CLR to clear any errors.
ENTER	NUM LED goes out. TM display value entered into memory.	
▲	Upper display shows SP-TYPE# PV display shows blinking output type code.	# = Channel 1 or 2 Code 0 = Linear Code 1 = Characterized
ENTER	NUM LED is blinking.	Indicates numbered keys allowed
Number Keys 0 through 9	Until PV display shows desired output type 0 = Linear 1 = Characterized	
ENTER	NUM LED goes out. PV display value entered into memory.	
▲ or ▼	To cycle through the 3 parameters if any changes need be made.	
◀	Upper display will return to RANGE-#	# = Channel 1 or 2
DISP or RESET	To return to control.	

Appendix

Introduction

This appendix explains the difference between PID-A equation and PID-B equation. It also gives the definition of the modes of control to help you determine which combination of Proportional Band, Rate, and Reset you'll need.

PID Equations

The DCP 700 Controller has two PID equations from which to choose: PID-A or PID-B. Additional control variations are possible including PI with A or B equation and proportional only. Duplex PID control is automatically selected whenever a PID equation is used with dual output relays.

PID-A Equation

In the "A" form of the PID equation, all three modes—proportional, integral, and derivative—act on the error. Figure 6-2 shows the implementation of the difference equations for the three modes when all three are active. As shown in the diagram, the process variable (PV) is compared with the present set point to obtain the present error (EM). The present error is then applied to the derivative block which contributes a rate action component to the signal. The present output from the derivative block (DM) is then simultaneously applied to the gain and integral blocks to separately include these functions in the signal. The present changes in output from the gain block (ΔK_m) is then combined with the present change from the integral block (ΔI_m) to obtain the present output error (GM) from the PID algorithm.

The PID-A equation is also used in various forms to provide additional variations of control, including PI control, PID or PI duplex control.

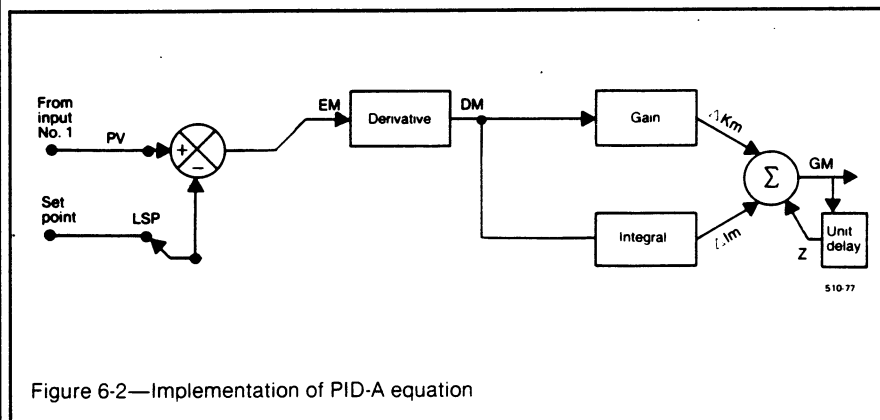


Figure 6-2—Implementation of PID-A equation

PID-A Control Variations

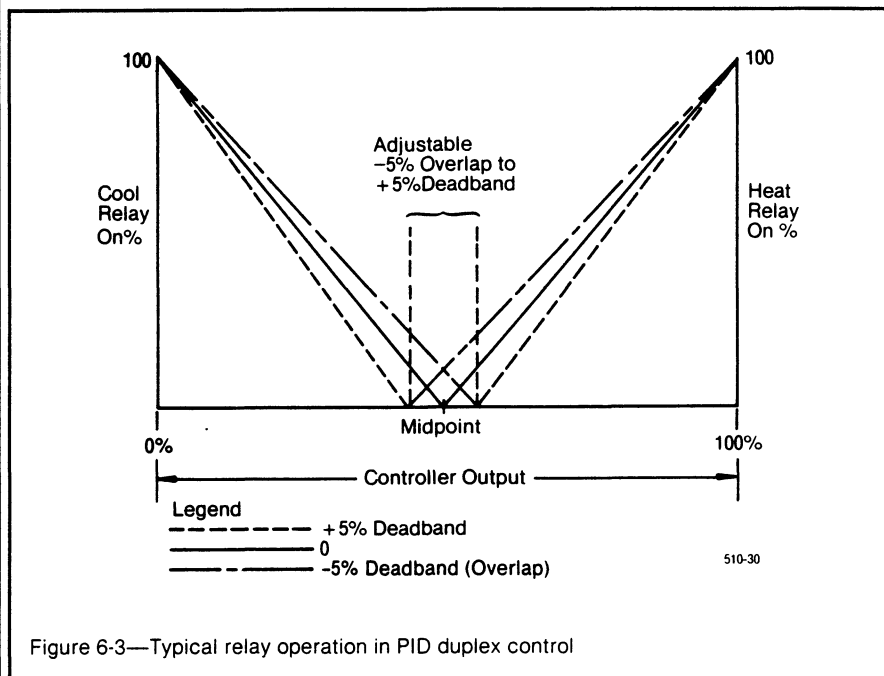
The two-mode PI control variation is obtainable from the PID-A equation. The desired control variation can be implemented by adjusting the unwanted derivative tuning constant to zero. This will cause the derivative contribution to be removed from the error by adjusting the rate time to zero. Integral action may be removed by adjusting the integral time to zero. This is not recommended except for making process response measurements for determination of optimum tuning constants.

PID Duplex Control

The PID-A equation will provide PID-a duplex control whenever the output is configured for duplex operation. In the duplex application, the PID algorithm will operate the two control relays for variations of the heat and cool applications. The PID algorithm computation is the same, but independent PB, integral time and rate tuning constants entered by the operation are used for calculation during the respective heating or cooling cycle. As shown in Figure 6-3, the heat and cool cycles are centered on the 50% output point (midpoint). Without a deadband-overlap setting, 50% output represents a static condition; no heat or cool action. The PID equation uses the normal tuning constant for outputs above 50%, but uses the cool tuning constants for outputs below 50%. 0% output represents full

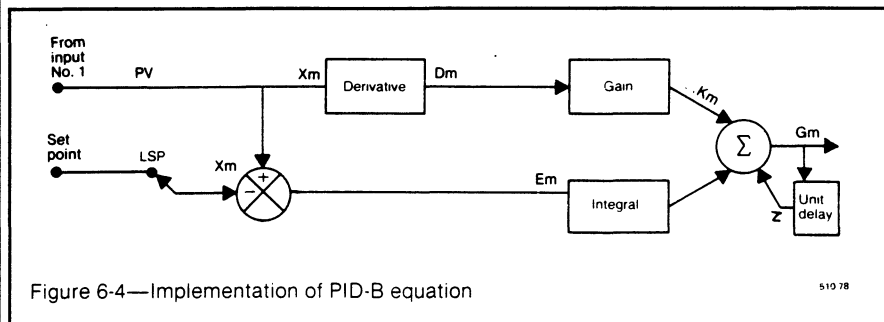
PID Duplex Control (Continued)

cool action (output relay number 2 full ON), and 100% output represents full heat action (output relay number 1 full ON). The deadband-overlap is operator-adjustable from 5% around midpoint to 25% deadband between the heat and cool cycles.



PID-B Equation

In the "B" form of the PID equation, the proportional and derivative modes act on the process variable and the integral mode acts on the error. In this equation, set point changes do not cause bumps in the output. Figure 6-4 shows the implementation of the difference equations for the three modes when all three are active. As shown in the diagram, the present process variable (X_m) is simultaneously applied to the derivative block and also compared with the present set point (Y_m). The derivative block contributes a rate action component to the process variable and routes its output (D_m) to the gain block where the appropriate proportional action is computed for the signal. The comparison of the present process variable with the present set point produces the present error (E_m) which is then routed to the integral block where the appropriate reset action from the error is computed. The present change in output from the gain block (ΔK_m) is then combined with the present change from the integral block (ΔI_m) to obtain the present output error (G_m) from the PID algorithm.

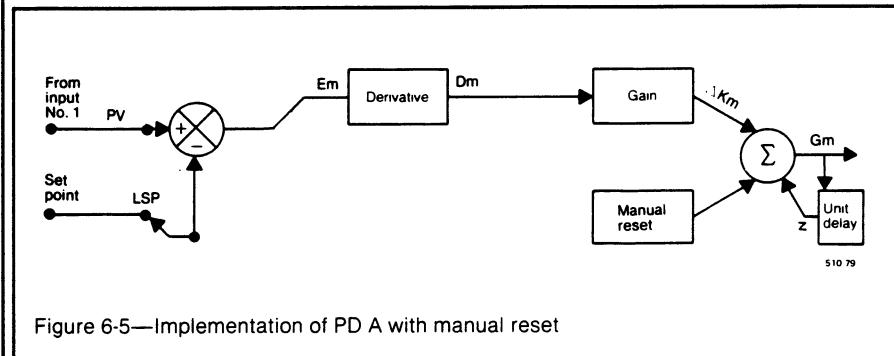


PID-B Control Variations

Control variations for PID-B are identical with those available for PID-A.

PD-A With Manual Reset Equation

This algorithm is a variation of the PID A equation (see Figure 6-5). The equation is computed with no integral contribution. The manual reset, which is operator adjustable, (value set by High reset limit), is then added to the present output error (k_m) to form the controller output (G_m).



Modes of Control

The basic characteristic of an automatic controller is the manner in which it acts to restore the controlled variable to the desired value: this is called the mode of control. The modes of control are

- Two Position (On/Off)
- Proportional
- Proportional plus reset
- Proportional plus rate
- Proportional plus reset plus rate

Two Position (On/Off)

In Two-Position Control, whenever the controlled variable deviates from the set point, the controller output relay is either activated or deactivated. Figure 6-6 shows the action graphically for a REVERSE acting controller where the relay is activated above the set point and deactivated below the set point. (The reverse of this is called a DIRECT acting controller.) This causes the controlled element to supply full energy or no energy to the process regardless of the size or rate of the deviation. This in turn causes the controlled variable to cycle.

A neutral zone is usually included between relay activation and deactivation to prevent relay "chattering."

Two-Position Control will not accommodate large or rapid load changes because the controlled variable will start to cycle at a higher rate and magnitude or develop a permanent offset from the set point.

In general, Two-Position Control is satisfactory for processes which have a slow reaction rate, minimum transfer lag or dead time, and the two extreme positions of the controlled element are adjusted to permit an input just slightly above and slightly below requirements for normal operation.

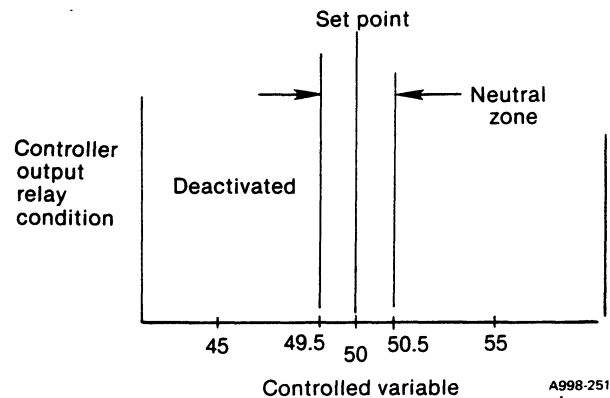


Figure 6-6—Controlled variable action with two-position control

Proportional

In proportional control, there is a fixed linear relationship between the value of the controlled variable (PV), and the value of the controller output (CO). The proportional controller changes CO a preset amount for a change in PV. The portion of the PV span within which the control action takes place is called the proportional band (PB). The PB is usually expressed as a percentage of the PV span. The three curves in Figure 6-7 graphically portray the action of a controller output within a proportional band. Curve A is based on the following:

Given

1. Input range = 50 to 250°F
2. Proportional band range = 100 to 200°F
3. Control set point (SP) = 50%
4. CO at which PV equals SP is 50%

Then

$$5. PB = \frac{200 - 100}{250 - 50} \times 100 = 50\%$$

6. A change in the PV of + 10% causes a change in CO = $(SP - PV)/PB = (50 - 60)/50 = -10/50 \times 100 = 20\%$ decrease in CO.

7. A change in the PV of - 10% causes a change in CO = $(SP - PV)/PB = (50 - 40)/50 = +10/50 \times 100 = 20\%$ increase in CO.

Curves B and C assume the same PB and SP as Curve A, but Curve B assumes the CO at which the PV equals the SP is 25%, while Curve C assumes the CO at which the PV equals the set point is 75%. You can see that the change in CO per change in PV (slope of the curve) is the same for the three curves.

On most controllers, the proportional band is adjustable from some minimum value up to 100%. The larger the proportional band, the smaller the change in the controller output will be for a given change in the PV.

Electric proportional controllers have two types of output. One type, called a current proportioning controller, has a current output, usually 4 to 20 mA. The other type, called a time proportioning controller, has a relay output.

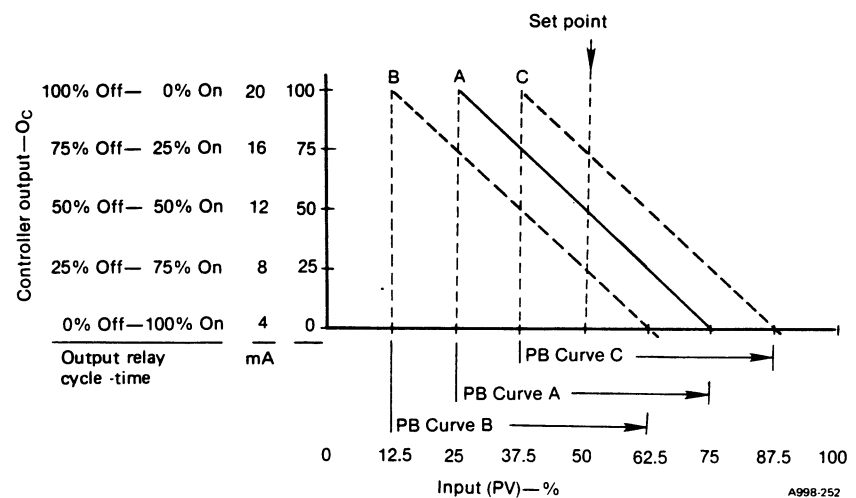


Figure 6-7—Controller output action with proportional action

Current Proportioning	The current proportioning controller action is the same as that for the example shown in Figure 6-7. The controller output is in milliamperes.
Time Proportioning	In time proportioning control, the output is a relay which is cycled on and off at a constant rate. However, the ratio of the on-time to the off-time varies in proportion to the amount the PV deviates from the SP. Figure 6 shows how the relay position is controlled under the same process conditions and controller settings as used for the current proportional controller. Notice that CO is the percentages of the output relay on-time and off-time, and that these percentages change in proportion to the change of the PV from the SP. In Figure 6-7, the CO at which the PV equals the set point is 50% (50% off-50% on). If the PV increases 25%, the proportions become 75% off-25% on. If the PV decreases 25%, the proportions become 25% off-75% on.
Proportional plus Reset	Proportional control reduces the PV cycling rate and amplitude below that of two position control, even with moderate load changes. However, sometimes the change in the output with proportional control is not enough to make the PV equal the set point. This continuing difference between the PV and the SP is called "offset." How to eliminate offset is discussed here. The offset present with proportional control only is due to the fact that, under the new load conditions, the controlled element is not positioned to supply the proper amount of energy or other medium to cause the PV to equal SP. Curve B in Figure 6-7 illustrates the Proportional control action line when the load is too large and the energy or medium is insufficient, and Curve C illustrates the Proportional Control action line when the load is too small and the energy or medium is too great. To correct these conditions it is necessary to "reset" the controller output so that it will move the controlled element to the proper position. The reset can be done manually or automatically. If done manually, the operator controls when the reset is made, the amount of the reset, and the time it takes to make the reset. If done automatically, the operator has to predetermine the time for the reset and set the value in the controller; then the controller will automatically make the reset. In the proportional plus reset mode, as soon as there is a difference between the SP and the PV, the reset action starts to change the output beyond that made by the proportional action, and at a rate proportional to the difference between the SP and the PV. For example, a four percent difference between the SP and the PV causes a change in controller output to be added to that caused by the proportional action at twice the rate as that for a two percent difference. Reset action causes an additional change in output as long as a difference exists. The amount of change for each minute the difference between the SP and PV exists is equal to that made by the proportional action, and is called the Reset Time in minutes per repeat. For example, if you set a Reset Time of 2, the reset action adds a change in controller output equal to 200% of the proportional action every 2 minutes the difference between the SP and the PV exists. Figure 6-8 illustrates reset action. T_r is the time per repeat, E is the difference between the Sp and PV, CO_p is the change in output due to proportional action, and CO_r is the change in output due to reset action. With proportional control, even with the reset action added, the length of time it takes to make the output change is sometimes too great. To correct this condition, a control action called "rate" can be added to a proportional controller. The affect of the rate action on the controller output is discussed next under the next heading "Proportional plus Rate."

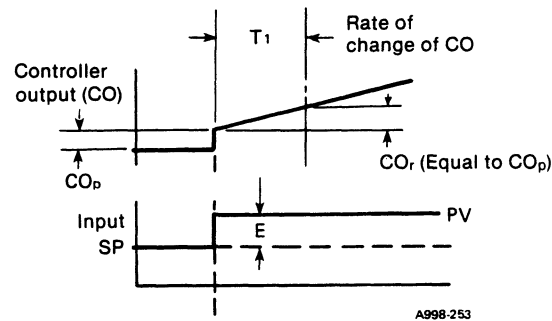


Figure 6-8—Controller output action with proportional plus reset

Proportional plus Rate

Rate action provides an immediate change in the controller output when the SP and PV start to deviate from one another. This change causes the output change being made by the proportional action to be reached sooner. In Figure 6-9, Curve A illustrates the time for CO to change with proportional action with or without reset, and Curve B the time when rate action is added.

Proportional controllers are usually provided with a "Rate Time" adjustment. Rate Time is the decrease in the time it takes the proportional control action to change the controller output when Rate Action is added. Rate Time is shown in Figure 6-10 as the difference in time between Curves A and B at the Final Controller Output Position.

The amount of the immediate change in controller output due to Rate Action depends on the amount of Rate Time, and the rate of change of the output. Figure 6-10 shows how proportional plus rate action with the same Rate Time setting causes a different immediate output change when the rate of change of CO is different. In 6-10A, CO is changing slowly, so the immediate output change is smaller than that in 6-10B, where CO is changing rapidly.

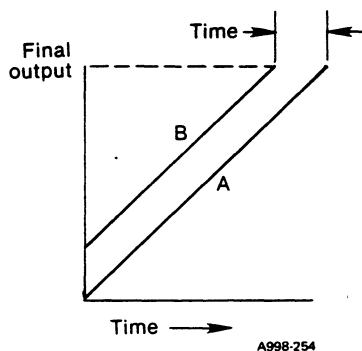


Figure 6-9—Illustration of rate time

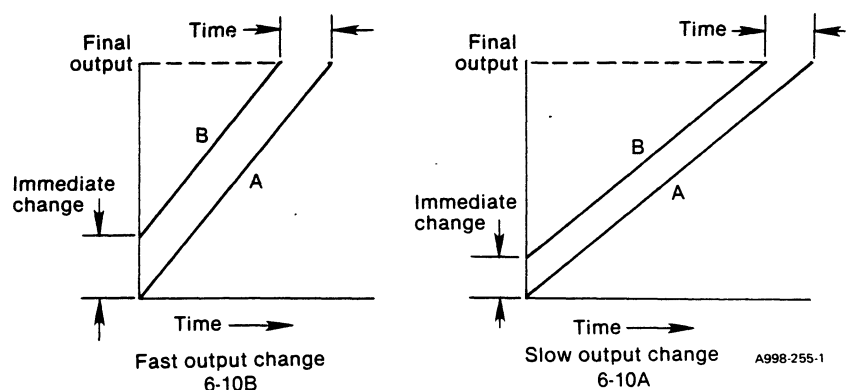


Figure 6-10—Variations in the immediate output change caused by rate action

Using the Modes

Each mode of control is applicable to processes having combinations of certain basic characteristics. It is important to remember that the simplest mode which will do the job is the best one to use, both for economy and for best results.

Considerable experience is required in the selection of the mode best suited to the process. Table 6-6 summarizes the principle process characteristics each of the control modes discussed here is best suited for.

TABLE 6-6—Summary of Principle Process Characteristics

Control Mode	Rate of PV Change	Transfer Lag (Lead Time)	Process Load Changes
Two-Position	Slow	Slight	Small and slow
Proportional	Slow or Moderate	Small or Moderate	Small
Proportional plus Reset	Fast	Small or Moderate	Slow but any amount
Proportional plus Rate	Slow or Moderate	Moderate	Small
Proportional plus Reset plus Rate	Fast	Moderate	Fast

**DCP 700
Digital Control Programmer****Operation**

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Introduction

It is assumed that the DCP has been wired and programmed in accordance with the prior sections of this manual.

Operation of the DCP requires you to do the following:

- supply power,
- add filtering to the PV if the PV is unstable enough to require it,
- use **SYNC** to turn on the SYNC LED when doing operations that you want to occur simultaneously on both channels of a two channel programmer,
- select and start the program required for the process,
- monitor the program, controller output and process variable,
- interact with the process during program run.

Add Filtering to the PV

You can add filtering to the inputs any time the DCP is powered. The amount of filtering is controlled by the time constant of the filter. This can be changed by using the procedures in the Configuration section under the heading "How to Configure"

On Two Channel DCPs, Synchronize Operations

CAUTION: Check both channels after pressing **SYNC** to make sure the programs are in the condition you expected.

Alternate presses of **SYNC** turn the SYNC LED on and off. When the LED is on, the following operations are done simultaneously on the program in each channel, regardless of the channel number displayed, and when the LED is off, the operations are done only to the program in the channel number displayed.

- Select
- Start
- Hold
- Advance
- Fast Forward
- Segment select
- Reset

The SYNC LED can be turned on or off when the RESET, RUN or HOLD LED is lit. Changing the condition of the SYNC LED does not automatically change the operating condition of the programs in the channels. The condition of the programs does not change until you do one of the above operations.

For example: if the SYNC LED is off and the program in CH 1 is running and that in CH 2 is held, pressing **SYNC** while CH 1 is displayed, does not cause the program in CH 2 to start or the program in CH 1 to stop. The next time one of the above operations is done, the programs in both channels will do the same thing.

Select and Start a Program

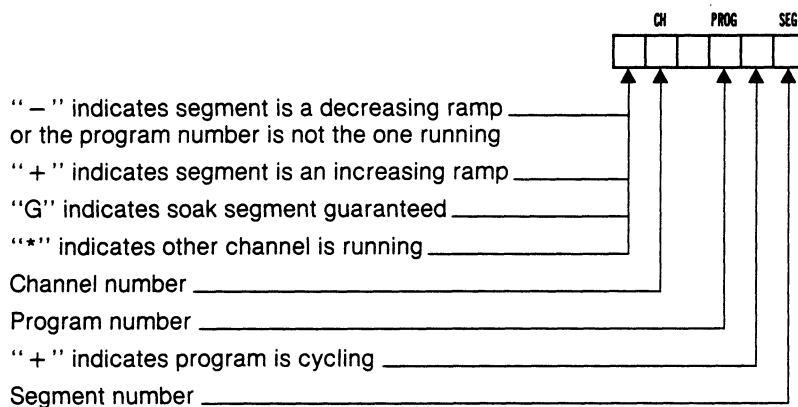
The following are the procedures to start a program, and a description of the displays and indicators occurring during the procedures.

Operation	Procedure	Displays
Program select	1. Press PROG .	The number in the PROG becomes "0," and the NUM LED blinks.
	2. Press the number keys required.	Your program number appears in the PROG display.
	3. Press ENTER .	NUM LED goes out.
Program Start At the first segment	1. Press RESET .	RESET LED lights.
	2. Press RUN .	RUN LED lights. Program starts at Segment 1.
At a chosen segment	1. Press HOLD .	HOLD LED lights
	2. Press SEG .	NUM LED blinks.
	3. Press the number keys required.	Your segment number appears in the SEG display.
	4. Press ENTER .	NUM LED goes out.
	5. Press RUN .	RUN LED lights. Program starts at chosen segment.
PV (Hot) start At the next segment equal to or greater than the PV.	1. Press ▲ RUN .	HOLD LED lights, and the starting segment number appears in the SEG display.
	2. Press RUN .	RUN LED lights. Program starts at displayed segment.
At the next segment equal to or less than the PV.	1. Press ▼ RUN .	HOLD LED lights, and the starting segment number appears in the SEG display.
	2. Press RUN .	RUN LED lights. Program starts at displayed segment.

Monitor Data During a Program Run

You can monitor the program and process status by observing the DCP's displays and LEDs.

- The SP display shows the present set point.
- With alternate presses of **DISP**, the TM display shows the remaining time in a segment or the controller output.
- The PV display shows the process variable value.
- When **DISP** and **←** are pressed at the same time, the cycle number for the complete program appears in the SP display and that for portions of the program in the TM display.
- The upper display shows the error and alarm messages listed in Tables 7-1 and 7-2, and additional data according to the following diagram:



Interact with the Process During a Program Run

After selecting and starting a program, you can do the following operations by using procedures listed with them. For information, the displays and LEDs are described.

Operation	Procedure	Displays and LEDs
Hold the program.	1. Press HOLD .	HOLD LED lights and the program is held at the set point that appears in the SP display.
Advance the program to the next segment.	1. Press ← ADV .	The next segment number appears in the SEG display, and the program continues running from there.
Fast forward the program.	1. Press ADV .	The program and event timers speed-up at the advance-rate previously stored. Guaranteed soak segments become plain soak segments. See the Configuration section.
Select a segment.	1. Press SEG .	The NUM LED blinks and "0" appears in the SEG display.
	2. Press number keys to enter new segment.	New segment number appears in the SEG display.
	3. Press ENTER .	NUM LED goes out.

Operation	Procedure	Displays
Reset the program to the starting segment.	1. Press ; then hold it and press .	The program returns to the starting segment, and the RESET LED lights. If you want the program to return to Segment 1, select Segment 1 by using the Operation procedure "Select a segment."
Change the controller to manual operation with the program held	NOTE: No verifying/editing can be done when controller is in "manual" operation. Press and .	MAN LED lights, the controller output appears in the TM display, and the first digit blinks.
without holding the program	Press .	MAN LED lights, the controller output appears in the TM display, and the first digit blinks.
Change the output in manual operation		
One digit at a time	1. Press to select a digit to change. 2. Press or to change the digit.	Digit blinks. Digit increases or decreases.
All digits	1. Press .	NUM LED lights. New value shows in the TM display.
Change the controller to automatic operation.	NOTE: If the NUM LED is lit, press .	MAN LED goes out and the program data appears in the SP and TM display.
Change program data.	1. Press .	Channel number being verified and time unit of program
NOTE: Changing data values for the following items in the current segment of a running program (or one on HOLD) will take affect immediately: SOAK SP GUARANTEED SOAK SP GUARANTEED SOAK BAND RAMP-T RAMP-E PV OR DEV EVENT (except event no.) PV RATIO/BIAS	2. Press .	Program data prompts are displayed sequentially.
Changes of other items in the current segment will not become effective until the current segment is completed. Thus, if a program is cycling, the change will be effective in the next cycle.	3. When correct prompt appears, press .	Value in SP display blinks. If you don't want to change this value or code, go to step 4d to change data in TM display or go to step 4m to return to segment prompt.
	4. To change the data in the SP, TM, or PV display, make sure memory protect switch is in far right position (M.E.).	
	a) Press .	NUM LED blinks.
	b) Press the number keys for the new value or code.	New value or code shows in the display
	c) Press .	NUM LED goes out.
	d) Press .	Value in TM display blinks. If you don't want to change this value or code, go to step 4h to change data in PV display or step 4m to return to segment prompt.
	e) Press .	NUM LED blinks
	f) Press the number keys for the new value or code.	New value or code shows in TM display.
	g) Press .	NUM LED goes out.
	h) Press .	Value in PV display blinks. If you don't want to change this value or code, go to step 4m to return to segment prompt.
	i) Press .	NUM LED blinks.
	j) Press .	New value or code shows in PV display
	k) Press the number keys for the new value or code.	
	l) Press .	NUM LED goes out
	m) Press .	Return to segment prompt
Display and change Parameters Tuning constants	See Page 5-1. See Page 6-1 (Manual Reset discussed on Page 6-3).	
Programmer output	See Page 6-8	

Remote Operation

The following operations can be actuated through external, user-supplied contacts:

- Run
- Hold
- Reset
- PV start and segment advance
- Ramp-E or IF EXT/ON EXT
- Program Select

See the installation section for wiring of external operation contacts.

Table 7-1 provides a summary of remote operation data for reference.

Table 7-1 Remote Operation Data

Operation Signal	Remote Contact Connected Across Terminals	Contact Function
Run	17-33	Pulse
Hold	17-34	Pulse
Reset	17-35	Pulse
PV Start and Seg. Adv.	17-36	Pulse
Synchronous Mode	17-37	Status**
Channel Set	17-38	Pulse
Channel 1	17-39	Status**
Channel 2	17-40	Status**
External 1	17-41	Pulse
External 2	17-42	Pulse
Program No. 1	17-43*	Status**
Program No. 2	17-44	Status**
Program No. 4	17-45	Status**
Program No. 8	17-46	Status**
Program No. 10	17-47	Status**
Program No. 20	17-48	Status**

*See Figure 2-14 in the Installation section for list of contact combinations for other program numbers.

**Status Contact(s) must be closed when used in conjunction with a pulse contact.

Remote Program Select

Be sure contact closures are made as follows to select program number before initiating RUN, PV START, etc. remotely.

1. Program number contact or combination of number contacts closes which is equal to the desired program number.
2. Do you want to use the same program numbers in both channels?
 Yes—Go to Step 3
 No—Go to Step 5
3. Close Synchronous contact
4. Pulse the Channel Set Contact. The REM LED lights, program number is selected, and DCP is in the synchronous mode.
5. Close Channel 1 or Channel 2 contact, as required. Be sure the synchronous signal is off.
6. Pulse the Channel Set Contact. The REM LED lights, program number is selected, and the DCP is in the asynchronous mode.

NOTE:

1. If the Channel Set Contact is pulsed while the synchronous signal is off and both the channels are on or off, the message ERROR 22 blinks in the upper display and the program number is not selected.
2. You can not select Program number 0 remotely.

Remote Run

1. Be sure the correct program number is selected. See the Remote Program Select procedure.
2. Do you want both channels to run?
Yes—Go to Step 3
No—Go to Step 5
3. Be sure the synchronous signal is on, or confirm that DCP is in the synchronous mode.
4. Pulse the run contact. Program run starts at segment 1, REM LED is on, and DCP is in the synchronous mode.
5. Close Channel 1 or Channel 2 contact, as required.
6. Be sure the synchronous signal is off, or confirm that DCP is in the asynchronous mode.
7. Pulse the Run contact. Program run starts at segment 1 of selected channel, REM LED is on, and DCP is in the asynchronous mode.

NOTE: If the Run contact is pulsed while the synchronous signal is off and both the channels are on or off, the message ERROR 22 blinks in the upper display and the run is not started.

Remote PV Start and Segment Advance

If DCP is in the ready mode (RESET LED is on), this operation will initiate a PV Start in the direction (upward or downward) previously configured. If DCP is in the RUN or HOLD mode, this operation will initiate an advance to the next segment in a program.

1. Be sure the correct program number is selected. See the Remote Program Select procedure.
2. Do you want a PV Start/Segment Advance on both channels?
Yes—Go to Step 3
No—Go to Step 5
3. Be sure the synchronous signal is on, or confirm that DCP is in the synchronous mode.
4. Pulse the PV Start/Segment Advance Contact. Program starts at desired set point or advances to next segment, REM LED is on, and DCP is in synchronous mode.
5. Close Channel 1 or Channel 2 contact, as required.
6. Be sure the synchronous signal is off, or confirm that DCP is in the asynchronous mode.
7. Pulse the PV Start/Segment Advance Contact. Program starts at desired set point in designated channel or advances to next segment in designated channel, REM LED is on, and DCP is in asynchronous mode

NOTE: If the PV Start/Segment Advance Contact is pulsed while the synchronous signal is off and both the channels are on or off, the message ERROR 22 blinks in upper display and the PV Start/Segment Advance is not initiated.

Remote Hold

1. Do you want to put both channels on HOLD?

Yes—Go to Step 2

No—Go to Step 4

2. Be sure the synchronous signal is on, or confirm that DCP is in the synchronous mode.
3. Pulse the HOLD Contact. Program is halted, REM LED is on, and DCP is in synchronous mode.
4. Close Channel 1 or Channel 2 contact, as required.
5. Be sure the synchronous signal is off, or confirm that DCP is in the asynchronous mode.
6. Pulse the HOLD Contact. Designated channel is halted, REM LED is on, and DCP is in the asynchronous mode.

NOTE: If the HOLD Contact is pulsed while the synchronous signal is off and both channels are on or off, the message ERROR 22 blinks in upper display and program is not halted.

Remote Reset

1. Do you want to Reset both channels?

Yes—Go to Step 2

No—Go to Step 4

2. Be sure the synchronous signal is on, or confirm that DCP is in the synchronous mode.
3. Pulse the RESET Contact. Program is RESET, REM LED is on, and DCP is in the synchronous mode.
4. Close Channel 1 or Channel 2 contact, as required.
5. Be sure the synchronous signal is off, or confirm that DCP is in the asynchronous mode.
6. Pulse the RESET Contact. Designated channel is RESET, REM LED is on, and DCP is in the asynchronous mode.

NOTE: If the RESET Contact is pulsed while the synchronous signal is off and both channels are on or off, the message ERROR 22 blinks in upper display and program is not reset.

**Remote Ramp-E or
IF EXT/ON EXT**

If segment being run has been programmed as a RAMP-E type or to include IF EXT/ON EXT function, each pulse of external 1/external 2 Contact increases or decreases the set point by the previously programmed ramp rate, or initiates segment change as programmed for IF EXT/ON EXT function.

TABLE 7-1—Programming Error Messages

Message	Meaning	Corrective action
ERROR 1	You are trying to enter data in, or copy data to, a program number that has been used.	Select a new program number or delete the program data.
ERROR 2	You are trying to verify or copy an unused program number.	Select a new program number.
ERROR 3	You are trying to run an unused program number.	Select a new program number.
ERROR 4	You did not enter the program number and segment number to run when IF EXT/ON EXT was programmed.	Enter the program number and segment number.
ERROR 5	You are trying to use a 2-channel Relative Humidity model (DCP722) as a single channel for RH only.	You can use channel 1 for "Dry Bulb" temperature only, but you cannot use channel 2 alone for % RH since the dry bulb set point (for channel 1) is needed to calculate the Wet Bulb set point for channel 2.
ERROR 6	The external signal has not been detected by the RAMP-E segment running in Channel 1.	Verify program data for Ramp-E segment. Check appropriate external contact (1 or 2), and repair or replace as required
ERROR 7	The external signal has not been detected by the Ramp-E segment running in Channel 2.	Verify program data for Ramp-E segment. Check appropriate external contact (1 or 2) and repair or replace as required.
ERROR 8	Signals that too many time event calculation blocks are being used at the same time. The software provides 12 time event calculation blocks. Therefore, that is the maximum number of time events that can be "active" at the same time.	Reconfigure events to allocate space in the event calculation blocks. Note that one time event number (e.g. Time Event No. 1) can use up all 12 blocks if it is turned on and off 12 times in the same segment.
ERROR 10	The low limit you are entering is greater than the high limit.	Decrease low limit or increase the high limit.
ERROR 11	You are using RAMP-T or RAMP-E as the last segment in the program.	Use RAMP-X or a soak segment.
ERROR 14	Data you are entering for the event is incorrect for the type.	Select event number with the correct type or change type for that number.
ERROR 15	Insufficient memory remaining to store the data.	Delete one or more stored programs.
ERROR 16	Your program has more than ninety-nine segments.	Reduce the number of segments to ninety-nine or less.

TABLE 7-1—Programming Error Messages (continued)

Message	Meaning	Corrective action
ERROR 17	The program you are trying to change is protected.	Change the position of the Program-Protection-Switch.
ERROR 18	The set point is greater than the high range limit.	Reduce the set point as required.
ERROR 19	The set point is lower than the low range limit.	Increase the set point as required.
ERROR 20	There is no set point that meets the PV Start requirements.	Select another method of starting.
ERROR 21	Incorrect remote program number	Select another program number.
ERROR 22	Incorrect remote channel number	Select another channel number.
ERROR 23	You are trying to delete a program that is running or held.	Reset the program or wait for the program to finish.
ERROR 24	You made an error in the dumping procedure.	Repeat the dumping procedure.
ERROR 25	Program number you are trying to load is not in the cartridge.	Select the correct program number.
ERROR 26	Program number you are trying to load already has data stored under it.	Delete data under the program number and repeat the loading procedure.
	Insufficient memory available.	Delete one or more programs.
ERROR 27	You made an error in the loading procedure.	Repeat loading procedure.
ERROR 28	Illegal remote operation actuated during program loading operation.	Disable remote operation and repeat loading procedure.
ERROR 29	You cannot download data while check-sum is being run.	Download data after check-sum run.
ERROR 30	DCP detects an EEPROM write error	Check that PV input actuation is correct. Must short the EEPROM protect connector pins to complete the attempted operation.
ERROR 31	Incorrect zero and/or span calibration values.	Check calibration values and change as required.

TABLE 7-2—Alarm Messages

Message	Meaning	Corrective action
ALARM 1	Process variable for Channel 1 is at or above the upper range limit.	Check that control action is correct. Check the input circuit for failure.
ALARM 2	Process variable for Channel 1 is at or below the lower range limit.	Check that control action is correct. Check the input circuit for failure.
ALARM 3	Process variable for Channel 2 is at or above the upper range limit.	Check that control action is correct. Check the input circuit for failure.
ALARM 4	Process variable for Channel 2 is at or below the lower range limit.	Check that control action is correct. Check the input circuit for failure.
ALARM 61	The communications board failed.	Replace the communications board.
ALARM 71	Current proportional output for CH 1 failed.	See the Troubleshooting procedures in the Service section.
ALARM 72	Current proportional output for CH 1 failed.	See the Troubleshooting procedures in the Service section.
ALARM 73	The Auxiliary Output Circuit failed.	See the Troubleshooting procedures in the Service section.
ALARM 74	D/A converter calibration failed.	See the Troubleshooting procedures in the Service section.
ALARM 81	The A/D converter correction calibration point for Channel 1 is incorrect.	See the Troubleshooting procedures in the Service section.
ALARM 82	The A/D converter range calibration for Channel 1 is incorrect.	See the Troubleshooting procedures in the Service section.
ALARM 83	The A/D converter correction calibration point for Channel 2 is incorrect.	See the Troubleshooting procedures in the Service section.
ALARM 84	The A/D converter range calibration for Channel 2 is incorrect.	See the Troubleshooting procedures in the Service section.
ALARM 92	Content of EEPROM lost.	See the Troubleshooting procedures in the Service section.
ALARM 93	Data check point lost.	See the Troubleshooting procedures in the Service section.
ALARM 94	Parameter data lost	Re-enter data.
ALARM 95	PID data lost	Re-enter data.
ALARM 96	Program data lost	Re-enter data.
ALARM 97	Battery power low	Replace the battery.
ALARM 98	RAM failed	See the Troubleshooting procedures in the Service section.
ALARM 99	ROM failed	See the Troubleshooting procedures in the Service section.

**DCP 700
Digital Control Programmer****Calibration**

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INPUT CALIBRATION

When and How to Calibrate PV Input

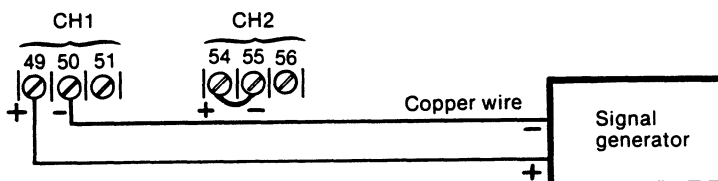
You do not have to calibrate the DCP on a periodic basis. Use this procedure to calibrate your DCP any time you believe the PV input is in error, when you replace boards in your DCP, when calibration is required for certification, or when required after changing the input type.

Equipment Required

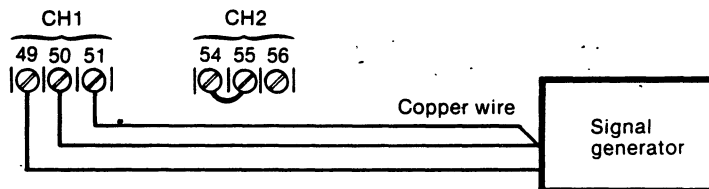
Medium-blade, Phillips-head screwdriver
Signal generator with accuracy of 0.01% for input type to be calibrated.
Copper wire to connect the generator to the DCP.

Set Up

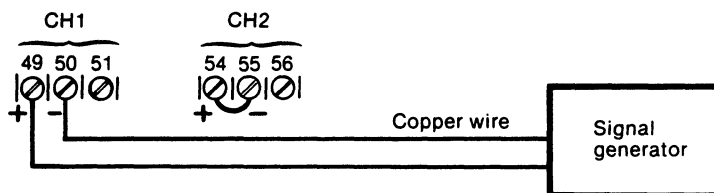
1. Remove power from DCP. Tag and disconnect any wiring from terminals 49 through 56 at the rear of the case.
2. Connect the signal generator to the input terminals for channel 1 and short-circuit the input terminals for channel 2 as shown.



Channel 1 calibration connections for thermocouple, millivolt, or volt input.



Channel 1 calibration connections for Resistance Temperature Detector input including dry bulb (CH1) and wet bulb (CH2) for RH input.



Channel 1 calibration connections for current input

3. Pull the latches, one on either side of the bezel, forward. Pull on the bezel, alternating sides, until the bezel starts to come away from the case. Slide the chassis out of case slowly until you see the connector on the CPU board (third from front).
4. Locate the two vertical pins on the EEPROM-protect-connector which is located to the left of the battery on the top right side of the CPU board. Wrap a piece of wire around the pins to short them together and be sure the wire doesn't contact any other components or the case.
5. Slide the chassis back into the case and push in the latches.
6. Open the plug-in module door and move the memory protect switch lever to the far right position (M.E.). Close the module door.
7. Apply power to the DCP and allow it to warm-up for two hours; then, proceed with the steps in the Tables 8-1 to calibrate the PV input.

TABLE 8-1—Calibration Procedure

Step	Action	Comment
1.	If the input type of your DCP is current, voltage, thermocouple with a °C range, or RTD, go to Step 12; otherwise go to Step 5.	
2.	Press ▼ and while pressing, press RESET .	"ADJUST" appears in the upper display, "0" appears in the SP display, and the NUM LED lights.
3.	Press 4 , ENTER .	"4" appears in SP display until ENTER is pressed, then "PASSWORD" appears in the upper display, and "0" in the SP display.
4.	Press 2 , 2 , 5 , 5 .	"2255" appears in the SP display.
5.	Press ENTER four times.	"PV-TYPE1" appears in the upper display, and a code appears in the SP display to identify the input type for CH 1.
6.	■ If you are not going to calibrate CH 1, go to Step 8. ■ If you are going to calibrate CH 1, and the code in the SP display corresponds to the one in the Table 8-2, °C (Calib) Code column go to Step 8. ■ If you are going to calibrate CH 1 and the code in the SP display corresponds to the one in the Table 8-2, °F Code column, go to Step 7.	All calibrations are done in °C.
7.	Select the corresponding code from the °C Calib. Code column and key it in. For example, if code 22 was being displayed, you would enter code "3" per Table 8-2.	The new code appears in the SP display.
8.	Press ENTER .	"PV-TYPE2" appears in the upper display, and a code (0 if the DCP doesn't have a CH 2) appears in the SP display to identify the input type for CH 2.
9.	■ If you are not going to calibrate CH 2, go to Step 11. ■ If you are going to calibrate CH 2 and the code in the SP display corresponds to the one in the Table 8-2, °C (Calib) Code column for the input type you are calibrating, go to Step 11; otherwise go to Step 10.	
10.	Select the corresponding code from the Table 8-2, °C (Calib) column for the °F code being displayed and key it in.	The new code appears in the SP display.
11.	Press ENTER .	Displays show RESET condition.
12.	Press ▼ , and while pressing; press RESET .	"ADJUST" appears in the upper display, "0" in the SP display, and the NUM LED blinks.
13.	Press 2 .	"2" appears in the SP display.
14.	Press ENTER .	"RG-C1-Z" appears in the upper display.
15.	Set the signal generator to the zero value from Table 8-2 for your DCP's input type. Wait until the reading in the SP display stabilized ± 1 count. This should be less than one minute. Then go to Step 16.	For thermocouple input, do <i>not</i> compensate for ambient temperature. Use the exact zero and span values given in Table 8-2 for the input.
16.	Press ENTER .	"RG-C1-S" appears for CH 1 in the upper display.
17.	Set the signal generator to the span value from Table 8-2 for your DCP's input type. Wait until the reading in the SP display stabilized ± 1 count. This should be less than one minute. Then go to Step 18.	
18.	Press ENTER .	"RG-C2-Z" appears in the upper display for CH 2 calibration. If you want to calibrate CH 2, go to Step 19; otherwise press RESET and go to Step 24.

TABLE 8-1 (continued)

Step	Action	Comment
19.	Refer to Step 2 in Set Up and reverse the connections to short-circuit the input terminals for channel 1 and connect a signal generator to input terminals for channel 2.	
20.	Set the signal generator to the zero value from Table 8-2 for your DCP's input type. Wait until the reading in the SP display stabilizes ± 1 count. This should be less than one minute. Then go to Step 21.	
21.	Press ENTER .	"RG-C2-S" appears for CH 2 in the upper display.
22.	Set the signal generator to the span value from Table 8-2 for your DCP's input type. Wait until the reading in the SP display stabilizes ± 1 count. This should be less than one minute. Then go to Step 23.	
23.	Press ENTER .	Displays show the RESET condition.
24.	If you changed a code in Steps 7 through 10, go to Step 25; otherwise calibration is completed.	
25.	Press ▼ and while pressing, press RESET .	"ADJUST" appears in the upper display, "0" appears in the SP display, and the NUM LED lights.
26.	Press 4 , ENTER .	"4" appears in SP display until ENTER is pressed; then "PASSWORD" appears in the upper display, and "0" in the SP display.
27.	Press 2 , 2 , 5 , 5 .	"2255" appears in the SP display.
28.	Press ENTER four times.	"PV-TYPE1" appears in the upper display, and a code appears in the SP display to identify the input type for CH 1.
29.	If the code in the SP display defines the temperature unit ($^{\circ}\text{F}$ or $^{\circ}\text{C}$) you want for CH 1, go to Step 31; otherwise go to the Step 30.	
30.	Select from Table 8-2 the code for the temperature unit you want and key it in.	The code appears in the SP display.
31.	Press ENTER .	"PV-TYPE2" appears in the upper display, and a code (0 if your DCP doesn't have a CH 2) appears in the SP display to identify the input type for CH 2.
32.	If the code in the SP display defines the temperature unit ($^{\circ}\text{F}$ or $^{\circ}\text{C}$) you want for CH 2, go to Step 34; otherwise go to Step 33.	
33.	Select from Table 8-2 the code for the temperature unit you want and key it in.	The new code appears in the SP display.
34.	Press ENTER .	Displays show the RESET condition.
35.	Remove power from DCP and reverse Steps (1-6) in Set Up procedure to return DCP to operating condition.	

TABLE 8-2—Calibration codes and values

Input Type	$^{\circ}\text{F}$ Code	$(^{\circ}\text{C})$ (Calib) Code	Zero Value	Span Value	Range Limits	
					Lower	Upper
T	20	1	- 5.603 mV	14.860 mV	- 200	300 $^{\circ}\text{C}$
J	21	2	0.000	45.498	0	800
E	22	3	0.000	61.022	0	800
K	23	4	0.000	48.828	0	1200
S	24	6	0.000	16.771	0	1600
R	25	7	0.000	18.842	0	1600
B	26	8	0.000	13.585	0	1800
W5W26	27	9	0.000	36.922	0	2300
Ni/Ni Moly	28	11	0.000	67.190	0	1300
RTD	29	15	18.49 ohm	280.9 ohm	- 300	900 $^{\circ}\text{F}$
		15	18.49 ohm	175.84 ohm	- 200	200 $^{\circ}\text{C}$
Current		19	4.0 mAdc	20.0 mAdc		
Volts		19	1.0 Vdc	5.0 Vdc		
Millivolt		19	0 mVdc	10 mVdc		
RH	Same as RTD for dry bulb on channel 1 and wet bulb on channel 2					

A/D, D/A AND C/J CALIBRATION

Summary

This procedure tells you how to check the analog-to-digital (A/D) converter, digital-to-analog (D/A) converter, and cold junction compensation correction calibration points. Note that these points are primarily for factory calibration and normally don't require readjustment in the field. There is also a procedure for making a General Calibration Reset.

Equipment Required

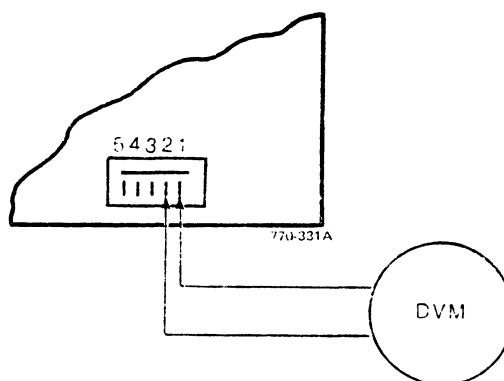
- Medium Phillips screwdriver
- Ammeter to read 4-20 mA output
- Digital voltmeter (DVM) capable of reading at least 5-digit values
- Calculator is recommended

Set Up

1. Remove power from DCP. Tag and disconnect any input and output wiring from terminals 49 through 62 at the rear of the case.
2. Short-circuit the input terminals for Channel 1 (49 and 50) and 2 (54 and 55).
3. Pull the latches, one on either side of the bezel, forward. Pull on the bezel, alternating sides, until the bezel starts to come away from the case. Slide the chassis out of case slowly until you see the connector on the CPU board (third from front).
4. Locate the two vertical pins on the EEPROM-protect-connector which is located to the left of the battery on the top right side of the CPU board. Wrap a piece of wire around the pins to short them together and be sure the wire doesn't contact any other components or the case.
5. Slide the chassis back into the case and push in the latches.
6. Remove the adhesive backed cover from the cutout near the bottom right-hand corner of the case. This lets you access the pins of connector P164. The P164 pin assignments are:

Pin No.	Signal
1	Channel 1 A/D ground
2	Channel 1 A/D multiplexer
3	Channel 2 A/D ground
4	Channel 2 A/D multiplexer
5	Cold junction compensation resistance

connect a DVM to pins 1 and 2



DVM connection for Channel 1 A/D calibration

7. Open the plug-in module door and move the memory protect switch lever to the far right position (M.E.). Close the module door.
8. Apply power to the DCP and allow it to warm-up for two hours; then, proceed with the steps in the following procedure.

Calibration Procedure

■ A/D Connection Points

Step	Action	Result
1.	Press and hold  and then, press RESET .	"ADJUST" appears in the upper display "0" appears in the SP display, and NUM LED blinks.
2.	Press 1 and then, press ENTER .	"1" appears in the SP display until ENTER is pressed, then "AD-C1-N1" appears in upper display and a 5-digit value (e.g. 1.0000) in the SP display. NUM LED blinks.
3.	Read 5-digit value measured by DVM and record value as V1 on the Calibration Data Chart in this section.	Maintain record of calibration data for possible future use.
4.	If necessary, use numerical keys to adjust value in SP display to equal 1.0000.	Value in SP display equals 1.000.
5.	Press ENTER .	"AD-C1-N2" appears in upper display and 5-digit value (e.g. 0.8000) in SP display.
6.	Read 5-digit value measured by DVM and record value as V2 on the Calibration Data Chart.	
7.	Divide V2 by V1; and if necessary use numerical keys to adjust value in SP display to equal the quotient.	Value in SP display equals the quotient of V2/V1.
8.	Press ENTER .	"AD-C1-N3" appears in upper display and 5-digit value (e.g. 0.6000) in SP display.
9.	Read 5-digit value measured by DVM and record value as V3 on the Calibration Data Chart.	
10.	Divide V3 by V1; and if necessary, use numerical keys to adjust value in SP display to equal the quotient.	Value in SP display equals the quotient of V3/V1.
11.	Press ENTER .	"AD-C1-N4" appears in upper display and 5-digit value (e.g. 0.4000) in SP display.
12.	Read 5-digit value measured by DVM and record value as V4 on the Calibration Data Chart.	
13.	Divide V4 by V1; and if necessary, use numerical keys to adjust value in SP display to equal quotient.	Value in SP display equals the quotient of V4/V1.
14.	Press ENTER .	"AD-C1-N5" appears in upper display and 5-digit value (e.g. 0.2000) in SP display.
15.	Read 5-digit value measured by DVM and record value as V5 on the Calibration Data Chart.	
16.	Divide V5 by V1; and if necessary, use numerical keys to adjust value in SP display to equal quotient.	Value in SP display equals the quotient of V5/V1.
17.	Press ENTER .	"AD-C1-N6" appears in upper display and 5-digit value (e.g. 0.0000 in SP display.
18.	Read 5-digit value measured by DVM and record value as V6 on the Calibration Data Chart.	
19.	Divide V6 by V1; and if necessary, use numerical keys to adjust value in SP display to equal quotient.	Value in SP display equals the quotient of V6/V1.

Step	Action	Result
20.	- If you have a two-channel DCP, disconnect the DVM from pins 1 and 2, and connect it to pins 3 and 4. Go to Step 21. - If you have a one-channel DCP, disconnect the DVM from pins 1 and 2. Press ENTER sequentially until "AD-C2-N6" appears in upper display. Go to D/A Correction Points procedure.	- Ready to check A/D correction points for Channel 2. - Ready for D/A correction points procedure.
21.	(1) Press ENTER. (2) Repeat Steps 3-19, but substitute "C2" for "C1" in the upper display prompts. After Step 19, go to D/A Correction Points Procedure.	(1) "AD-C2-N1" appears in upper display and a 5-digit value (e.g. 1.0000) in the SP display. (2) Check A/D correction points for Channel 2.

■ D/A Correction Points (Current Output Only)

Step	Action	Result
1.	Disconnect DVM from pins 3 and 4, if applicable. Connect ammeter to terminals 57 (+) and 58 (–) on rear terminal panel.	
2.	Press ENTER. NOTE: If you DCP doesn't have current output on Channel 1, press ENTER until "DA-C1-N6" appears in the upper display. Go to Step 20.	"DA-C1-N1" appears in the upper display and 5-digit value (e.g. 1.4812) in the SP display. NUM LED blinks.
3.	Read 5-digit value measured by ammeter and record value as A1 on the Calibration Data Chart in this section.	
4.	Use A1 in the formula given in the Calibration Data Chart to calculate value for SP display. If necessary, use numerical keys to adjust value in SP display to equal the quotient.	Value in SP display equals the quotient of formula from chart using A1 as ammeter reading.
5.	Press ENTER .	"DA-C1-N2" appears in the upper display and 5-digit value (e.g. 1.1458) in the SP display.
6.	Read 5-digit value measured by ammeter and record value on A2 on the Calibration Data Chart.	
7.	Use A2 in the formula given in the Calibration Data Chart to calculate value for SP display. If necessary, use numerical keys to adjust value in SP display to equal the quotient.	Value in SP display equals the quotient of formula from chart using A2 as ammeter reading.
8.	Press ENTER .	"DA-C1-N3" appears in the upper display and 5-digit value (e.g. 0.8109) in the SP display.
9.	Read 5-digit value measured by ammeter and record value as A3 on the Calibration Data Chart.	
10.	Use A3 in the formula given in the Calibration Data Chart to calculate value for SP display. If necessary, use numerical keys to adjust value in SP display to equal the quotient.	Value in SP display equals the quotient of formula from chart using A3 as ammeter reading.
11.	Press ENTER .	"DA-C1-N4" appears in the upper display and 5-digit value (e.g. 0.4763) in the SP display.

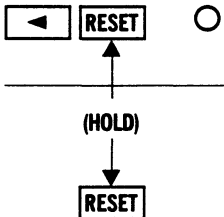
Step	Action	Result
12.	Read 5-digit value measured by ammeter and record value as A4 on the Calibration Data Chart.	
13.	Use A4 in the formula given in the Calibration Data Chart to calculate value for SP display. If necessary, use numerical keys to adjust value in SP display to equal the quotient.	Value in SP display equals the quotient of formula from chart using A4 as ammeter reading.
14.	Press ENTER .	"DA-C1-N5" appears in the upper display and 5-digit value (e.g. 0.1421) in the SP display.
15.	Read 5-digit value measured by ammeter and record value as A5 on the Calibration Data Chart.	
16.	Use A5 in the formula given in the Calibration Data Chart to calculate value for SP display. If necessary, use numerical keys to adjust value in SP display to equal the quotient.	Value in SP display equals the quotient of formula from chart using A5 as ammeter reading.
17.	Press ENTER .	"DA-C1-N6" appears in the upper display and 5-digit value (e.g. - .1895) in the SP display.
18.	Read 5-digit value measured by ammeter and record value as A6 on the Calibration Data Chart.	
19.	Use A6 in the formula given in the Calibration Data Chart to calculate value for SP display. If necessary, use numerical keys to adjust value in SP display to equal the quotient.	Value in SP display equals the quotient of formula from chart using A6 as ammeter reading.
20.	• If you have a two-channel DCP with current output on Channel 2, disconnect ammeter from terminals 57 and 58 (if applicable) and connect it to terminals 59 (+) and 60 (-). Go to Step 21. • If you have a one-channel DCP or your DCP doesn't have current output on Channel 2, press ENTER sequentially until "DA-C2-N6" appears in upper display. Go to Step 22.	Ready to check D/A correction points for Channel 2. Skip check of D/A correction points for Channel 2.
21.	(1) Press ENTER. (2) Repeat Steps 3-19, but substitute "C2" for "C1" in the upper display prompts. After Step 19, go to Step 22.	(1) "DA-C2-N1" appears in upper display and a 5-digit value (e.g. 1.4665) in the SP display. (2) Check D/A correction points for Channel 2.
22.	• If you have a DCP with the auxiliary Output option, disconnect ammeter from other output terminals (if applicable) and connect it to terminals 61 (+) and 62 (-). Go to Step 23. • If your DCP doesn't have the auxiliary output option, disconnect ammeter from output terminals (if applicable). Press ENTER sequentially until "DA-C3-N6" appears in upper display. Go to CJ Compensation Point Procedure.	Ready to check D/A correction points for auxiliary output. Ready to check CJ compensation point.
23.	(1) Press ENTER. (2) Repeat Steps 3-19, but substitute "C3" for "C1" in the upper display prompts. After Step 19 disconnect ammeter and go to CJ Compensation Point Procedure.	(1) "DA-C3-N1" appears in upper display and a 5-digit value (e.g. 1.4757) in the SP display. (2) Check D/A correction points for auxiliary output option.

■ CJ Compensation Point

Step	Action	Result
1.	Press ENTER .	"CJ-SHORT" appears in upper display.
2.	Short pin 1 to pin 5 on connector P164. Wait 2 minutes before going to Step 3.	
3.	Press ENTER . Remove short from pins 1 and 5. Wait 2 minutes before going to Step 4.	"CJ-TEMP1" (see note 1) appears in upper display. Ambient temperature value in degrees C appears in SP display. NOTE 1: On older units, only "CJ-TEMP" appears once. Complete steps 3, 4, 5 & 6 and continue at step 8.
4.	Use thermometer to measure ambient temperature at terminal 49 on rear terminal panel.	
5.	Compare ambient temperature (°C) measurement from Step 4 with value in SP display. If necessary, use numerical keys to adjust value in SP display to equal the measured temperature in °C.* Note that decimal point is entered automatically by the display. CAUTION: If it becomes necessary to recalibrate the cold junction reference temperatures (CJ-TEMP1 and CJ-TEMP2), BE EXTREMELY CAREFUL. The temperature for each step must be re-entered (using the number keys) even if it has not changed! Failure to do this by simply pressing ENTER may result in severe malfunction of the PV display.	Value in SP display equals the ambient temperature measurement in °C. *If CPU or analog board was just changed, you must key in the ambient temperature value even if the value in the SP display is the same.
6.	Press ENTER . Wait 2 minutes before going to the next step.	DCP returns to normal operation.
7.	Repeat steps 3 through 6 using "CJ-TEMP2" and measure the temperature at terminal 54. Then go to Step 8.	
8.	Replace adhesive backed cover over P164 connector cutout.	
9.	Do a PV input calibration. See Calibration Procedure in Product Manual.	
10.	After Calibration, be sure to remove short from EEPROM — connector pins, move lever to M.P. position and connect field wiring.	

How to Make a General Calibration Reset

Use this procedure to reset all calibration data to initial factory values. However, before you make a calibration reset, we recommend that you make a record of all calibration data presently stored in the DCP. See the Calibration and Appendix A sections in the Product Manual as well as this section for data to be recorded. Make this reset only if all else has failed to clear extraneous error messages.

Step	Key	Display	Comments
1.			Pull down module door on front of DCP and locate hole marked G.R. to left of memory protect switch. Be sure memory protect switch lever is in the M.E. position.
2.		SP 8.	While pressing  and RESET , use pen or pencil to press G.R. switch.
3.	RESET	All displays light for an instant.	Remove the pen or pencil and continue pressing, RESET for about 5 seconds until you hear a beep and the SYNC LED lights, if not lit before.
4.			All calibration data has been cleared from memory and reset to initial factory values. Recalibrate and/or enter calibration data.

Calibration Data Chart

For DCP 700 Model Number: _____

Serial Number: _____

■ A/D Corrective Points

Prompt	DVM Reading	Adjusted Value in SP display to Read
AD-C1-N1	(V1)	1.0000
AD-C1-N2	(V2)	(= V2/V1)
AD-C1-N3	(V3)	(= V3/V1)
AD-C1-N4	(V4)	(= V4/V1)
AD-C1-N5	(V5)	(= V5/V1)
AD-C1-N6	(V6)	(= V6/V1)
AD-C2-N1	(V1)	1.0000
AD-C2-N2	(V2)	(= V2/V1)
AD-C2-N3	(V3)	(= V3/V1)
AD-C2-N4	(V4)	(= V4/V1)
AD-C2-N5	(V5)	(= V5/V1)
AD-C2-N6	(V6)	(= V6/V1)

■ D/A Correction Points (Current Output Only)

Prompt	Ammeter Reading	Formula for determining value (V) in SP display	Adjusted Value in SP display to Read
DA-C1-N1 DA-C1-N2 DA-C1-N3 DA-C1-N4 DA-C1-N5 DA-C1-N6	(A1) (A2) (A3) (A4) (A5) (A6)	$V = \frac{\text{Ammeter Reading} - 4}{16}$	
DA-C2-N1 DA-C2-N2 DA-C2-N3 DA-C2-N4 DA-C2-N5 DA-C2-N6	(A1) (A2) (A3) (A4) (A5) (A6)	$V = \frac{\text{Ammeter Reading} - 4}{16}$	
DA-C3-N1 DA-C3-N2 DA-C3-N3 DA-C3-N4 DA-C3-N5 DA-C3-N6	(A1) (A2) (A3) (A4) (A5) (A6)	$V = \frac{\text{Ammeter Reading} - 4}{16}$	

■ CJ Compensation Point

Prompt	Ambient Temperature Reading in °C	Adjusted Value in SP display to read
CJ-TEMP		

DCP 700 Digital Control Programmer

Changing the Temperature Unit or Input Type

Changing the Temperature Unit

The unit can be changed to °F or °C on DCPs with thermocouple or RTD type inputs as follows:

Preparation

1. Remove power from the DCP.
2. Pull the latches, one on either side of the bezel, forward. Pull on the bezel, alternating sides, until the bezel starts to come away from the case. Slide the chassis out of the case slowly until you see the connector on the CPU board (third from front).
3. Locate the two vertical pins on the EEPROM-protect-connector which is located to the left of the battery on the top right side of the CPU board. Wrap a piece of wire around the pins to short them together and be sure the wire doesn't contact any other components or the case.
4. Slide the chassis back into the case and push in the latches.
5. Open the plug-in module door and move the memory protect switch lever to the far right position (M.E.). Close the module door.
6. Apply power to the DCP and follow steps in Table 9-1.

TABLE 9-1—Steps to Change the Temperature Unit

Step	Action	Comment
1	Press ▼ and while pressing, press RESET .	"ADJUST" appears in the upper display, "0" appears in the SP display, and the NUM LED lights.
2	Press 4 , ENTER .	"4" appears in SP display until ENTER is pressed; then "PASSWORD" appears in the upper display, and "0" in the SP display.
3	Press 2 , 2 , 5 , 5 .	"2255" appears in the SP display.
4	Press ENTER four times.	"PV-TYPE1" appears in the upper display, and a code appears in the display to identify the input type for CH 1.
5	Depending on the units you want, select a code from the °F Code or °C Code column in Table 9-2 and key it in.	The new code appears in the SP display.
6	Press ENTER .	"PV-TYPE2" appears in the upper display, and a code (0 if the DCP doesn't have a CH 2) appears in the SP display to identify the input type for CH 2.
7	If you are not going to change CH 2 go to Step 9; otherwise go to Step 8.	
8	Depending on the units you want, select a code from the °F Code or °C Code column in Table 9-2 and key it in.	The new code appears in the SP display.
9	Press ENTER .	Displays show the RESET condition.
10	Refer to the Preparation steps and repeat steps as required to remove the piece of wire from around the EEPROM-protect-connector pins.	
11	Perform a General Reset — See Page 4-14	This General Reset clears data from the Random Access Memory including the programming, configuration and tuning data. Configuration and tuning data revert to the initial values listed in Tables 5-2 and 6-1, respectively.
12	Enter new programs and related data with the new unit.	

TABLE 9-2—Unit Codes and Ranges

Input Type	°F Code	°F Range	°C Code	°C Range
T	20	– 300 to 700	1	– 200 to 300
J	21	0 to 1600	2	0 to 800
E	22	0 to 1800	3	0 to 800
K	23	0 to 2400	4	0 to 1200
S	24	0 to 3100	6	0 to 1600
R	25	0 to 3100	7	0 to 1600
B	26	0 to 3300	8	0 to 1800
W5W26	27	0 to 4200	9	0 to 2300
Ni/Ni Moly	28	0 to 2500	11	0 to 1300
RTD	29	– 300 to 900	15	– 200 to 200

NOTE: The °F range is calibrated as a °C range — See the Calibration section in this manual

Changing the Input Type

Change thermocouple, millivolt or current input types as follows:

1. Complete steps 1 through 5 in Preparation procedure on Page 9-1.
2. In Table 9-3, locate the present input type of your DCP combined with the input you want to change to; then procure any parts listed and follow the procedure required.

TABLE 9-3 Types of Input you can change from and to

Type of Input on Your DCP	Types of Inputs You Can Change to	Parts to Procure	Procedure Required
Thermocouple Any Type	Current	2.5 ohm 1/4W Resistor	Procedure B in Table 9-6
E	Thermocouple-Ni-NiMoly	None	Procedure A in Table 9-4
	Any thermocouple type other than Ni-Ni Moly or millivolt	None	Procedure B in Table 9-6
Ni-NiMoly	Thermocouple-Type E	None	Procedure A in Table 9-4
	Any thermocouple type other than Type E or millivolt	None	Procedure B in Table 9-6
K	Thermocouple-Types J and W5W26	None	Procedure A in Table 9-4
	Any thermocouple type other than Types J and W5W26 or millivolt	None	Procedure B in Table 9-6
J	Thermocouple-Types K and W5W26	None	Procedure A in Table 9-4
	Any thermocouple type other than Types K and W5W26 or millivolt	None	Procedure B in Table 9-6
W5W26	Thermocouple-Types K and J	None	Procedure A in Table 9-4
	Any thermocouple type other than Types K and J or millivolt	None	Procedure B in Table 9-6
R	Thermocouple-Types B and S	None	Procedure A in Table 9-4
	Any thermocouple type other than Types B and S or millivolt	None	Procedure B in Table 9-6
B	Thermocouple-Types R and S	None	Procedure A in Table 9-4
	Any thermocouple type other than Types R and S or millivolt	None	Procedure B in Table 9-6
S	Thermocouple-Types R and B	None	Procedure A in Table 9-4
	Any thermocouple type other than Types R and B or millivolt	None	Procedure B in Table 9-6
T	Any thermocouple type	None	Procedure B in Table 9-6
Current	Millivolt	None	Procedure B in Table 9-6
	Volts	2.49K ohm, 1/4W Resistor*	Procedure B in Table 9-6
	Any Thermocouple	None	procedure B in Table 9-6
Millivolt	Current	2.5 ohm, 1/4W Resistor	Procedure B in Table 9-6
	Thermocouple Types R, B and S	None	Procedure A in Table 9-4
	Any thermocouple type other than Types R, B, and S	None	Procedure B in Table 9-6
Volts	Current	2.5 ohm, 1/4W Resistor	Procedure B in Table 9-6

*Plus one or two 249K ohm, 1/4W resistors

TABLE 9-4—Procedure A

Step	Action	Comment
1	Press ▼ and while pressing, press RESET .	"ADJUST" appears in the upper display, "0" appears in the SP display, and the NUM LED lights.
2	Press 4 , ENTER .	"4" appears in SP display until ENTER is pressed; then, "PASSWORD" appears in the upper display, and "0" in the SP display.
3	Press 2 , 2 , 5 , 5 .	"2255" appears in the SP display.
4	Press ENTER four times.	"PV-TYPE1" appears in the upper display, and a code appears in the SP display to identify the input type for CH 1.
5	Depending on the input type and units you are changing to, select a code from Table 9-5 and key it in.	The new code appears in the SP display.
6	Press ENTER .	"PV-TYPE2" appears in the upper display, and a code (0 if your DCP doesn't have a CH 2) appears in the SP display to identify the input type for CH 2.
7	If you are not going to change CH 2, go to Step 9; otherwise go to Step 8.	
8	Depending on the input type and unit you are changing to, select a code from Table 9-5 and key it in.	The new code appears in the SP display.
9	Press ENTER .	Displays show the RESET condition.
10	Refer to the Preparation procedure and repeat steps as required to remove the piece of wire from around the EEPROM-protect-connector pins.	
11	Enter new programs and related data with the input type.	

TABLE 9-5—Input Type Codes

Input Type	°F Code	°F Range	°C Code	°C Range
T	20	– 300 to 700	1	– 200 to 300
J	21	0 to 1600	2	0 to 800
E	22	0 to 1800	3	0 to 800
K	23	0 to 2400	4	0 to 1200
S	24	0 to 3100	6	0 to 1600
R	25	0 to 3100	7	0 to 1600
B	26	0 to 3300	8	0 to 1800
W5W26	27	0 to 4200	9	0 to 2300
Ni-Ni Moly	28	0 to 2500	11	0 to 1300
Current	19	4 to 20 mA		
Millivolt	19	0 to 10 mV		
Volts	19	1 to 5 Vdc		

Equipment Required for Procedure B

- Wire-wrap tool (Optional. Can use soldering iron.)
- Small soldering iron
- Small wire cutter

Material Required for Procedure B

- Small diameter copper wire to fit wire-wrap tool or small diameter copper wire for soldering connections.
- If soldering iron is to be used, solder and flux
- If changing to current input type, a 2.5 ohm $\pm 1\%$, 1/4W resistor
- If changing to volts input type, a 2.49K ohm $\pm 1\%$, 1/4W resistor and two 249K ohms $\pm 1\%$, 1/4W resistors

TABLE 9-6—Procedure B

Step	Action
1.	Use the instructions in the Service section (Page 10-12) to remove the analog board from the chassis.
2.	Use Figure 9-1 as a guide to locate the connections and resistors listed in Table 9-7 for the input type you are changing to, and change their status (INSTALLED or Removed) as required. ■ Remove connections and resistors by cutting at both ends. ■ Install connections by wire wrapping or soldering to the common pin and the pin identified by a W and a number ■ For the Current Input Type, solder the 2.5 ohm resistor in the holes provided for R216 or R258. For Volts input type, solder in a 2.49K ohm resistor. For all other input types in Table 9-7, leave the hole open. ■ For Volts Input Type, solder 249K ohm resistor in the holes provided for R122 (channel 1) and/or R123 (channel 2).
3.	Replace the analog board and push the chassis into the case.
4.	Use Procedure A in Table 9-4 to enter the code for the new input type.
5.	Calibrate the DCP (see Page 8-1)
6.	Refer to the Preparation procedure and repeat steps as required to remove the piece of wire from across the EEPROM-protect-connector.
7.	Enter new programs and related data based on the new input type.

TABLE 9-7—Connection and Resistor Status

Channel		Connections and Status					Resistor Status	
		W10	W11	W13	W14	W15	R216	R122*
	1 ———→	W10	W11	W13	W14	W15	R216	R122*
	2 ———→	W16	W17	W19	W20	W21	R258	R123
Input Type	E	Removed	INSTALLED	Removed	INSTALLED	INSTALLED	Open	Open
	NI-Ni Moly	Removed	INSTALLED	Removed	INSTALLED	INSTALLED	Open	Open
	K	INSTALLED	Removed	Removed	Removed	INSTALLED	Open	Open
	J	INSTALLED	Removed	Removed	Removed	INSTALLED	Open	Open
	W5W26	INSTALLED	Removed	Removed	Removed	INSTALLED	Open	Open
	Current	INSTALLED	Removed	Removed	Removed	INSTALLED	INSTALLED (2.5Ω)	Removed
	Volts	INSTALLED	Removed	Removed	Removed	INSTALLED	INSTALLED (2.49KΩ)	INSTALLED (249KΩ)†
	R	Removed	Removed	INSTALLED	Removed	Removed	Open	Open
	S	Removed	Removed	INSTALLED	Removed	Removed	Open	Open
	B	Removed	Removed	INSTALLED	Removed	Removed	Open	Open
	Millivolt	Removed	Removed	INSTALLED	Removed	Removed	Open	Open
	T	INSTALLED	Removed	Removed	INSTALLED	Removed	Open	Open

*R122 and R123 are not used on earlier version of the analog board.

†Also, remove jumpers W135 and W136.

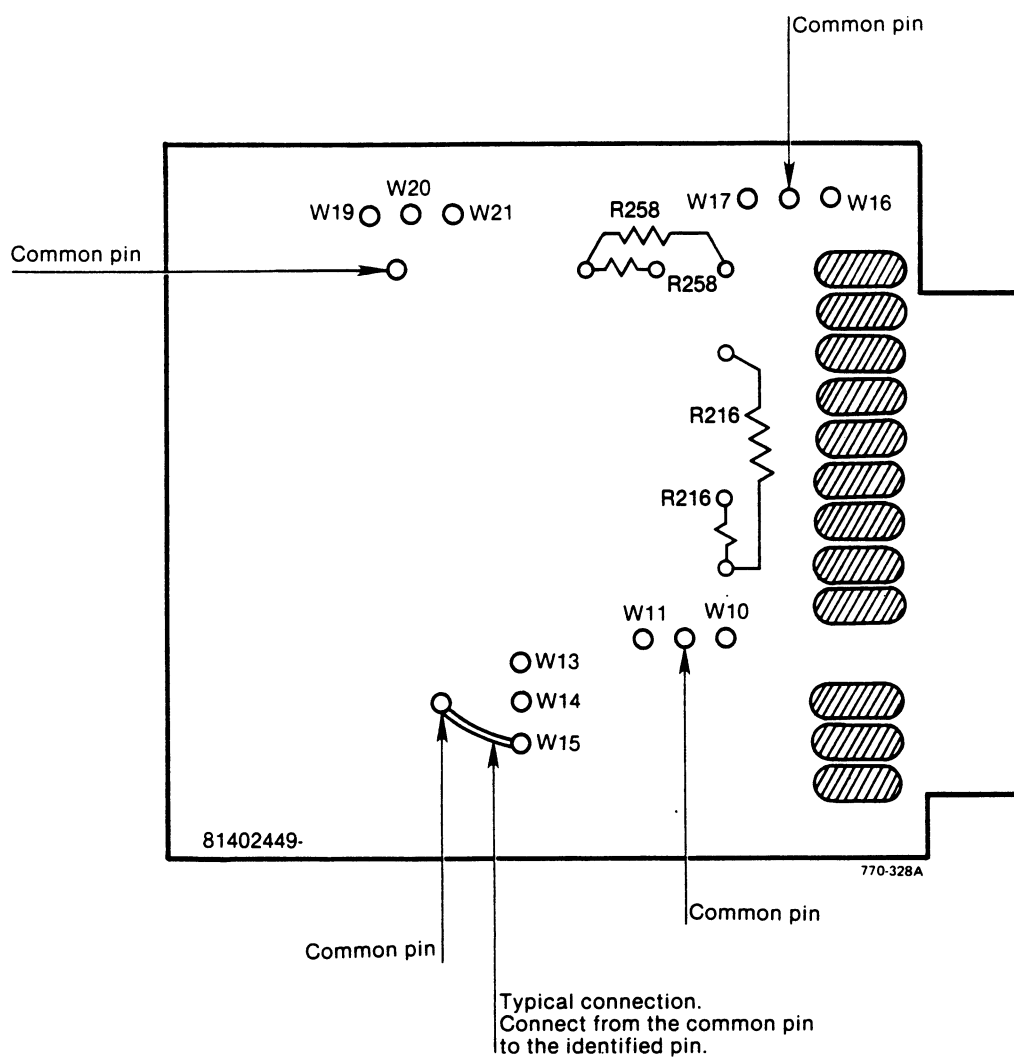


Figure 9-1—Connector locations on the analog board

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TROUBLESHOOTING

Introduction

Instrument performance can be adversely affected by installation and application problems as well as hardware problems. It is recommended that you investigate the problems in the following order:

- Installation related problems—Read the installation section in this manual to make sure the DCP 700 has been properly installed. The installation section provides information on protection against electrical noise, connecting external equipment to the programmer, and shielding and routing external wiring.
- Application related problems—Review the application of the programmer; then, if necessary, direct your questions to the Customer Marketing Manager, Honeywell, Process Control Division, 1100 Virginia Drive, Ft. Washington, Pa.
- Hardware related problems—Use the Troubleshooting diagnostics, programmer failure symptoms, and troubleshooting procedures to identify typical failures which may occur in the controller. Use the replacement procedures as well as the input and output calibration procedures to correct these failures.

Diagnostics

The DCP 700 Programmer's diagnostic self-tests let you know that the programmer is operating properly. Your window to recognizing diagnostic test failures are the displays and indicators. They are the quickest way for you to locate a problem. Begin troubleshooting by carefully noting which displays and indicators are working and which are not. Compare the symptoms with those given in Table 1. For a given set of symptoms, table 1 will refer you to an alarm table (Table 2) or a troubleshooting procedure that progresses from the most likely cause for the failure to the least likely.

Use the procedures in this section (page 10-5) to replace the failed part. While it is not stated in the troubleshooting procedure, it should be understood that after each repair action or board replacement, power should be applied to the chassis to see if the same failure symptom is present.

If a suspected faulty board is replaced during servicing and the replacement board does not correct the problem, replace the replacement board with the original board before continuing the procedure. If a set of symptoms, other than the ones you started with, appears while troubleshooting, go back to Table 1 and re-evaluate the symptoms. This may lead you to a different troubleshooting procedure. If the symptom is still present after the last troubleshooting step, contact your local Honeywell Process Control Division branch office.

TABLE 1 — Programmer Failure Symptoms

What/Where Observed										When Observed						Probable Cause	Troubleshooting Reference
Controller Output		Display							Key Sound		Operation Mode						
		7 Seg & LED			16 Segment												
None	OK	Blank	Some LED lit	OK	Blank	Some bar lit	Alarm Flashing	OK	None	OK	Ready	Run	Hold	Any	Unk		
	●			●			●			●				●		Abnormal condition of PV or hardware	Alarm Table 2
●		●			●				●						●	Power Failure	Procedure #1
	●	●	●		●	●			●	●					●	Display Failure	Procedure #2
Does not agree with displayed output		Does not agree with Programmer Output						●		●	●	●	●			Current output Failure	Procedure #3
															Time proportioning, event, alarm, status output failure	Procedure #4	
Does not agree with Event LED		Event LED does not agree with event Output						●		●		●				Event Output Failure	Procedure #4
	●	Remote LED does not agree with remote input						●		●				●		Remote Function Failure	Procedure #5
	●	PV display does not agree with PV input						●		●				●		Analog Input Failure	Procedure #6

Alarm Messages | Upon the occurrence of an abnormal condition of the Process Variable or Hardware, the following alarm messages are displayed:

TABLE 2 — Alarm Messages

Message	Meaning	Reference Troubleshooting Procedure
ALARM 1	Process Variable for Channel 1 is at or above the upper range limit.	Procedure #6
ALARM 2	Process variable for Channel 1 is at or below the lower range limit	
ALARM 3	Process variable for Channel 2 is at or above the upper range limit	
ALARM 4	Process variable for Channel 2 is at or below the lower range limit.	
ALARM 71	Current proportional output for CH 1 failed.	Procedure #3
ALARM 72	Current proportional output for CH 2 failed.	
ALARM 73	The auxiliary output circuit failed.	
ALARM 74	D/A Converter calibration failed	Procedure #7
ALARM 81	The A/D converter correction calibration point for Channel 1 is incorrect.	Procedure #8
ALARM 82	The A/D converter range calibration for Channel 1 is incorrect.	
ALARM 83	The A/D converter correction calibration point for Channel 2 is incorrect.	
ALARM 84	The A/D converter range calibration for Channel 2 is incorrect.	
ALARM 92	Content of EEPROM lost.	Procedure #12
ALARM 93	Data check point lost.	Procedure #9
ALARM 94	Parameter data lost.	Procedure #10
ALARM 95	PID data lost.	
ALARM 96	Program data lost	
ALARM 97	Battery power low	Procedure #11
ALARM 98	RAM failed	Procedure #12
ALARM 99	ROM failed	

Troubleshooting Procedures

Procedure 1 — Troubleshooting Power Failure Symptom

What to Do	How to do it or where to find the reference data
1. Reset the programmer.	1. Press the reset key on the keyboard and reset switch behind the front door.
2. Check all the connectors.	2. Withdraw the chassis and visually inspect all the connectors. Make sure they are seated properly.
3. Replace the following boards, one at a time, in the order listed. Power Supply Board Display Control Board CPU Board* Display Board I/O Control Board *Reconfigure and reprogram data.	3. See page 10-8 in this section for replacement procedures. *See configuration and programming sections.
4. Recalibrate the programmer.	4. See the calibration section.

Procedure 2 — Troubleshooting Display Failure Symptom

What To Do	How to do it or where to find the reference data
1. Run a check on displays and LEDS.	1. Run Program 0.
2. Replace the following boards, one at a time, in the order listed. Display Board Display Control Board	2. See page 10-8 in this section for replacement procedures.
3. Replace the CPU Board.	3. See page 10-8 in this section for replacement procedures.
4. Reconfigure, reprogram and recalibrate the programmer.	4. See calibration section. See configuration and programming sections.

Procedure 3 — Troubleshooting Current Proportional Output Failure Symptom

What to do	How to do it or where to find the reference data
1. Check all the connectors.	1. Withdraw the chassis and visually inspect all the connectors. Make sure they are seated properly.
2. Check the Output load resistance. Is it greater than 600 ohms?	2. Replace the Output load.
3. Replace the following boards, one at a time, in the order listed. Output Board CPU Board** I/O Control Board Analog Board Sequencer Board **Reconfigure and reprogram data.	3. See page 10-8 in this section for replacement procedures.
4. Recalibrate the programmer.	4. See the calibration section.

Procedure 4 — Troubleshooting Event, alarm, status, time proportioning output failure (open collector output failure)

What to do	How to do it or where to find the reference data
1. Reconfigure Event Configuration Data. (Event output failure only)	1. See the configuration section.
2. Check the 24V power wiring source.	2. Repair the wiring.
3. Replace the following boards, one at a time, in the order listed: Terminal Board I/O Control Board CPU Board* *Reconfigure and reprogram the data.	3. See page 10-8 in this section for replacement procedures.
4. Recalibrate the programmer.	4. See the calibration section.

Procedure 5 — Remote Function Failure Symptom

What to do	How to do it or where to find the reference data
1. Check all the connectors.	1. Withdraw the chassis and visually inspect all the connectors. Make sure they are seated properly.
2. Check to see if the RAM LED lights.	2. Input the remote signal.
3. Check the remote wiring.	3. Repair the wiring.
4. Replace the following boards, one at a time, in the order listed: Sequencer Board Power Board CPU Board* I/O Control Board *Reconfigure and reprogram the data.	4. See page 10-8 in this section for replacement procedures.
5. Recalibrate the programmer.	5. See the calibration section.

Procedure 6 — Analog Input Failure Symptom

What to do	How to do it or where to find the reference data
1. Check all the connectors.	1. Withdraw the chassis and visually inspect all the connectors. Make sure they are seated properly.
2. Check the input sensor. Replace if necessary.	2. See manufacturers instructions for the input source.
3. Replace the following boards, one at a time, in the order listed: Analog Board CPU Board* I/O Control Board Terminal Board *Reconfigure and reprogram the data.	3. See page 10-8 in this section for replacement procedures.
4. Recalibrate the programmer.	4. See the calibration section.

Procedure 7 — Troubleshooting Alarm 74 — D/A Converter

What to do	How to do it or where to find the reference data
1. Check all the connectors.	1. Withdraw the chassis and visually inspect all the connectors. Make sure they are seated properly.
2. Recalibrate the programmer.	2. See the calibration section.

Procedure 8 — Troubleshooting Alarm 81 thru 84 — A/D Converter

What to do	How to do it or where to find the reference data
1. Check all the connectors.	1. Withdraw the chassis and visually inspect all the connectors. Make sure they are seated properly.
2. Recalibrate the programmer.	2. See the calibration section.
3. Replace the following boards, one at a time, in the order listed: Analog Board CPU Board* I/O Control Board *Reconfigure and reprogram the data.	3. See page 10-8 in this section for replacement procedures.
4. Recalibrate the programmer.	4. See the calibration section.

Procedure 9 — Troubleshooting Alarm 93 — Data Check Point Lost

What to do	How to do it or where to find the reference data
1. Check the configuration data.	1. See the configuration section.
2. Push the reset key.	2. The reset key is on the keyboard.
3. Replace the CPU board.* *Reconfigure and reprogram the data.	3. See page 10-8 in this section for replacement procedures.
4. Calibrate the programmer.	4. See the calibration section.

Procedure 10 — Troubleshooting Alarms 94, 95, and 96 — Parameter, P/D, or Program Data Lost

What to do	How to do it or where to find the reference data
1. Re-enter parameter, PID or program data.	1. See the programming or configuration section.
2. Press the reset key.	2. The reset key is on the keyboard.

Procedure 11 — Troubleshooting Alarm 97 — Battery Power Low

What to do	How to do it or where to find the reference data
1. Replace the battery.	1. See page 10-8 in this section for replacement procedure.

**Procedure 12 — Troubleshooting Alarm 92 — Content of EEPROM Lost
Alarm 98, 99 — ROM or RAM Failed**

What to do	How to do it or where to find the reference data.
1. Replace the CPU.* *Reconfigure and reprogram the data.	1. See page 10-8 in this section for replacement procedure.
2. Recalibrate the programmer.	2. See the calibration section.

REPLACEMENT PROCEDURES

Introduction

These replacement procedures allow you to remove and replace the parts or printed wiring boards shown in the parts list (page 11-1).

CAUTION: When removing and installing board assemblies, turn off power to the programmer. Insert the boards completely into the programmer, and reseal any connector before restoring power.

Equipment Needed

- Small flatbladed screwdriver
- Medium flatbladed screwdriver
- Medium phillips head screwdriver

How to replace the FILTER

Procedure

1. Insert a small flatbladed screwdriver into the slot at the top of the filter and pry out (Figure 1).
2. Place the new filter into the bottom of the opening and press into place at the top.

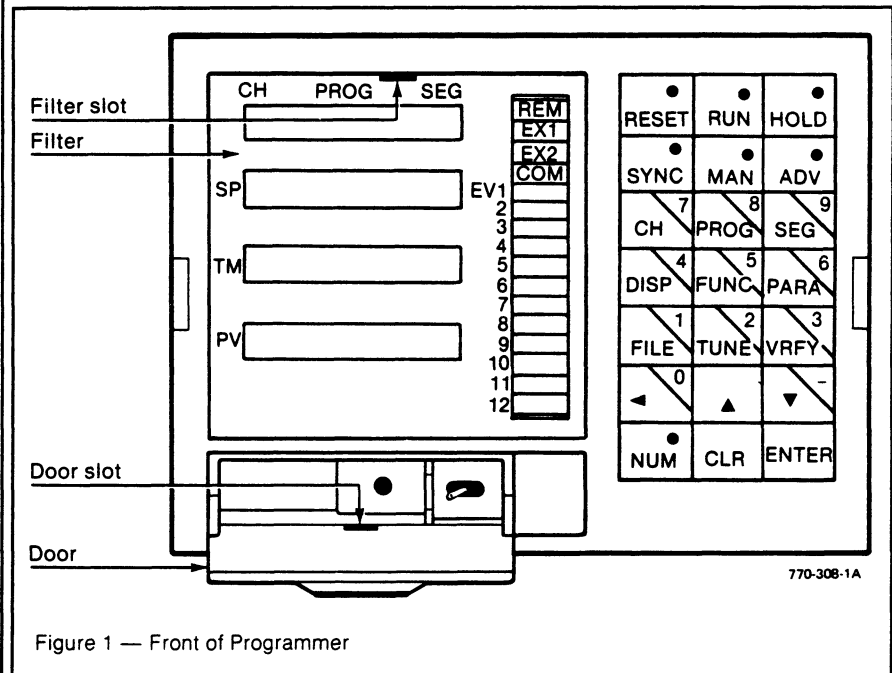


Figure 1 — Front of Programmer

How to replace the DOOR

Procedure

1. Pull the door down (Figure 1).
2. Insert a small flatbladed screwdriver into the slot in the middle of the door and twist.
3. Pull door down and out.
4. To insert the new door, put one side into place, bend door slightly, and push the other side into place.

How to Remove the CHASSIS Procedure

1. Push the brackets on the side of the bezel forward (Figure 2).
2. Pull on the bezel, alternating sides, until the chassis starts to come away from the case.
3. Slide the chassis out of the case slowly until you see the connector on the CPU board (third from the front).
4. Remove the connector from its socket.
5. Continue to slide the chassis out slowly until it is almost all the way out of the case.
6. Remove the two connectors from the terminal board assembly.
7. Slide the chassis all the way out of the case.
8. When you return the chassis to the case, reverse the above procedure.

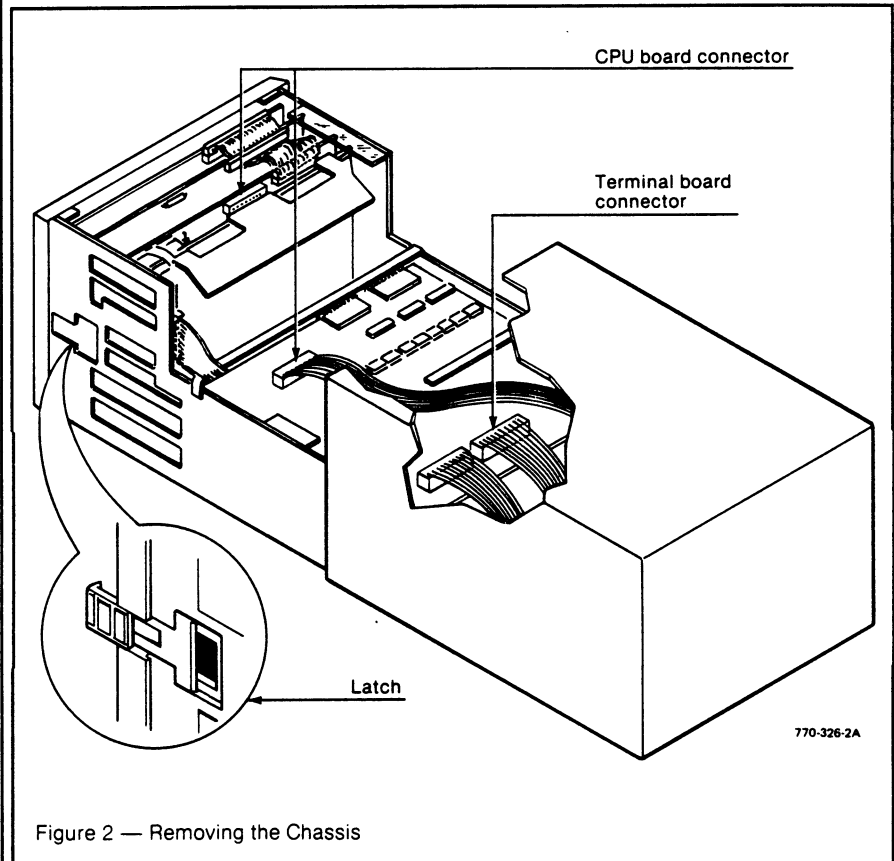


Figure 2 — Removing the Chassis

**How to Replace the
REAR TERMINAL PANEL,
POWER SUPPLY,
CASE**

Procedure

1. Remove the chassis from the case (see page 10-9).
2. Loosen the two screws on the top of the rear terminal panel (Figure 3).
3. Pull the top of the terminal panel slightly away from the case and remove the connector on the power supply board. (squeeze the connector and pull up)
4. Pull the terminal panel away from the case and replace if necessary.
5. Remove the five screws on the top of the case that hold the power supply unit. Keep one hand under the unit when removing the last screw.
6. Remove the power supply through the rear of the case and replace if necessary.
7. Replace the case if necessary.
8. Reverse this procedure to replace any or all of these parts.

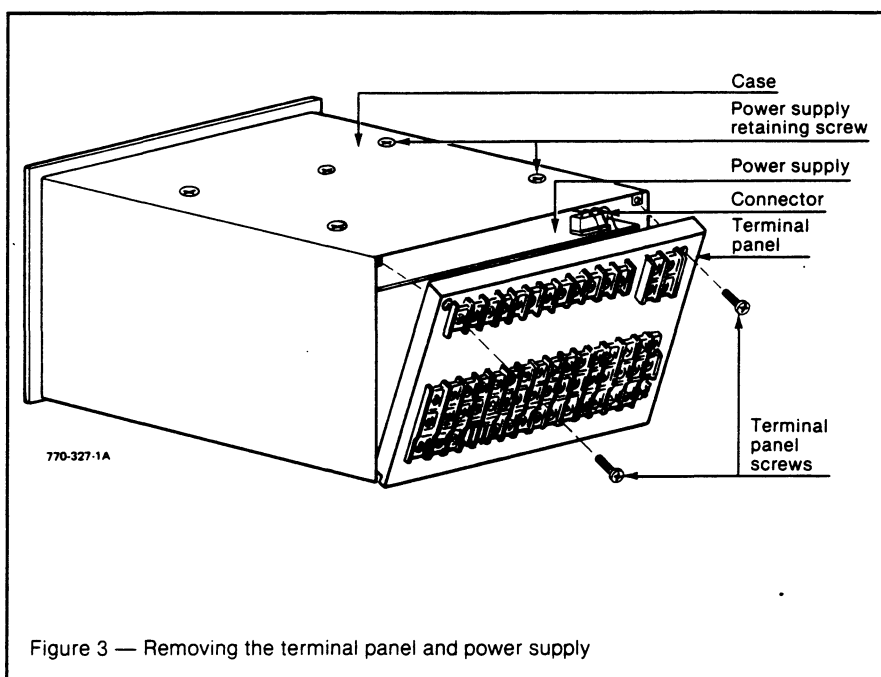


Figure 3 — Removing the terminal panel and power supply

**How to Replace the
TERMINAL BOARD,
SEQUENCER BOARD,
ANALOG BOARD**

Procedure

1. Use a screwdriver to lift the four tabs on the side of the rear bracket and remove the bracket (Figure 5).
2. Loosen the screws slightly on each side of the chassis.
3. Remove the two connectors on the top of the terminal board assembly.
4. Pull the terminal board out slightly.
5. Remove the two connectors on the right rear top of the sequencer board and the connector on the output board assembly.
6. Pull the sequencer board out slightly.
7. Remove the connector on the left top rear of the analog board.
8. Slide all three boards out together. The three boards are held together by various connectors on the left side of the boards.
9. Disconnect whatever connectors are necessary to replace any of the boards.

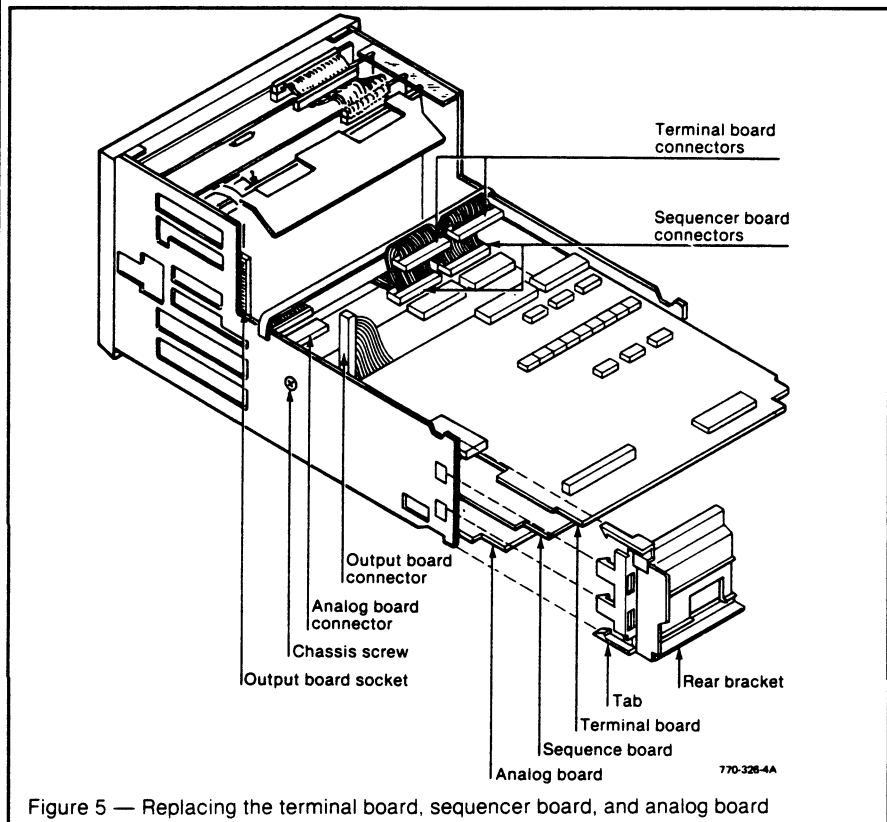


Figure 5 — Replacing the terminal board, sequencer board, and analog board

10. Reattach the connectors to the new board(s) and reposition the boards in the side brackets.
11. Turn the chassis on its side and slide the boards into the grooves in the side brackets. Slide the boards almost all the way in until you reach the connectors that have been unattached.
12. Attach the connector to the bottom board (analog board).
13. Reattach the connector to the output board assembly.
14. Attach the two connectors to the sequencer board then to the terminal board assembly.
15. Slide the boards all the way up to the stop. Make sure the boards are positioned properly in the top and bottom guide.
16. Tighten the screws on either side of the chassis.
17. Attach the bracket to the rear of the boards.

**How to Replace the
OUTPUT BOARD,
CPU BOARD,
BATTERY,
COMMUNICATIONS BOARD,
I/O CONTROL BOARD**

Procedure

1. Lift the tabs on each side of the chassis that hold the plastic stoppers over the boards (Figure 6).
2. Remove the metal shield.

To remove the

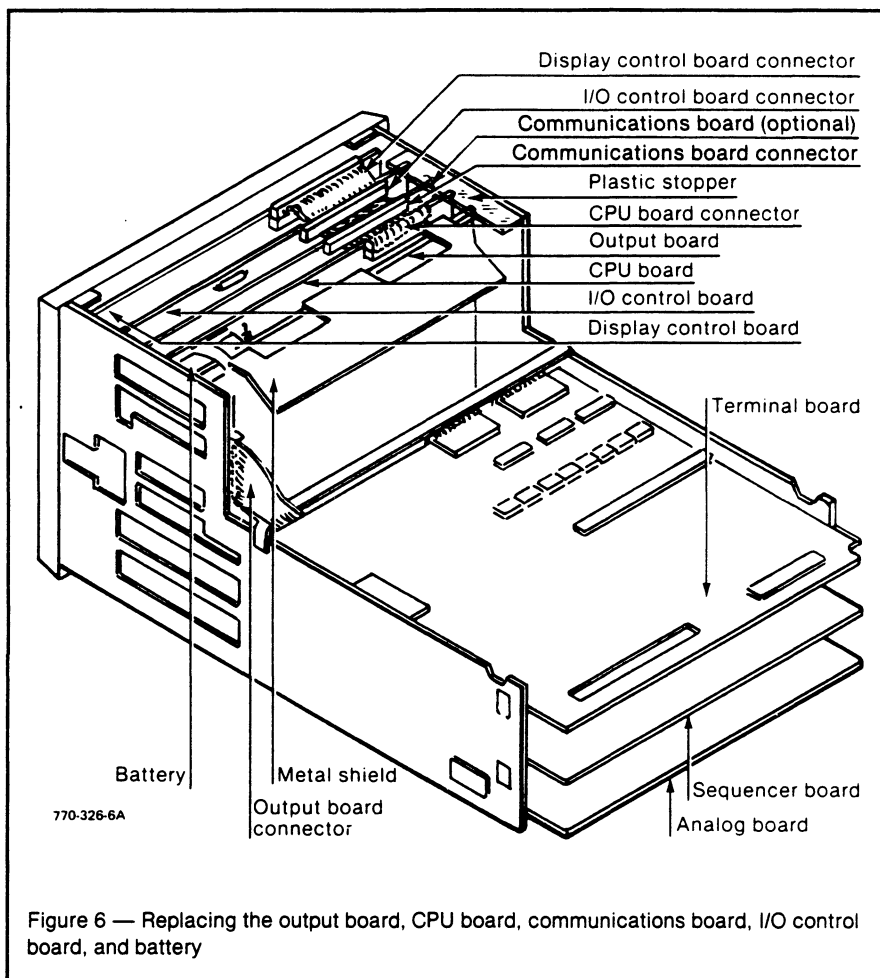
Output board, go to step 3

CPU board, go to step 7

Battery, go to step 7

Communications board, go to step 12

I/O Control board, go to step 16



OUTPUT BOARD

3. Remove the connector on the left side of the board (Figure 6).
4. Turn the chassis on its side and remove the connector at the bottom of the board.
5. Slide the board up and out.
6. Put in new board and replace the connectors.

CONTINUED NEXT PAGE

***How to replace the
OUTPUT BOARD,
CPU BOARD,
BATTERY,
COMMUNICATIONS BOARD,
I/O CONTROL BOARD
(CONTINUED)***

CPU BOARD AND BATTERY

7. Remove the connector on the top of the CPU board (Figure 6).
8. Slide the board up and out. Replace if necessary.
9. To replace the battery, disconnect the leads from the two phillips head screws and pull the battery from the bracket.
10. Install the new battery and return the CPU board to the chassis.
11. Replace the connector.

COMMUNICATIONS BOARD

12. Remove the connectors from the top of the CPU board and the Communications board.
13. Turn the chassis on its side and remove the connector from the bottom of the Communications board.
14. Slide the board up and out.
15. Put in the new board and replace all the connectors.

I/O CONTROL BOARD

16. Remove the CPU board (see step 7), and the Communications board (see step 12), if supplied.
17. Remove the connector on the display control board (Figure 6).
18. Turn the chassis over and remove the connectors from the analog board and output board.
19. Remove the two connectors from the sequencer board and two from the terminal board.
20. Loosen the screws that hold the front bezel to the chassis and pull the bezel forward slightly.
21. Lift the I/O Control board and cables out and replace.
22. Reverse the procedures to replace all of the parts.

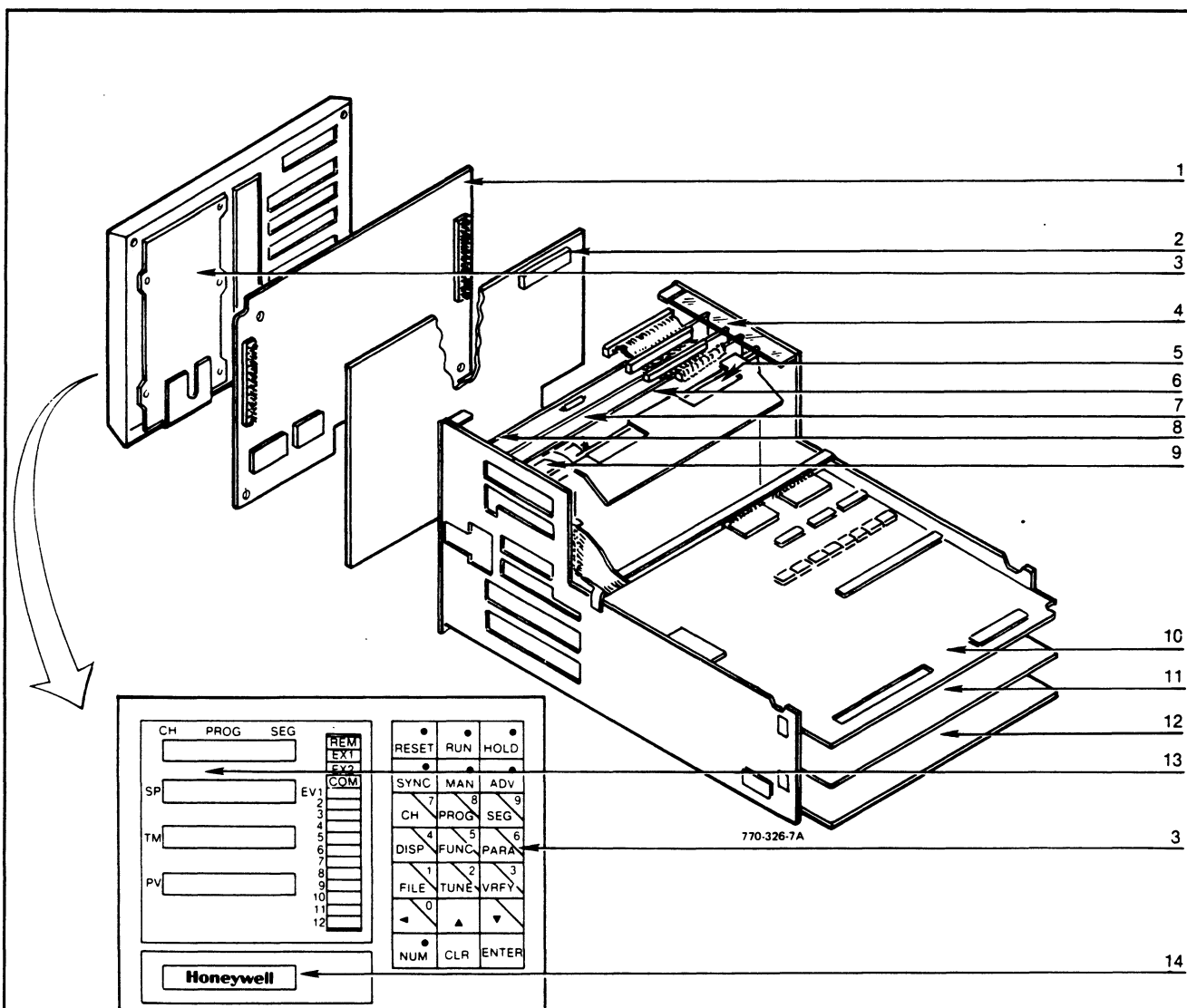
DCP 700

Digital Control Programmer

Parts List

Component Identification

- All saleable parts are indicated in each figure by a callout 1, 2, 3 and so on.
- Note that the part numbers listed for printed circuit boards are the correct ones to use when ordering a replacement board. However, the part number printed on the board may have a different "tab" number. For example, the correct part number for a CPU board assembly without communications is 81402439-101, but the number printed on the board may be 81402439-001-3. In most cases, the correct tab number (101) will be located elsewhere on the board as a separate number.



NOTE: Part number for plug-in memory module (not shown) is QN700A105.

Figure 1—Chassis Assembly

Figure 1
Key No.

Figure 1 Key No.	Part Number	Description	Spares for 100 Instrument		
			Spares for 10 Instruments	Quantity per Instrument	
1	81402442-101	Display Board Assembly	1	1	3
2	81402441-101	Display Control Board Assembly	1	1	3
3	81402462-001	Touch Sheet Switch	1	1	3
4	81402413-001	Plastic Stopper	1	1	3
5		Output Board Assembly	1	1	3
	81402451-501	4-20 mA / No Aux. Out.			
	81402451-511	4-20 mA / 4-20 mA Aux. Out.			
	81402451-521	4-20 mA / 1-5 Volt Aux. Out.			
	81402451-701	1-5 Volt / No Aux. Out.			
	81402451-711	1-5 Volt / 4-20 mA Aux. Out.			
	81402451-721	1-5 Volt / 1-5 Volt Aux. Out.			
6		CPU Board Assembly	1	1	3
	81402439-101	Without Communications			
	81402439-111	With Communications			
7	81402443-111	Communications Board Assembly (Optional)	1	1	3
8		I/O Control Board Assembly	1	1	3
	81402440-101	Without Communications			
	81402440-111	With Communications			
9	83975073-001	Battery	1	1	3
10		Terminal Board Assembly	1	1	3
	81402447-101	Without Communications			
	81402447-121	With Communications			
11	81402448-101	Sequencer Board			
12		Analog Board	1	1	3
	81402449-301	RTD (One Channel)			
	81402449-C01	4-20 mA (One Channel)			
	81402449-V01	1-5V (One Channel)			
	81402449-TAB	Any T/C, One Channel (Specify T/C Type)			
	81402449-331	RTD (Two Channel)			
	81402449-CC1	4-20 mA (Two Channel)			
	81402449-VV1	1-5V (Two Channel)			
	81402449-TAB	Any T/C, Two Channel (Specify T/C Type)			
	81402449-MO1	0-10 mV (One Channel)			
	81402449-MM1	0-10 mV (Two Channel)			
13	81402463-001	Filter	1	1	3
14	81402412-001	Door	1	1	3

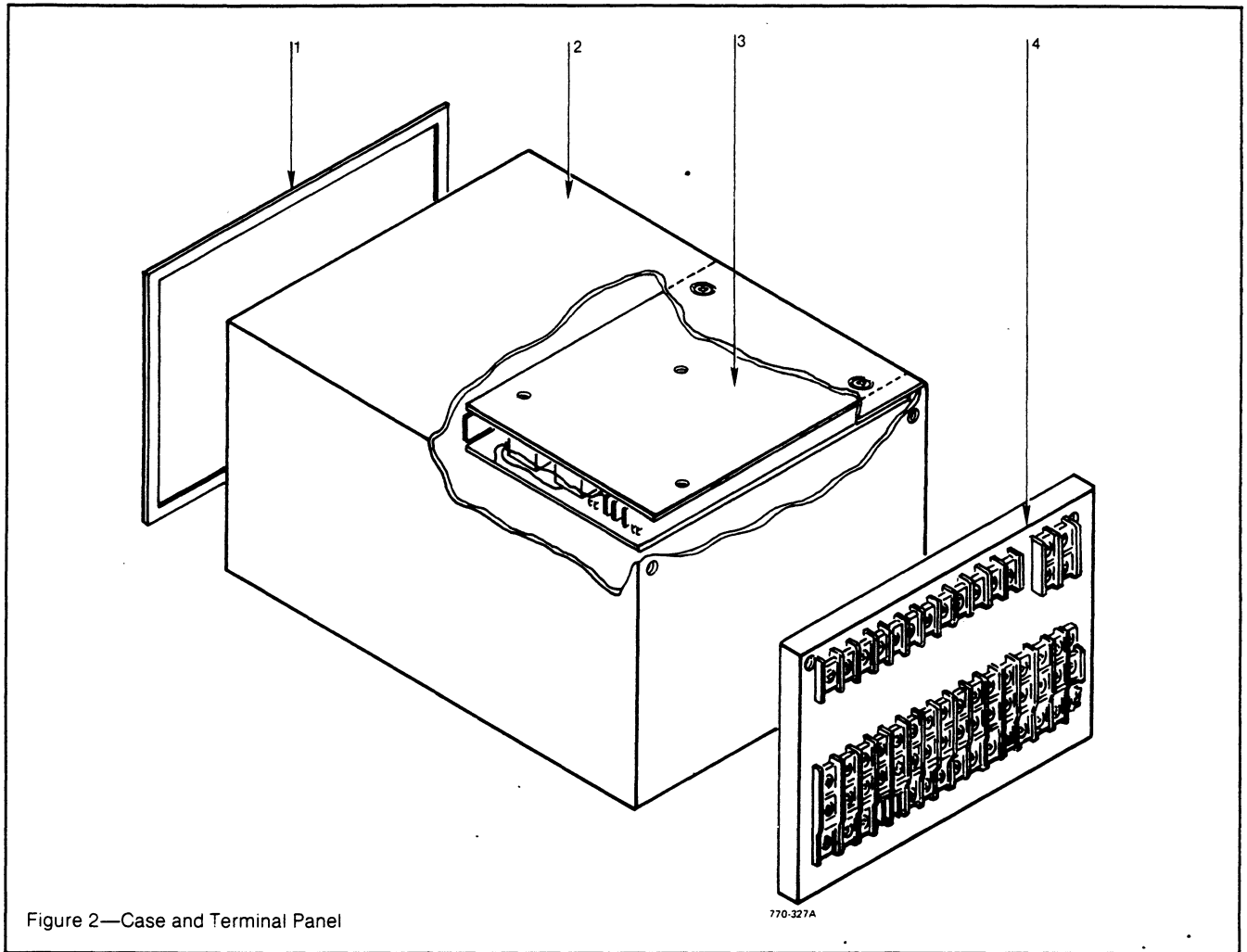


Figure 2—Case and Terminal Panel

Figure 2 Key No.	Part Number	Description	Spares for 100 Instrument Spares for 10 Instruments Quantity per Instrument		
1	81402464-001	Dust Seal	1	1	3
2	81402417-001	Case	1	1	3
3		Power Supply Board Assembly	1	1	3
	81402418-501	Power for CPU and 1 Channel Only			
	81402418-511	2 Channel Power — No Options			
	81402418-521	2 Channel Power — Auxiliary Output			
	81402418-531	2 Channel Power — Aux. Out., Communications			
	81402418-541	1 Channel Power — Communications			
4	81402409-101	Rear Terminal Panel Assembly	1	1	3
5	80237440	Mounting Bracket (Not Shown)			

DCP 700 Digital Control Programmer

Specification

Function

The DCP 700 Digital Control Programmer is really a family of microprocessor-based programmer only and programmer controllers that carry out predetermined processing or testing schedules on a time and variable basis. The variable can be temperature, pressure, relative humidity or any other measurable and controllable quantity. Both the programmer only and the programmer controller come in one- or two-channel versions to accommodate single- and two-loop applications. Plus, there is a two-channel, programmer controller dedicated to the measurement and control of Relative Humidity using a proven wet bulb and dry bulb temperature measurement technique. Choice of output type for controller and output characterization form for programmers makes it easy to match a DCP with most final control devices and process controllers.

The DCP's clearly labeled keys, easy-to-read displays, and prompt messages quickly guide you through the steps for establishing programs that change the magnitude of the variable, regulate the rate of change, and precisely time the beginning, duration and termination of these changes. When initiated, program runs take a process through a simple or very complex series of steps, that we call segments, without attention from an operator. The DCP can automatically repeat a program identically time after time. The ability to download and upload programs to and from a plug-in memory module means you can create a limitless library of programs for every application.

Programs can also include event settings to trigger alarms and/or control auxiliary equipment at specific points in the program. The settings can be made in time, process variable, size of the difference between the process variable (PV) and set point (SP), commonly called deviation, or all three.

External contacts can be used to supplement certain keyboard operations for the added flexibility of performing some operations from a remote location. Light Emitting Diodes (LEDs) and digital displays let you know at a glance

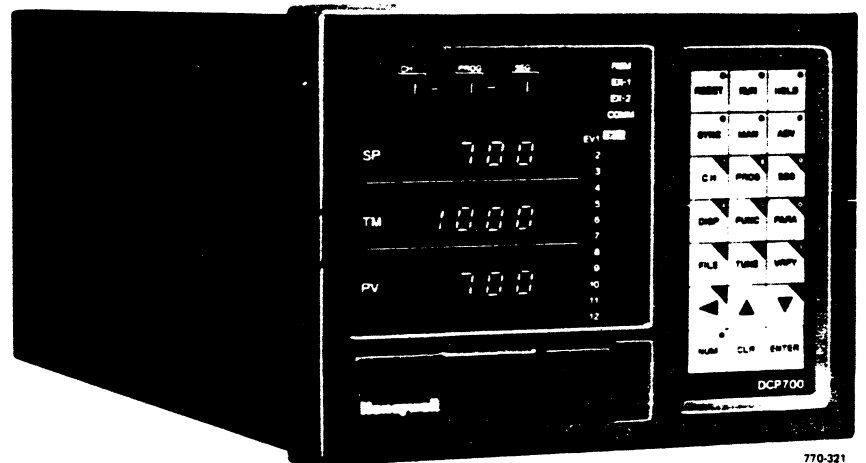


Figure 1

the status of the process, program, and control operation. The DCP includes many features to provide the flexibility and adaptability needed in any control method.

Features

- **Plug-in Memory Modules** supply an inexhaustible capacity for program storage and an easy means for changing programs to meet varying application needs.
- **Sizable Memory Capacity** lets you enter up to 39 programs with as many as 99 segments each; then you can download the 39 programs to a memory module for external storage and enter another 39 programs in the DCP.
- **Automatic Cycling** lets you select a program or a portion of a program to be cycled (repeated) up to 9999 times.
- **Configurable Tuning Constants** and other control-related parameters let you establish up to nine different combinations of constants and parameters which can be assigned on a per segment basis in every program to optimize controller response.
- **Versatile Operator Interface** lets you initiate such actions as start, hold, jump, restart, reset, advance segment, hot start, verify program contents, and copy or delete a program. Many of these actions can also be initiated through the closure of a remote contact.

- **Integral Auto-Manual Station** lets you switch to manual mode to adjust output manually through the keyboard.
- **Scaleable Range** for linear current and voltage inputs lets you set the desired low and high range values within a range of - 19999 to 19999 with or without decimal point.
- **Choice of PID-A or PID-B** control algorithm lets you tailor control response to application needs.
- **Selectable synchronous or asynchronous mode** gives you a choice of applying operator actions to one or both program/control channels.
- **Optional Auxiliary Output** lets you configure a current output to represent any of up to three DCP parameters* from each channel for recording or control purposes.
- **Optional DMCS Communication Interface** lets you establish a secure communication link with a host computer through Honeywell's Gateway interface.

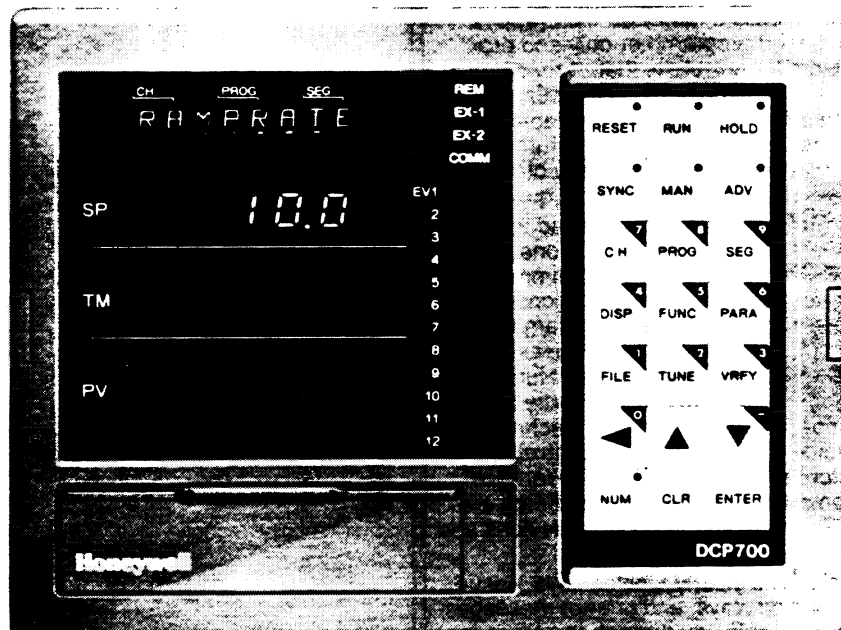
*Number and type of parameter differ with DCP for Relative Humidity.

Programs

Creation

Through simple keystroke sequences and prompts you can quickly create time and variable based programs to meet your specific requirements. Each program can have a time base of minutes or hours and includes two channels with up to 99 segments. Channel numbers correspond with programmer/controller outputs one and two, respectively. Segments are either soak or ramp type with duration and rate of change set as integers of the base time, and set points established as integers of the value of the process variable (PV). Soak segments can be guaranteed or unguaranteed. Ramps can be set on a variable value per unit of time basis (ramp rate) instead of the length of time (duration). Other program data includes.

- Value of guaranteed soak band
- On and off times for time event
- High and low values for PV event
- High and low values for Deviation event
- Number of cycles
- Process variable (PV) ratio value/PV bias value
- Number of PID combinations
- Number of program and segment that operation should divert to if external signal is on
- Number of program and segment that operation should divert to when external signal comes on



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Figure 2

Storage

You can permanently store up to 39 programs in the DCP's memory. However, plug-in memory modules let you create a file for storing 39 programs externally by simply downloading them to a module. To recall the filed programs, you upload them from the module into the DCP's memory. An internal memory protect switch inhibits access to stored programs.

Configuration

For optimum control operation, you set the values for the tuning constants and other operating parameters as well as select certain operating functions through the keyboard. This means you can configure the operation of the DCP to meet your specific requirements. The configurable tuning constants include up to nine different combinations of these parameters:

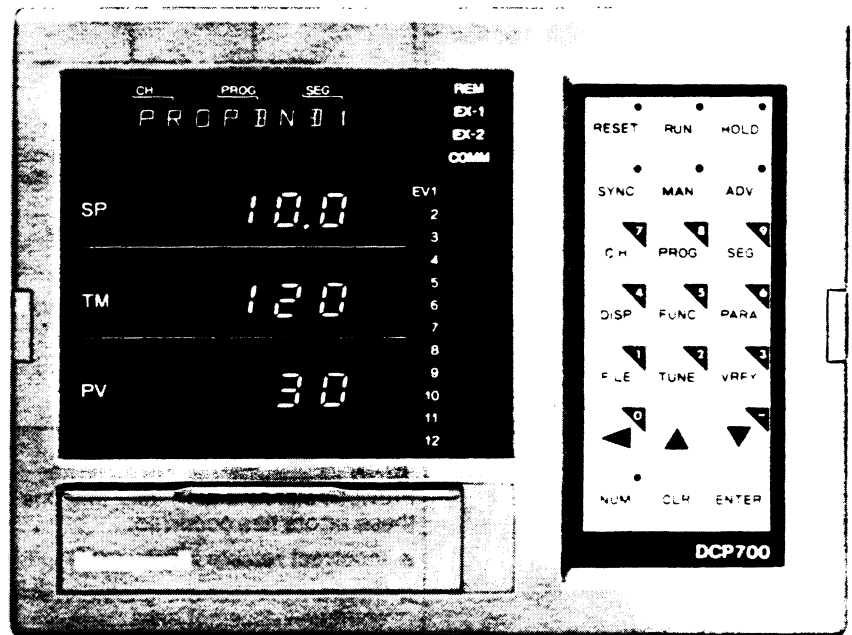
- Proportional Band
- Reset Time
- Rate Time
- High and Low Output Limits
- Rate Limit
- High and Low Reset Limits
- Manual Reset, as required

For programmer only, these parameters are supplied in place of the above tuning constants:

- High and Low Set Point (SP) Output Limits
- SP Output Type (selected on per channel basis)

Other operating parameters which are configurable include:

- Time Unit (selected on per program basis)
- Square Root Operation plus Drop-Out Values for Channels 1 and 2
- Input Filtering
- Hysteresis for PV and Deviation Events
- Supervising Time for External Ramp Signal Input
- Time Reduction Rate for Advance Function
- Decimal Point plus High and Low Range Values for Linear PV Input
- Pressure Compensation and Wind Velocity Code for RH Operation
- Set Point Bias Value
- Type of Events
- PV ratio and bias values
- Direction for Remote PV (Hot) Start Operation
- Direct or Reverse Acting Control Operation
- Time Proportional Output Interval (cycle) Time
- PID-A or PID-B Control Algorithm
- Initial value for control operation
- Output Value in READY Mode
- Dead Band Value for Duplex Control
- Optional Communication Address
- Optional Auxiliary Output Representation



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Figure 3

Control Algorithms

Whether you choose control algorithm PID-A or PID-B you can vary the control action through the configurable tuning constants. Each algorithm combines Proportional (P), Integral (I), and Derivative (D) control actions to provide PID, or three mode, control. But, by configuring tuning constants, single mode (P) and two mode (PI or PD) control action variations are possible.

The main difference between the algorithms is in their response to changes in set point. As shown in Figure 4, PID-A provides the full control action response to set point changes while PID-B responds with only integral control action. As a result, the operator doesn't have to inch along the set point to a new value with type B; even a large step-change can be entered, and controller output will move the process gradually to the new control point. When a change in process variable occurs, however, type B control response is the same as type A.

Output

Control

Programmer Controller

Models are available with combinations of these controller output types:

- Time Proportional (transistor open collector)—Cycles a transistor on and off with the ratio of on-time to total cycle time proportional to the output of the control algorithm. Requires an external 24 Vdc power supply.
- Current Proportional—Supplies direct or reverse acting 4-20 mA dc current proportional to the output of the control algorithm. Current proportional duplex is also available.
- Duplex Time Proportional (transistor open collector)—Provides two time proportional outputs, one above the 50% output and one below. This is ideal for processes which require both heating and cooling to maintain a set point.

Programmer Only

Models are available with a 4 to 20 mA dc set point output characterized as a thermocouple type, resistance temperature detector, or linear (0 to 100%) range.

Events

Provide on and off action in accordance with user configurable time, deviation, and process variable values. The transistor open collector output requires an external 24 Vdc power supply.

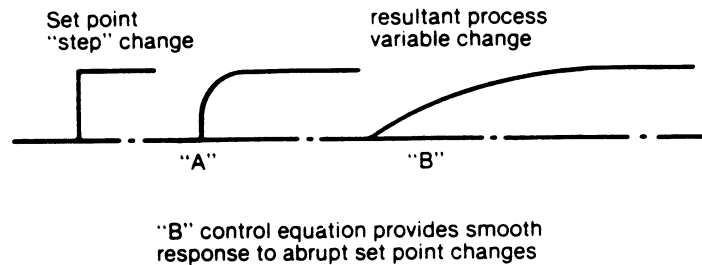


Figure 4

Error

Provides signal to alert user that one of these errors has occurred:

- Incorrect remote input.
- Program number designated in JUMP operation does not exist.
- External signal to set channel 1 or channel 2 ramp rate has not been detected.
- Time event has not been executed.
- Communications with host has been interrupted.

The transistor open collector output requires an external 24 Vdc power supply.

Status

Provides signals to indicate that channel 1 and channel 2 are in the RUN/HOLD status of program execution. The transistor open collector output requires an external 24 Vdc power supply.

CPU Failure

Provides a signal that is driven by the watchdog timer to monitor CPU operation. The signal is turned off, indicating that CPU operation has ceased, if the watchdog timer circuit times out. The transistor open collector output requires an external 24 Vdc power supply.

Alarm

Provides signal to activate external alarm if process variable is up- or downscale. The transistor open-collector output requires an external 24V, 1A power supply.

Operator Interface

Digital Displays

The upper display normally shows the active channel number, program number, and segment number. The "SP" display normally indicates the set point value in engineering units. The "TM" display normally shows, as selected, either the time remaining in the active segment or the output in percent (%). The PV display indicates the value of the process variable in engineering units.

You can pre-empt displays to show prompts and programs or configuration parameters for entering, verifying and changing operating data.

Light Emitting Diodes (LEDs)

The labeled LEDs provide pertinent operating information at a glance. For example, when EV5 is lit, event number 5 is on.

Keyboard

Eleven of the twenty-one keys are dual function type so you can input integers for program or configuration values in the numerical mode. Almost half of the keys let operators initiate dedicated functions to interact with the operation of the DCP or the control of the process. These functions include:

- Switching between manual and auto operating modes
- Switching between synchronous and asynchronous modes
- Putting a program on hold
- Resetting a program
- Raising or lowering the controller output manually
- Selecting channel, program or segment number
- Advancing a segment

The remaining keys are used to aid the program and configuration functions to enter, file, change, and verify data.

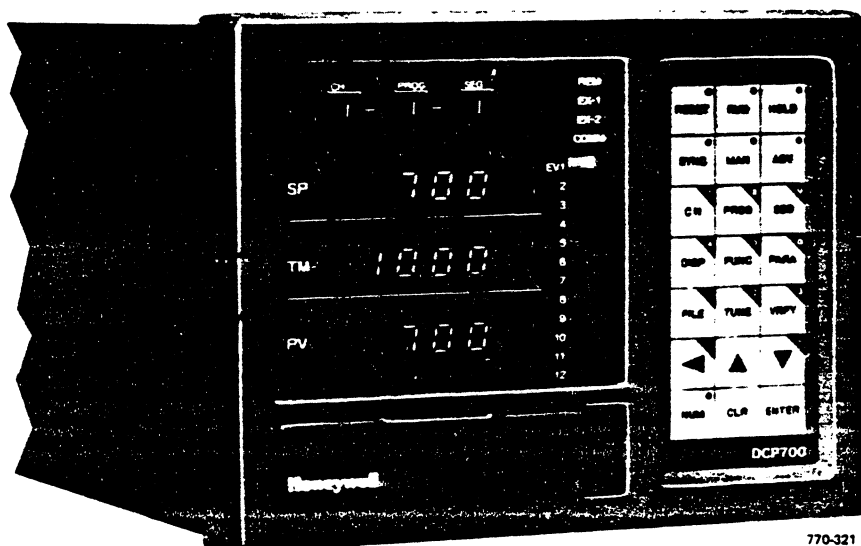


Figure 5

Diagnostics

The DCP 700 runs self-tests to check input, memory and processor operation. An alarm is activated if the self-tests detect any faults.

In addition, a user initiated self-test lets you quickly check the operation of all displays, LEDs and keys.

Process Interface

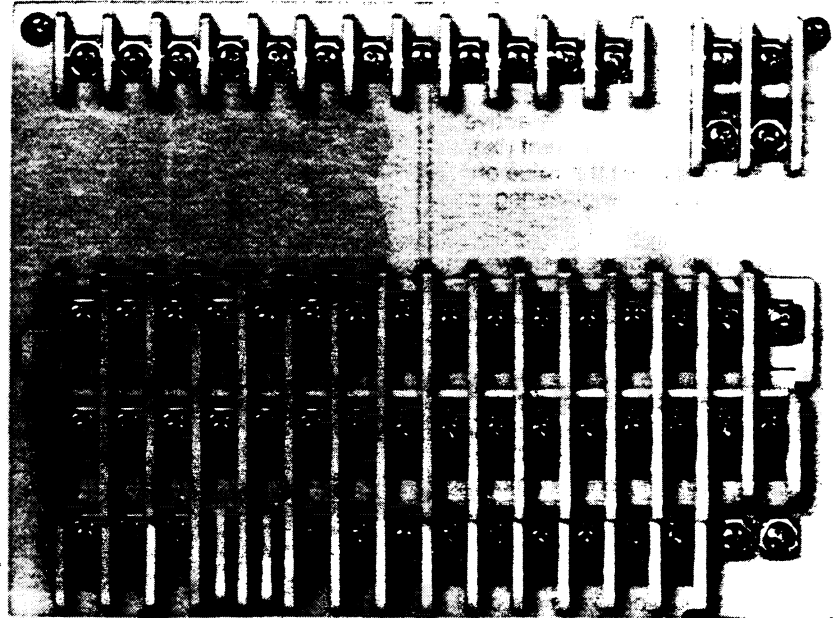
All power, input and output wiring are connected to screw terminals molded into the rear terminal panel, the terminals are labeled for easy identification.

Construction

The DCP is housed in a metal case that can be panel mounted. The chassis is attached to the rear terminal panel by ribbon cable and plugs. The modular construction of the chassis partitions major functions among individual printed wiring boards.

Options

- **Auxiliary Output**—A 4 to 20 mA dc output can be configured to represent one of these parameters: the set point for channel 1 or channel 2, the process variable for channel 1 or channel 2, or the output for channel 1 or channel 2. Plus, on the relative humidity DCP, can configure one of these parameters: relative humidity set point, relative humidity process variable, wet bulb set point, or wet bulb process variable.
- **DMCS Communications Interface**—Equips the DCP for use on Honeywell's Distributed Manufacturing Control System (DMCS) bus so a host computer can communicate with it through Honeywell's Gateway. Gateway gives a host computer data acquisition and remote control capabilities. For example, the host computer can retrieve set point values from a DCP and change the ramp rate of a programmed segment in a DCP.



770-324

Figure 6

Specifications

Performance

PV Indication Accuracy	$\pm 0.1\%$ of span at reference conditions*
Indication Resolution	<i>Thermocouple or RTD:</i> $\pm 0.1^\circ\text{C}$ (1°F) <i>Current or Voltage:</i> Depends on the scaled range

*Accuracy is not specified for 0 to 100°C (0- 200°F) portion of the type B thermocouple range.

Design

Number of Inputs	Model DCP71X—one input Model DCP72X—two inputs		
PV Input and SP Output Characterization¹	Type	Set Point Range	
		$^\circ\text{C}$	$^\circ\text{F}$
Thermocouple	T	– 200 to 300	– 300 to 700
	J	0 to 800	0 to 1600
	E	0 to 800	0 to 1800
	K	0 to 1200	0 to 2400
	Ni Ni Moly	0 to 1300	– 0 to 2500
	S	0 to 1600	0 to 3100
	R	0 to 1600	0 to 3100
	B	0 to 1800	0 to 3300
	$\text{W}_5\text{W}_{26}(\text{W})$	0 to 2300	0 to 4200
Platinum RTD	100 ohms (IEC)*	– 200 to 200	– 300 to 900
Millivolts	0 to 10 mVdc		– 19999 to 19999*
Milliamperes	4 to 20 mA dc		– 19999 to 19999*
Volts	1 to 5 Vdc		– 19999 to 19999**
Relative Humidity ²	Dry Bulb (100 Ω RTD)	– 200 to 200	– 300 to 900
	Wet Bulb (100 Ω RTD)		0 to 100% RH***

*IEC Alpha = 0.00385

**Decimal point can be moved up to four places to the left.

***From 0 to 100°C (32 to 212°F)

Sampling Rate	<i>Input:</i> 500 milliseconds <i>Output:</i> renewed every 500 milliseconds
Input Filtering	Single pole low pass section
Input Impedance	<i>Thermocouple:</i> 22 megohms upscale burnout <i>4-20 mA dc:</i> 2.5 ohms <i>1-5 Vdc:</i> 250K ohms
Sensor Break Protection	<i>Thermocouple and RTD:</i> Upscale
Power Consumption	40 VA
Stray Rejection	<i>Common Mode:</i> 120 db at 240 Vac, 50 Hz <i>Normal Mode:</i> 60 db (Voltage two times span maximum)
Controller Output	<i>Time Proportioning:</i> Open-collector transistor <i>Supply Voltage:</i> 24 Vdc $\pm 10\%$, 1 ampere, user-supplied (One supply only for this, events, and alarms. Honeywell supply 30683383-001 meets the requirements.) <i>Current Load:</i> 75 mA dc maximum <i>Time Resolution:</i> 1/1000 of cycle time <i>Cycle Time:</i> 4 to 600 seconds user configurable in four second increments <i>Duplex Time Proportioning:</i> Two open-collector transistors. See time proportioning for specifications. <i>Current Proportioning:</i> 4 to 20 mA dc into a negative or positive grounded or non-grounded load of 0 to 600 ohms. <i>Accuracy:</i> $\pm 0.1\%$ of span at reference conditions <i>Resolution:</i> 1/16000 <i>Output Ripple:</i> 4 mV peak-to-peak at less than 50 Hz with 100 to 600 ohms load resistance.
Programmer Output	4 to 20 mA dc—Linear or characterized (600 ohm load maximum). Output is isolated from input. <i>Accuracy:</i> $\pm 0.1\%$ of span

¹ SP Output Characterization only applies for programmer only models.

² Only available on Relative Humidity Programmer Controller.

Design (Continued)

Events, Error, Status, CPU Failure and Alarm Outputs	Open collector transistor. <i>Supply Voltage:</i> 24 Vdc $\pm$ 10%, 1 ampere, user-supplied (One supply only for these and time proportioning. Honeywell supply 30683383-001 meets the requirements.) <i>Current Load:</i> 75 mA dc maximum <i>Relays:</i> User-supplied. [Honeywell relays 30731674-001 (30731675-001, with socket) and 30736332-001 (solder-in) meet the requirements.]	
Control Action	Direct or reverse can be selected internally.	
Program Storage Capacity	<i>DCP Memory:</i> 39 programs with up to 99 segments. <i>Plug-in Memory Module:</i> 39 programs with up to 99 segments in each module.	
Program Cycles	Up to 9999 repeats	
Program Protection	Battery (3 year life from date of manufacture) backed memory; internal switch locks programs in memory and prevents accidental change through the keyboard.	
Guaranteed Soak Band	1 to 100 units	
Ramp Rate	<i>Thermocouple or RTD Input:</i> 1 to 999.9 <i>Linear Input:</i> 1 to 9999	
Soak or Ramp Duration	0.00 to 255.59 (minutes or hours)	
Events	1 to 12 — Configurable as time, PV, or deviation type.	
Remote Switching	External dry contacts are all that is needed to activate certain DCP operations from remote locations.	
Tuning Constants	Parameter	Setting Range
	Proportional Band	0.1 to 1000%
	Reset Time	0.1 to 1800 seconds/repeat (0 = no reset action)
	Rate Time	0.1 to 600 seconds (0 = no rate action)
	High and Low Output Limits	– 5 to 105% of span
	Output Rate Limit	0 to 100% of span
	High and Low Reset Limits	– 5 to 105% of span (must be within configured range of high and low output limits)
	Manual Reset	0 to 100% of span (only effective with no reset action)
	Can store up to nine groups of the above and assign any group to any segment of any program. Also, can establish different groups as HEAT and COOL settings for time proportional duplex models.	
Other Parameters	Parameter	Setting Range
	Time Unit	Minutes or hours with setting as 0.00 to 255.59
	Advance Rate	1 to 120 (Modifies time axis to speed up program execution)
	Digital Filtering	0 to 255 seconds (each channel)
	Event Hysteresis	1 to 200 units for PV and Deviation events
	Supervising Time for External Ramp Signal	0 to 255.59
	Set Point Bias	Within low and high range limits
	PV Ratio/Bias	<i>Ratio:</i> 0.000 to 9.999 <i>Bias:</i> Within low and high range limits
	Drop-Out Value	0 to 10% (each channel for square root operation)
	High and Low Linear Range Limits	– 19999 to 19999
	Time Proportional Output Interval (Cycle) Time	4 to 600 seconds
	Ready Mode Output	0 to 100%
	Dead Band	– 5.0 to 5.0% (Duplex Control)
	RH Compensation	
	<i>Pressure</i>	500 to 1000 mm hg
	<i>Wind Velocity</i>	Three ranges: 0 to 50 meter/sec. 0.6 to 2.4 2.5 and higher

Options

Auxiliary Output 4-20 mA dc into a negative or positive grounded or non-grounded load of 0 to 600 ohms.
Accuracy: $\pm 0.1\%$ full scale at reference conditions
Sampling Rate: Renewed every 500 milliseconds

DMCS Communications *Baud Rate:* 19.2K bauds
Length of Link: 2000 ft. maximum
Link Characteristics: Two wire, multi-drop proprietary protocol.

Physical Considerations

Weight 13.2 pounds (6 kilograms)

Mounting Panel (mounting hardware supplied)

Wiring Connections Screws accept terminals with 0.125 inch (3,2 mm) holes or slot.

Dimensions See Figure 7.

Color *Case:* Black
Bezel: Caribbean blue

Operating Conditions

	Reference Conditions	Rated Conditions	Operative Limits	Transportation and Storage
Ambient Temperature	23 $\pm$ 2°C 73 $\pm$ 4°F	10 to 40°C 50 to 104°F	0 to 50°C 32 to 122°F	- 20 to 60°C 4 to 140°F
Relative Humidity (%)	60 $\pm$ 5	0 to 90*	10 to 95	10 to 95
Vibration				
Frequency (Hz)	0	10 to 60	10 to 60	
Acceleration (g)	0	0.02	0.2	0.5
Mechanical Shock				
Acceleration (g)	0	1	5	50
Voltage (Vac)	120 $\pm$ 1	108 to 132	108 to 132	—

*The maximum rating only applies up to 40°C (104°F). For higher temperatures, the RH specification is derated to maintain constant moisture content.

Dimensions: $\frac{\text{mm}}{\text{in.}}$

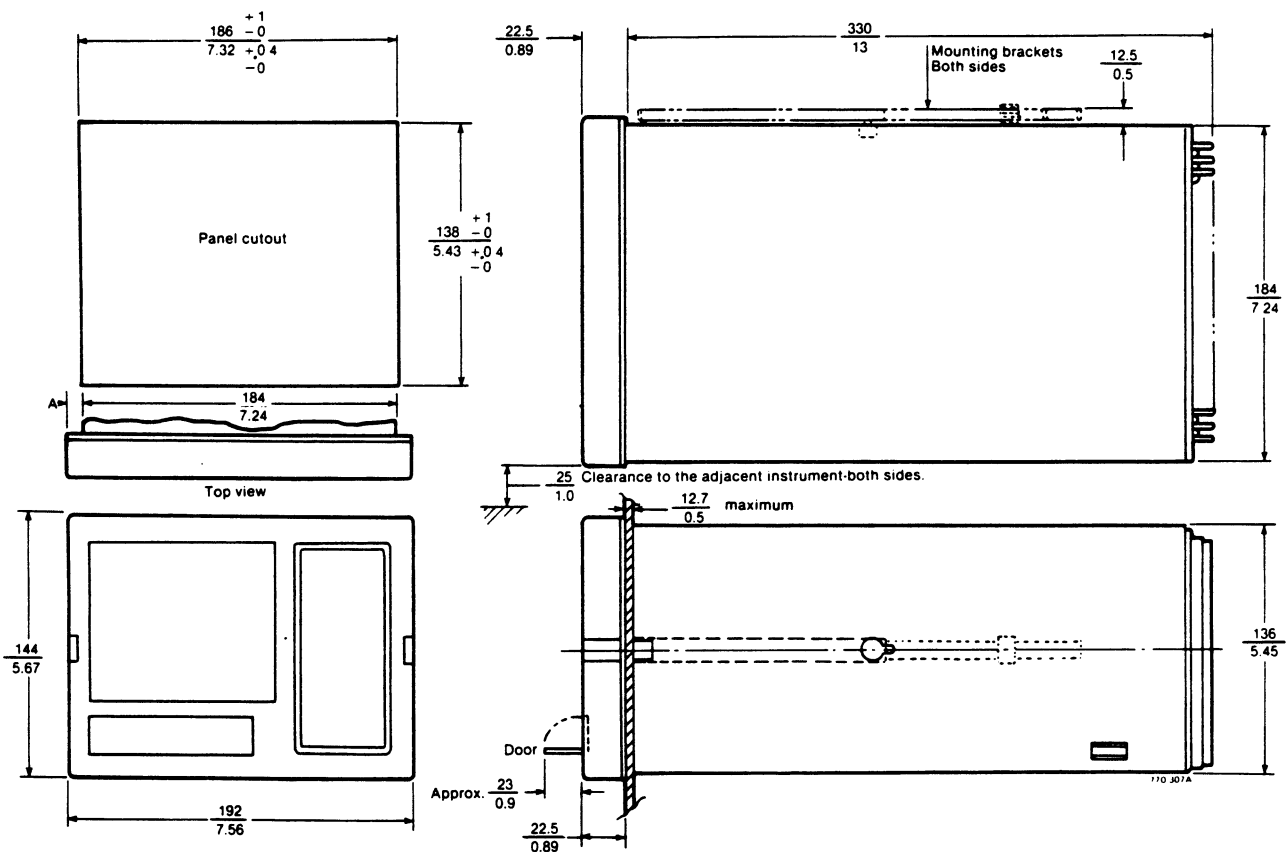


Figure 7

Ordering Information

For complete ordering information, request Model Selection Guide 57-77-16-10 DCP 700 Digital Control Programmer

Honeywell offers a full line of sensors, transmitters, process controllers and final control devices for use with DCP. These devices include:

- Thermocouples,
- RTDs,
- Pressure Transmitters,
- Flow Transmitters,
- Liquid Level Transmitters,
- Valves,
- Actuators,
- Electric Motors.

For more information contact one of the following:

Your nearest Honeywell Branch Office

Honeywell Inc.
1100 Virginia Drive
Fort Washington, PA 19034

Specifications are subject to change without notice.

DCP 700 Digital Control Programmer Appendix A

Model, Option, Output, and PV Input Data

Summary

This section tells you how to check your DCP's basic configuration codes which define what model it is, what options it has, what output type it has, and what PV inputs it will accept. It also explains how to change codes, if necessary.

How To Check Basic Configuration Codes

Use the following steps in conjunction with Tables A 1 through A4 to check the stored basic configuration codes.

Step	Key	Displays	Numerical Mode	Comments
1				Be sure RESET LED lit (ready mode) and MAN LED is out.
2	▼ RESET	A-N Adjust		NUM LED blinks
3			Key in 4	4 appears in SP display
4	ENTER	A-N Password SP O		NUM LED blinks
5			Key in 2255	2255 appears in SP display
6	ENTER	A-N Model SP X		X must equal appropriate code from Table A1 for your DCP model. NOTE: NUM LED blinks, but you cannot key in a different code without performing the steps in the How to change Code paragraph first.
7	ENTER	A-N Option SP X		X must equal sum of appropriate option codes from Table A2. See "Note" in Step 6.
8	ENTER	A-N Out Type SP X		Only applicable on Programmer Controller model. X must equal appropriate code from Table A3 for your DCP's output type. See "Note" in Step 6.
9	ENTER	A-N PV-Type 1 SP X		X must equal appropriate code from Table A4 for your DCP's Channel 1 input type. See "Note" in Step 6.
10	ENTER	A-N PV Type 2 SP X		Only applicable for two channel models. X must equal appropriate code from Table A4 for your DCP's Channel 2 input type. See "Note" in Step 6.
11	ENTER			Return to normal display.

Table A1 — Model Number Basic Configuration Codes

Code	Key Number*	Description
0	DCP721	2 — Channel Programmer Controller
1	DCP720	2 — Channel Programmer Only
2	DCP722	Relative Humidity Programmer Controller
3	DCP711	1 — Channel Programmer Controller
4	DCP710	1 — Channel Programmer Only

*Key number is first six digits from DCP's model number.

Table A2—Option Basic Configuration Codes

Code*	Type of Optional Function
0	None
1	DMCS Communications
4	Duplex Control
8	Auxiliary Output

*Code that appears in SP display will equal the sum of all the optional functions supplied. For example, your DCP was supplied with DMCS Communications, duplex control, and auxiliary output so the code in the SP display must be 13 ($1 + 4 + 8 = 13$).

Table A3—Output Type Basic Configuration Codes

Code	Number of Channels	Output Type
0	1	Current Proportional (4-20 mA dc)
1		Time Proportional (Open Collector Transistor)
2		Time Proportional Duplex (open Collector Transistor)
3		Current Proportional Duplex (4-20 mA dc)
4	2	Current Proportional (4-20 mA dc) Channels 1 and 2
5		Current Proportional (4-20 mA dc) Channel 1 Time Proportional (Open Collector Transistor) Channel 2
6		Time Proportional (Open Collector Transistor Channel 1 Current Proportional (4-20 mA dc) Channel 2
7		Time Proportional (Open Collector Transistor Channels 1 and 2
8		Time Proportional Duplex (Open Collector Transistor) Channel 1 Time Proportional (Open Collector Transistor) Channel 2
9		Time Proportional Duplex (Open Collector Transistor) Channel 1 Current Proportional (4-20 mA dc) Channel 2
10		Current Proportional Duplex (4-20 mA dc) Channel 1 Time Proportional (Open Collector Transistor) Channel 2

Table A4 — Input Type Basic Configuration Codes

Input Type	Code	°F Range	Code	°C Range
T	20	– 300 to 700	1	– 200 to 300
J	21	0 to 1600	2	0 to 800
E	22	0 to 1800	3	0 to 800
K	23	0 to 2400	4	0 to 1200
S	24	0 to 3100	6	0 to 1600
R	25	0 to 3100	7	0 to 1600
B	26	0 to 3300	8	0 to 1800
W ₃ W ₂₈	27	0 to 4200	9	0 to 2300
Ni-Ni Moly	28	0 to 2500	11	0 to 1300
RTD PT 100Ω (IEC)	29	– 300 to 900	15	– 200 to 200
Current	19	4 to 20 mA		
Millivolt	19	0 to 10 mV		
Volts	19	1 to 5 Vdc		

How To Change A Basic Configuration Code

Use the following steps in conjunction with the steps in the How to Check . . . Codes procedure to change a basic configuration code. However, be sure that any code changes reflect the actual hardware configuration of your DCP. In most cases, you can not get a function change without a corresponding change in hardware. See the Parts List section for a breakdown of hardware variations versus DCP functions.

1. Remove power from the DCP.
2. Pull the latches, one on either side of the bezel, forward. Pull on the bezel, alternating sides, until the bezel starts to come away from the case. Slide the chassis out of the case slowly until you see the connector on the CPU board (third from front).
3. Locate the two vertical pins on the EEPROM-protect-connector which is located to the left of the battery on the top right side of the CPU board. Wrap a piece of wire around the pins to short them together and be sure the wire doesn't contact any other components or the case.
4. Slide the chassis back into the case and push in the latches.
5. Open the plug-in module door and move the memory protect switch lever to the M.E. position. Close the module door.
6. Apply power to the DCP and follow steps in the How To Check . . . Codes Procedure to key in different codes, where applicable.
7. Remove power from the DCP, remove wire from EEPROM connector pins, and move memory protect switch lever to the M.P. position.
8. Return DCP to normal operation.

DCP 700 Digital Control Programmer Appendix B

DMCS Communications Option

Summary

This instruction provides the procedures for adding the DMCS Communications option to your DCP in the field. It assumes that the following components (included in kit 30755128-001) are on hand:

Component	Part Number	Qty.
CPU board assembly	81402439-111	1
Communications board assembly	81402443-111	1
I/O control board assembly	81402440-111	1
Terminal board assembly	81402447-121	1
Power supply board assembly	81402418-531	

Conversion Procedures

Use the following procedures to install the components listed above. These procedures assume that you are familiar with the DCP. If you are not familiar, you may need to supplement these procedures with data from the Service section in the DCP's product manual (57-77-25-07).

Equipment Needed

- small flatbladed screwdriver
- medium flatbladed screwdriver
- medium phillips head screwdriver

Removing the Chassis

1. Remove power from DCP.
2. Pull latches, one on either side of bezel, forward.
3. Pull on bezel, alternating sides, until chassis starts to come away from case.
4. Slide chassis out of case slowly until you see connector on CPU board (third from front). Remove connector from its socket.
5. Continue to slide chassis out slowly until it is almost out of case. Remove two connectors from terminal board assembly.
6. Remove chassis from case.

Removing the CPU Board

1. Remove plastic stoppers, one on each side of chassis, over boards by lifting appropriate tab.
2. Remove connector from top of CPU board.
3. Slide board up and out.
4. Slide metal shield up and out.

Removing the I/O Control Board

1. Remove connector from top of Display Control Board.
2. Remove two connectors from top left side of terminal board.
3. Turn chassis over. Remove two connectors from sequencer board, one connector from analog board, and one connector from bottom of output board.
4. Turn chassis top side up. Loosen screws that hold front bezel to chassis, and pull bezel forward slightly.
5. Lift I/O control board and cables out.

Installing New CPU, Communications and I/O Control Boards

1. Slide new I/O Control board and cables into its slots on chassis. Push bezel back into place and tighten screws.
2. Slide new communications board* into next chassis slots behind I/O control board.
*NOTE: If communications board has a DIP switch on its top right side (as installed), be sure that switch 3 is "ON" and all the other switches are "OFF."
3. Slide new CPU board into its slots on chassis.
4. Plug connector on short cable on top of I/O control board into connector on top of display control board.
5. Plug center connector on longer cable on top of I/O control board into connector on top left side of communications board. Plug connector on end of long cable into connector on top left side of CPU board.
6. Turn chassis over. Plug large center connector on right hand cable from I/O control board into connector on bottom of communications board. Leave the four smaller connectors on end of this cable loose for now.
7. Plug large connector on left hand cable from I/O control board into connector on bottom of output board. Leave smaller connector on end of this cable loose for now.
8. Turn chassis top side up.

Removing and Installing the Terminal Board

1. Use a screwdriver to lift four tabs (two on each side) on rear bracket and remove bracket.
2. Loosen chassis screws, one on either side of chassis, slightly.
3. Remove connector from component side of output board.
4. Slide all three boards (terminal, sequencer, and analog) out together until connectors on top right side of terminal board can be disconnected. Unplug connectors and remove terminal board.
5. Slide in new terminal board to point where connectors on top right side can be plugged in. Plug in connectors and slide all three boards forward.
6. Slide terminal board back slightly and connect first set of connectors on left hand cable to connectors on top left side of the sequencer board. Slide terminal board forward and connect remaining set of connectors to connectors on the top left side of the terminal board.
7. Slide all three boards forward all the way. Connect cable connector from sequencer board to connector on component side of output board.
8. Turn chassis front face down and snap rear bracket in place.
9. Turn chassis bottom side up and plug remaining connector into connector on analog board.
10. Turn chassis top side up. Replace metal shield between CPU and output boards. Install plastic stoppers over tops of boards.
11. Check that all connectors are seated properly.

Removing and Installing the Power Supply

1. Loosen two screws on top of rear terminal panel at back of case.
2. Pull top of terminal panel slightly away from case and squeeze connector on power supply board and pull it off.
3. Pull terminal panel away from case.
4. With one hand under power supply assembly, remove five screws on top of case that hold assembly in place and remove it from the rear of the case.
5. Reverse previous steps to install new power supply board assembly.

Installing the Chassis

1. Raise cable connectors in case and start to slide chassis into case. Plug set of dual connectors into connectors on top right rear of terminal board.
2. Hold remaining connector outside of case and slide chassis further into case until connector can be plugged into connector near top OF CPU board.
3. Slide chassis all the way into case and push in latches.

Changing the Option Code

1. Remove power from the DCP.
2. Pull the latches, one on either side of the bezel, forward. Pull on the bezel, alternating sides, until the bezel starts to come away from the case. Slide the chassis out of the case slowly until you see the connector on the CPU board (third from front).
3. Locate the two vertical pins on the EEPROM-protect-connector which is located to the left of the battery on the top right side of the CPU board. Wrap a piece of wire around the pins to short them together and be sure the wire doesn't contact any other components on the case.
4. Slide the chassis back into the case and push in the latches.
5. Open the plug-in module door and move the memory protect switch lever to the M.E. position. Close the module door.
6. Apply power to DCP. Be sure **RESET** LED is lit (ready mode) and **MAN** LED is out. Press **▼** and **RESET** keys. ADJUST appears in upper display and 0 in SP display. Key in **4** which appears in SP display. Press **ENTER** key, PASSWORD appears in upper display and 0 in SP display. Key in 2, 2, 5, 5 which appears in SP display. Press **ENTER** key twice. OPTION appears in upper display and code appears in SP display. Key in a value that equals the code plus one. For example, assume present code is 8, you would key in 9. Press **ENTER** key. Press **RESET** key to return to normal display.
7. Remove power from the DCP, remove wire from EEPROM connector pins, and move memory protect switch lever to the M.P. position.
8. Return DCP to normal operation.

Entering a Station Address

1. Apply power to DCP.
2. Open the plug-in module door and move the memory protect switch lever to the M.E. position. Close the module door.
3. Press **PARA** key and "TIMEUNIT" appears in upper display. Press **▼** key; "COMMPAR" appears in upper display and "0" shows in SP display.
4. Press **ENTER** key, NUM LED blinks, key in station address 1 to 31, and Press **ENTER**.
5. Press **DISP** key to exit parameter setting mode.
6. Move memory protect switch lever to the M.P. position.
7. Cycle DCP power off and then on.

Connecting DMCS Network Cable

Connect the positive (+) lead to terminal 18, negative (–) lead to terminal 19, and shield to terminal 29 on the rear terminal panel.

Entering Program and Configuration Data

See appropriate sections in the DCP product manual to re-enter and/or enter new program and configuration data.

Changing Model Number

Locate the last three digits in the model number listed on the nameplate attached to the side of the case. Cross out the zero that precedes the letter "H" in the last three digits and replace it with the number "2." This indicates that the DCP is equipped with the communications option.

DCP 700 Digital Control Programmer

Model Selection Guide

Instructions

- Select the desired Key Number. The arrow to the right marks the selections available.
- Make one selection each from Tables I through IV, using the column below the proper arrow. A dot (•) denotes unrestricted availability. A letter denotes restricted availability.
- A complete Model Number must have the designated number of digits in each table.

Key Number I II III IV

Key Number	Selection		Availability			
	Number of Programmer Channels	Number of Controllers				
1		0	DCP710	↓		
		1	DCP711		↓	
2		0	DCP720	↓		
		2	DCP721		↓	
		2 Relative Humidity	DCP722			↓

Table I Output	4-20 mA Output Programmer Only Models	5S	•			
	Current Proportional (4-20 mA DC) PID	5G		•	•	•
	Time Proportional (Open Collector Transistor) PID	8D		•	•	•
	Current Proportional Duplex (4-20 mA DC) PID	5K		•		
	Time Proportional Duplex (Open Collector Transistor) PID	8K		•		
	Current Proportional (4-20 mA DC) PID Channel 1	91			•	•
	Time Proportional (Open Collector Transistor) PID Channel 2					
	Time Proportional (Open Collector Transistor) PID Channel 1)	92			•	•
	Current Proportional (4-20 mA DC) PID Channel 2					
	Time Proportional Duplex (Open Collector Transistor) PID Channel 1	93			•	•
	Time Proportional (Open Collector Transistor) PID Channel 2					
	Time Proportional Duplex (Open Collector Transistor) PID Channel 1	94			•	•
	Current Proportional (4-20 mA DC) PID Channel 2					
	Current Proportional Duplex (4-20 DC mA) PID Channel 1	95			•	•
	Time Proportional (Open Collector Transistor) PID Channel 2					

Table II Controller Actuation or Programmer Output Characteristics

					DCP7 ☐ ☐											
					Selection		10 20	11	21	22						
Table II Selection Format	DCP720 } DCP 721 }	With Different Actuations Channel 1 Channel 2			A											
	DCP720 } DCP721 } DCP722 }	With same actuation														
	DCP710 } DCP711 }															
Thermocouple	T	- 300 to 700°F	- 200 to 300°C	TF1	T	c	•	c								
	J K W ₅ W ₂₆	0 to 1600°F 0 to 2400°F 0 to 4200°F	0 to 800°C 0 to 1200°C 0 to 2300°C	JF1 KF1 WF1	J K W	c	•	c								
	R B S Linear	0 to 3100°F 0 to 3300°F 0 to 3100°F 0 to 10 mV	0 to 1600°C 0 to 1800°C 0 to 1600°C 0 to 10 mV	RF1 BF1 SF1 M01	R B S M	c	•	c								
	E Ni Ni Moly	0 to 1800°F 0 to 2500°F	0 to 800°C 0 to 1300°C	EF1 ZF1	E Z	c	•	c								
Resistance Temperature Detector	100 Ohm (IEC 0.00385)	- 300 to 900°F	- 200 to 200°C	HF1	6	c	•	c								
Relative Humidity	Dry Bulb 100 ohm PT (IEC 0.00385)	- 300 to 900°F or - 200 to 200°C	HF1													
	Wet Bulb 100 ohm pt (IEC 0.00385)	0 to 100% RH From 0 to 100°C (32 to 212°F)														
Linear	Milliamps	4-20 mA DC		CO1	C	c	•	c								
	Volts	1-5 Vdc		VO1	V	c	•	c								

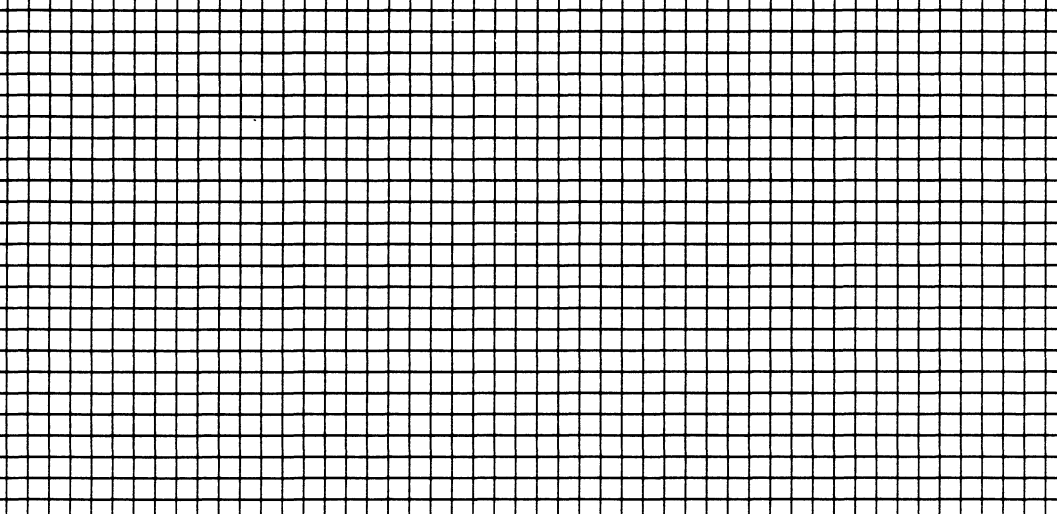
Table III	Power	120 VAC 50/60 Hz	5	•	•	•	•
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Table IV Options	None	000H0	•	•	•	•
	Auxiliary Output (4-20 mA DC)	010H0	•	•	•	•
	DMCS Communications and Auxiliary Output	012H0	•	•	•	•

Restrictions	Restriction Letter	Description
	c	
	■	If TC actuation is required on Channel 2, then TC actuation must also be specified on Channel 1. All temperature units will be shipped with Deg. F. (field selectable for Deg. C.).

Worksheet

	Channel 1	Channel 2
Input type		
Set point unit		
Set point range	_____ to _____	_____ to _____
Time unit	_____	

[illegible]A large grid of graph paper, consisting of 30 columns and 20 rows of small squares, intended for drawing a picture.

Parameters

Parameter	Your Selection		Prompt
	Code	Value	Upper Display
Time unit	_____	_____	TIMEUNIT
Advance rate	_____	_____	ADV-RATE
Auxiliary output	_____	_____	RECORDER
Input filter	_____	_____	FILTER
Channel 1	_____	_____	FILTER1
Channel 2	_____	_____	FILTER2
Event conf	_____	_____	EVT-CONF
Event 1	_____	_____	EVT1-4
Event 2	_____	_____	
Event 3	_____	_____	
Event 4	_____	_____	
Event 5	_____	_____	EVT5-8
Event 6	_____	_____	
Event 7	_____	_____	
Event 8	_____	_____	
Event 9	_____	_____	EVT9-12
Event 10	_____	_____	
Event 11	_____	_____	
Event 12	_____	_____	
Event hysteresis	_____	_____	EVT-PARA
Channel 1	_____	_____	EVT-HYS1
Channel 2	_____	_____	EVT-HYS2
External ramp monitoring time	_____	_____	RAMP-PAR
Channel 1	_____	_____	ALM-TM1
Channel 2	_____	_____	ALM-TM2
Set point bias	_____	_____	SP-BIAS
Channel 1	_____	_____	SP-BIAS1
Channel 2	_____	_____	SP-BIAS2
PV ratio & bias	_____	_____	PVBIAS1
Channel 1—Ratio	_____	_____	PVRATIO1
—Bias	_____	_____	PVBIAS1
Channel 2—Ratio	_____	_____	PVRATIO2
—Bias	_____	_____	PVBIAS2

Parameters (continued)

Parameter	Your Selection		Prompt
	Code	Value	Upper Display
External PV (Hot) start	_____	_____	PV-START
Channel 1	_____	_____	PVSTART1
Channel 2	_____	_____	PVSTART2
Range limits	_____	_____	RANGE-1
Channel 1	_____	_____	DEC-PT1
Limits	_____	_____	RANGE-H1
Hi-rng value	_____	_____	RANGE-L1
Lo-rng value	_____	_____	
Channel 2	_____	_____	RANGE-2
Limits	_____	_____	DEC-PT2
Hi-rng value	_____	_____	RANGE-H2
Lo-rng value	_____	_____	RANGE-L2
Square root	_____	_____	ROOT-CAL
Apply to PV	_____	_____	USE-CH
Dropout value	_____	_____	
Channel 1	_____	_____	DROP-OT1
Channel 2	_____	_____	DROP-OT2
Output relay interval	_____	_____	TP-INTVL
Channel 1	_____	_____	
Heat	_____	_____	INTAVL-1
Cool	_____	_____	INTAVL-2
Channel 2	_____	_____	INTAVL-3
Output direct.	_____	_____	CTL-ACT
PID Equation	_____	_____	PID-TYPE
Initial output	_____	_____	INIT-VAL
RH corrections	_____	_____	RH-PARA
Pressure	_____	_____	PRESSURE
Wind velocity	_____	_____	WIND-VEL
Time-prop relay deadband	_____	_____	DEADBAND
Communication address	_____	_____	COMMPAR

Controller tuning constants and limits

Write your selections under the PID number									
Tuning Constants									
PID Number	1	2	3	4	5	6	7	8	9
Proportional band									
Reset time									
Rate time									
Output limits									
High									
Low									
Output rate of change limit									
Reset limit									
High									
Low									

Programmer output limits and types

Channel 1	
High limit	_____
Low limit	_____
Output type	_____
Channel 2	
High limit	_____
Low limit	_____
Output type	_____

DCP 700 Digital Control Programmer

Errata for Product Manual

2-Channel Model w/ Time Proportional Duplex Outputs

Summary

NOTE: This Errata only applies for 2-channel DCP 700 Digital Control Programmers with time proportional duplex outputs - *Table 1 in model number = 99.*

This 2-channel DCP 700 Digital Control Programmer provides a time proportional duplex output on both channels 1 and 2. It can also have two auxiliary outputs as well as DMCS communications options, if specified on the order.

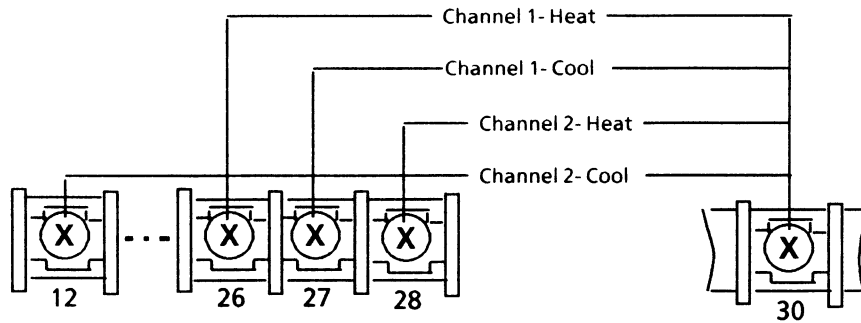
To provide two time proportional duplex outputs, this model DCP 700 features modified software, wiring connections, and configuration data that differs from other models covered in the DCP 700 Product Manual 57-77-25-07.

Use the data in this Errata in place of the specified data in the Product Manual to correctly wire and configure your DCP 700 with two time proportional duplex outputs.

■ **Installation Section, Page 2-7 - Replace list of wiring figures in first paragraph with the following data:**

- | | |
|---|----------------|
| ● Power and ground | Figure 2-6 |
| ● Input | Figure 2-7 |
| ● Current proportioning control and programmer output | Not Applicable |
| ● Time proportioning duplex output | Figure 2-9 |
| ● External power supply | Figure 2-10 |
| ● Events (1-11) | Figure 2-11 |
| ● Auxiliary outputs (2) | Figure 2-12 |
| ● External operation contacts | Figure 2-13 |
| ● Program number selection | Figure 2-14 |
| ● Alarm outputs | Figure 2-15 |
| ● Communication | Figure 2-16 |
| ● Composite | Figure 2-17 |

- **Installation Section, Page 2-8, Figure 2-9** - Delete Time proportional diagram and replace time proportional duplex diagram with the following diagram:



- **Installation Section, Page 2-9, Figure 2-11** - Delete "Event 12". There are only eleven events available with this DCP 700 model.
- **Installation Section, Page 2-10, Figure 2-12** - Replace auxiliary output wiring diagram with the following diagram and change caption as noted.

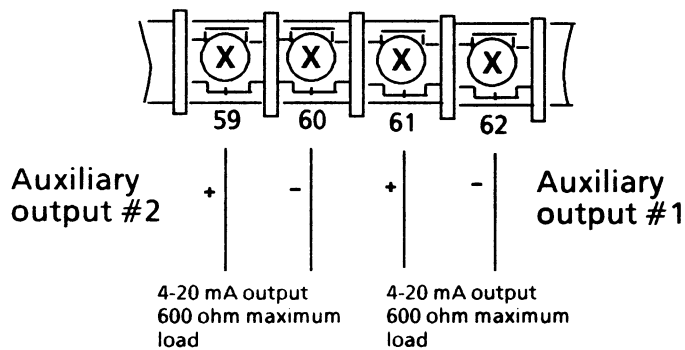


Figure 2-12 - Auxiliary (Recorder) outputs 1 and 2

- **Installation Section, Page 2-13, Figure 2-17** - Make the following changes in the wiring diagram:

<u>Terminal</u>	<u>Now</u>	<u>Change To</u>
1-12	Events 1 through 12	Events 1 through 11
12	Event 12	CH 2 Cool
28	CH 2	CH 2 Heat
59,60	Current prop output	Auxiliary output #2
61,62	Auxiliary (Recorder) output	Auxiliary output #1

- **Operator Interface Section, Page 3-5**, - Note that the LED for "EV 12" is not applicable.
- **Programming Section, Page 4-3**, - Change *Setting Range* for *Parameter Prompt* EVT-NO for *Function Prompts* TIME-EVT and PV-EVT from "1 to 12" to 1 to 11.
- **Programming Section, Page 4-4**, - Change *Setting Range* for *Parameter Prompt* EVT-NO for *Function Prompt* DEV-EVT from "1 to 12" to 1 to 11. Make the following changes under *Comments for Function Prompt* PID-NO.
 - For Duplex operation (heat and cool)
 - Channel 1 (heat) uses PID 1
 - Channel 1 (cool) uses PID 2
 - Channel 2 (heat) uses PID 5
 - Channel 2 (cool) uses PID 6
- **Programming Section, Page 4-6**, - Under the heading *Programming Checklist*, replace the "For Duplex operation (heat) and (cool)" data in paragraph six with the following:
 - For Duplex operation (heat and cool)
 - Channel 1 (heat) will use PID 1
 - Channel 1 (cool) will use PID 2
 - Channel 2 (heat) will use PID 5
 - Channel 2 (cool) will use PID 6

■ **Configuration Section, Page 5-2, Table 5-1 - Make the following changes in the Auxiliary output and Event configuration *Selections*:**

Auxiliary outputs (DCPs w/2 duplex outputs only) #1 and #2	CH 1 - Set point	1	Auxiliary output #1: Provides a 4-20 mA output on Terminals 61 and 62 to represent the variable you select.
	- Process Variable	2	
	- Output	3	
	CH 2 - Set point or RH set point	4	Auxiliary output #2: Provides a 4-20 mA output on Terminals 59 and 60 to represent the variables you select.
	- Process variable or RH process variable (CH2)	5	
	- Output	6	
	(With Relative Humidity)		
	Wet bulb set point (CH2) TwSP	7	
Event configuration	Process variable as RH TwPV	8	
	Not used	0	Establishes the type for each of the eleven events. Once the type has been selected for an event it is effective on all programs stored in the DCP. Select a code for each of the eleven events.
	Time-based	1	
	PV-based	2	
	DEV-based	3	

■ **Configuration Section, Page 5-3, Table 5-1 - Make the following changes in the Output relay interval and output direction *Selections*:**

Output relay interval	CH 1 - Heat -4 to 600 sec		On DCPs with a time proportional controller, sets the operating interval of the output relay. Set in 4 sec. intervals
	- Cool -4 to 600 sec		
	CH 2 - Heat -4 to 600 sec		
	- Cool -4 to 600 sec		
Output Direction	CH 1	CH 2	Reverse action results in a decrease in the output when the input is above the set point. Direct action results in an increase in the output when the input is above the set point.
	Reverse	---	
	Direct	---	
	Reverse	Reverse	
	Direct	Direct	
	Reverse	Direct	
	Direct	Reverse	

■ **Configuration Section, Page 5-4, Table 5-1 - Make the following changes in the Duplex-time output relay deadband *Selection*:**

Duplex-time	CH 1: - 5.0 to 5.0%	Sets the deadband between the heat and cool relays
	CH 2: - 5.0 to 5.0%	

- **Configuration Section, Page 5-6, Table 5-2** - Make the following changes in the Recorder output and Event conf *Parameters*:

Recorder output	RECORDER 1		1		
	RECORDER 2			1	
Event conf	EVT-CONF				
Events	EVT1-4		0		
Events	EVT5-8			0	
Events	EVT9-11				0

- **Configuration Section, Page 5-7, Table 5-2** - Make the following changes in the Output relay interval and Time-prop relay deadband *Parameters*:

Output relay interval	TP-INTVL 1			
CH 1 - Duplex				
Duplex Heat	INTAVL - 1	4		
Duplex Cool	INTAVL - 2		4	
CH 2 - Duplex	TP-INTVL 2			
Duplex Heat	INTAVL - 1	4		
Duplex Cool	INTAVL - 2		4	
Time-prop relay deadband	DEADBAND1	0.0		
	DEADBAND2		0.0	

- **Configuration Section, Page 5-8** - Replace paragraph 4 under heading "Example of Event Configuration" with the following:

- Events 9, 10 and 11 are to be reconfigured as deviation-based events (Code 3).

- **Configuration Section, Page 5-8** - Delete "12" under PV Display in action column for Step 2, and change Steps 11, 12, and 13 as follows:

11		"EVT9-11" appears in upper display, and the "0" in the PV display blinks.
12		"EVT9-11" appears in upper display. "0" appears in the PV display, and the NUM LED blinks.
13	 (4 times)	The three 3s appear in the PV display.

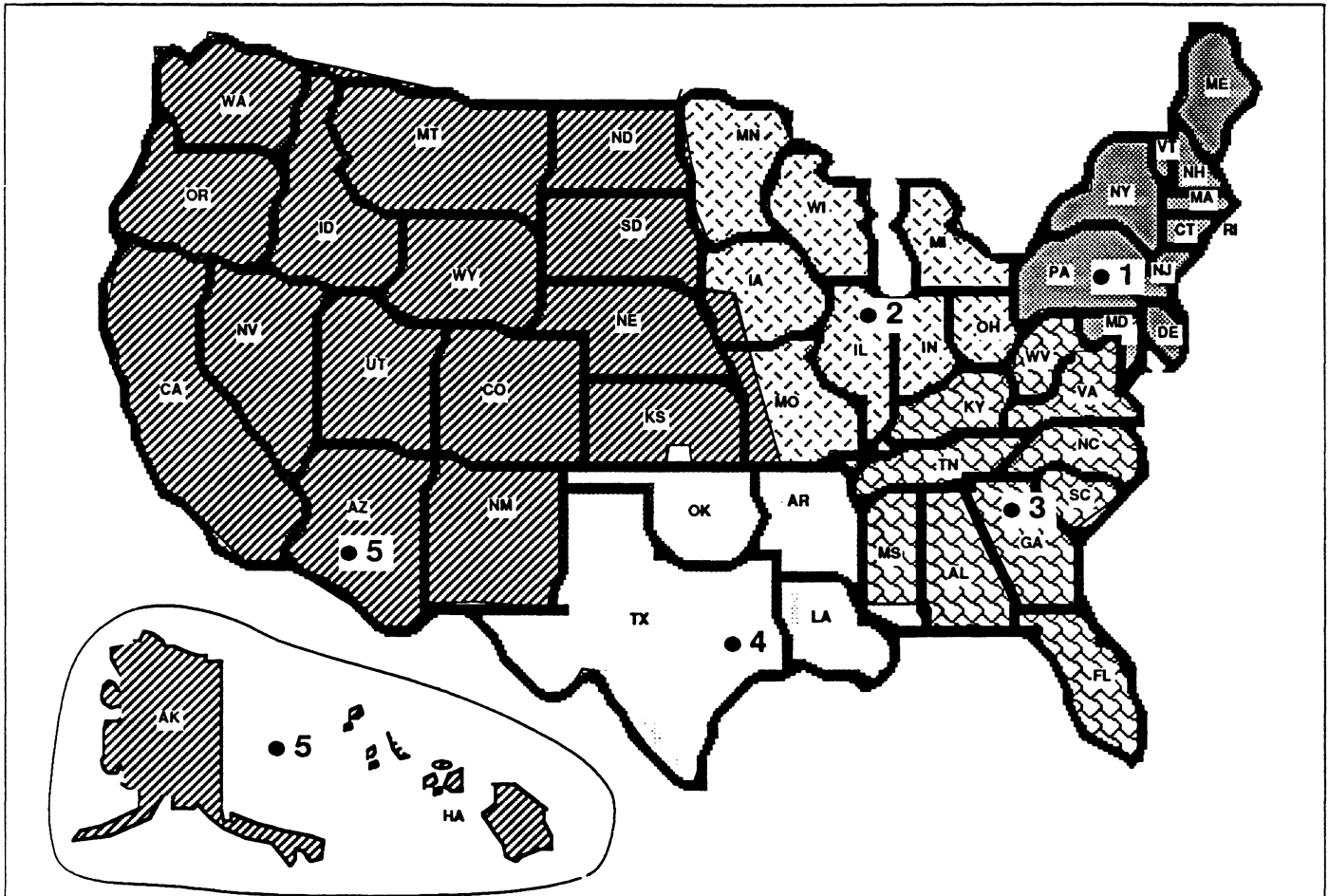
- Appendix A, Page A-3, Table A3 - Add the following code to the bottom of the table:

11	Time Proportional Duplex (Open Collector Transistor) Channel 1 Time Proportional Duplex (Open Collector Transistor) Channel 2
----	--

- Worksheet, 57-77-99-06, Page 2 - Make the following changes in Parameters:

Auxiliary output	_____	RECORDER 1	Output relay interval	TP-INTVL 1
	_____	RECORDER 2	Channel 1	
			Heat	_____
			Cool	_____
				INTAVL-1
				INTAVL-2
Event conf		EVT-CONF	Channel 2	TP-INTVL 2
Event 1	_____	EVT1-4	Heat	_____
Event 2	_____		Cool	_____
Event 3	_____			INTAVL-1
Event 4	_____			INTAVL-2
Event 5	_____	EVT5-8		
Event 6	_____			
Event 7	_____			
Event 8	_____			
Event 9	_____	EVT9-11		
Event 10	_____		Time-prop relay deadband	_____
Event 11	_____			DEADBAND 1
				DEADBAND 2

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